POWER ELECTRONICS

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Company / customer	POWER ELECTRONICS		
Project description	DC/DC GEN3		
Document designation	ESQ_FD1200U000002_000		
Project name	FD1200U000002		
Reference Project	FD1200U000001110		
Manufacturer	POWER ELECTRONICS	Power supply	
		Control voltage	
Product	DCDC	Part feature	
Type	HW1.0	Date	30/05/2024
Installation site	PAIS	Responsible for project	IP
Created on	29/05/2024	Creator	A.NIETO
Modification date	30/05/2024	Number of pages	181

Colour code

Power supply	N/A
Protective ground conductor	GNYE
AC Control voltage	BK
DC Control voltage	RD
External power voltage	N/A
Continuous presence of voltage	N/A

Colour abbreviations

	IEC 60757
Black	BK
Brown	BN
Red	RD
Orange	OG
Yellow	YE
Green	GN
Blue	BU
Violet	VT
Gray	GY
White	WH
Pink	PK
Turquoise	TQ
Green/Yellow	GNYE

Remarks

NC = No Wiring

Normalization

Wiring

IEC 60332-1
IEC 60204-1

Symbol
EN 81346-2

Symbol



Electrostatic discharge



Electric shock risk



Info

Structure identifier overview

POWER_IE_cambio_estrc

Plant identifier	Plant description	Mounting location identifier	Mounting location description
ZMI	INVERTER ASSEMBLY AREA	ADU	DU CABINET
MOI	INVERTER MODULE AREA	ADU.PADU	CUSTOMER BESS INPUT
PCTRL		ADU.PPCU	MAIN CUSTOMER PANEL
		ADU.PFU/BESS	FUSES PRE-ASSEMBLY BESS
		ADU.PFU/PV	FUSES PRE-ASSEMBLY PV OPTIONAL
		ADU.PDC/BESS	DC DISCONNECTOR PRE-ASSEMBLY BESS
		ADU.PDC/PV	DC DISCONNECTOR PRE-ASSEMBLY PV
		ADU.PDC	DC DISCONNECTORS PANEL MANUEVER
		ADU.PMV1/DU	FAN PRE-ASSEMBLY 1 DU
		ADU.PTA	SERVICES AUXILIARY TRANSFORMER ASSEMBLY
		ADU.PSA	POWER SUPPLY PANEL
		ADU.PCL	CONTROL PANEL
		ADU.PPU	PUSHBUTTONS PANEL
		ADU.PRC	HEATING RESISTOR PRE-ASSEMBLY
		ADU.PTH	THERMISTOR HEATING RESISTOR PRE-ASSEMBLY
		PMV2/BESS	FAN PRE-ASSEMBLY 2 BESS
		PMV3/PV	FAN PRE-ASSEMBLY 3 PV
		ETR_BESS	REMOVABLE DC/AC STAGE BESS
		ETR_PV	REMOVABLE DC/AC STAGE PV

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Superior level	Structure	Page	Page description
		1	Title page / cover sheet
		2	Information
		3	Structure identifier overview
		5	Table of contents
		6	Table of contents
		7	Table of contents
		8	Table of contents
		9	Table of contents
		10	Table of contents
		11	Table of contents
		3000	Plug overview : =+++ - =+++
		1	Track changes
		50	Single line diagram
		1	Title page / cover sheet
		11	Interconnection Areas ZMI - MOI
		100	Interconnection - Optical Fiber
		102	Interconnections - Comms RS485 Fans
		2000	Cable overview
	ADU	1	Title page / cover sheet
	ADU	10	Layout
	ADU.PADU	1	Title page / cover sheet
	ADU.PADU	10	Layout
	ADU.PADU	200	DC Input 1 (+) to PFU1
	ADU.PADU	201	DC Input 1 (-) to PFU1
	ADU.PADU	202	DC Input 2 (+) to PFU2
	ADU.PADU	203	DC Input 2 (-) to PFU2
	ADU.PADU	204	Earth connection
	ADU.PPCU	1	Title page / cover sheet

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Superior level	Structure	Page	Page description
ZMI	ADU.PPCU	10	Layout
	ADU.PPCU	11	Interconnection Areas PPCU-DU
	ADU.PPCU	20	Designation terminals and connectors
	ADU.PPCU	100	Interconnections PPCU/PSA
	ADU.PPCU	300	Interconnection Auxiliary Power Supply
	ADU.PPCU	301	Interconnection Customer Signals
	ADU.PPCU	1000	Bill of materials
	ADU.PPCU	2000	Cable overview
	ADU.PPCU	3000	Plug overview : =DCDC++ZMI+ADU.PPCU-XC01.1 - =DCDC++ZMI+ADU.PPCU-XC10.1
	ADU.PFU/BESS	1	Title page / cover sheet
	ADU.PFU/BESS	20	Designation terminals and connectors
	ADU.PFU/BESS	100	Interconnections SPD / Fuses
	ADU.PFU/BESS	200	DC Input 1 (+)
	ADU.PFU/BESS	201	Surge protector / Feedback fuse
	ADU.PFU/BESS	202	RFI DC
	ADU.PFU/BESS	1000	Bill of materials
	ADU.PFU/BESS	3000	Plug overview : =DCDC++ZMI+ADU.PFU/BESS-XC41.2 - =DCDC++ZMI+ADU.PFU/BESS-XC41.2
	ADU.PFU/PV	1	Title page / cover sheet
	ADU.PFU/PV	200	DC Input 1 (+)
	ADU.PFU/PV	202	RFI DC
	ADU.PFU/PV	1000	Bill of materials
	ADU.PDC/BESS	1	Title page / cover sheet
	ADU.PDC/BESS	20	Designation terminals and connectors
	ADU.PDC/BESS	200	Power DC Disconnector (DC+/-)
	ADU.PDC/BESS	300	Disconnector switch DC 1
	ADU.PDC/BESS	1000	Bill of materials
	ADU.PDC/BESS	3000	Plug overview : =DCDC++ZMI+ADU.PDC/BESS-XC470.1 - =DCDC++ZMI+ADU.PDC/BESS-XC471.1
	ADU.PDC/PV	1	Title page / cover sheet

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Superior level	Structure	Page	Page description
ZMI	ADU.PDC/PV	20	Designation terminals and connectors
	ADU.PDC/PV	200	Power DC Disconnector (DC+/-)
	ADU.PDC/PV	300	Disconnector switch DC 1
	ADU.PDC/PV	1000	Bill of materials
	ADU.PDC/PV	3000	Plug overview : =DCDC++ZMI+ADU.PDC/PV-XC475.1 - =DCDC++ZMI+ADU.PDC/PV-XC476.1
	ADU.PDC	1	Title page / cover sheet
	ADU.PDC	10	Layout
	ADU.PDC	20	Designation terminals and connectors
	ADU.PDC	100	Interconnection
	ADU.PDC	301	Distribution 120 VDC
	ADU.PDC	400	PCB´s - Disconnector - Module
	ADU.PDC	1000	Bill of materials
	ADU.PDC	3000	Plug overview : =DCDC++ZMI+ADU.PDC-E55-J2 - =DCDC++ZMI+ADU.PDC-XC425.2
	ADU.PMV1/DU	1	Title page / cover sheet
	ADU.PMV1/DU	10	Layout
	ADU.PMV1/DU	11	Layout
	ADU.PMV1/DU	20	Designation terminals and connectors
	ADU.PMV1/DU	200	DU - Fan 1
	ADU.PMV1/DU	2000	Cable overview
	ADU.PMV1/DU	3000	Plug overview : =DCDC++ZMI+ADU.PMV1/DU-X111 - =DCDC++ZMI+ADU.PMV1/DU-XC111.2
	ADU.PTA	1	Title page / cover sheet
	ADU.PTA	10	Layout
	ADU.PTA	20	Designation terminals and connectors
	ADU.PTA	101	Interconnection PTA - PPCU
	ADU.PTA	103	Interconnection PTA - PCL
	ADU.PTA	200	SPD
	ADU.PTA	201	SSAA Auto transformer
	ADU.PTA	1000	Bill of materials

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Superior level	Structure	Page	Page description
ZMI	ADU.PTA	3000	Plug overview : =DCDC++ZMI+ADU.PTA-XC51.1 - =DCDC++ZMI+ADU.PTA-XC51.1
	ADU.PSA	1	Title page / cover sheet
	ADU.PSA	10	Layout
	ADU.PSA	20	Designation terminals and connectors
	ADU.PSA	101	Interconnections PSA/PCL
	ADU.PSA	102	Interconnections PSA/PPCU
	ADU.PSA	201	Bridge Rectifier
	ADU.PSA	301	120Vdc Distribution
	ADU.PSA	400	PCB - Inputs/Outputs
	ADU.PSA	1000	Bill of materials
	ADU.PSA	3000	Plug overview : =DCDC++ZMI+ADU.PSA-E10-J2 - =DCDC++ZMI+ADU.PSA-XC20.2
	ADU.PCL	1	Title page / cover sheet
	ADU.PCL	10	Layout
	ADU.PCL	20	Designation terminals and connectors
	ADU.PCL	101	Interconnections PCL/PSA-PDC
	ADU.PCL	102	Interconnections PCL/PDC Voltage Measurement
	ADU.PCL	103	Interconnections PCL/PPU
	ADU.PCL	104	Interconnections PCL/PFU (SPD / Fuses)
	ADU.PCL	105	Interconnection PCL/SSAA
	ADU.PCL	106	Interconnections - RJ45 (I/O)
	ADU.PCL	107	Interconnections - ETHERNET/USB
	ADU.PCL	108	Interconnections - PCL / MOI (FANS)
	ADU.PCL	201	Fan distribution
	ADU.PCL	202	24Vdc Power Supplies
	ADU.PCL	204	24Vdc distribution
	ADU.PCL	250	BESS/PV Measure (+/-)
	ADU.PCL	300	120Vdc Distribution
	ADU.PCL	301	Ethernet Switch

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Superior level	Structure	Page	Page description
ZMI	ADU.PCL	400	PCB Control Panel
	ADU.PCL	401	PCB Inputs/Outputs
	ADU.PCL	1000	Bill of materials
	ADU.PCL	2000	Cable overview
	ADU.PCL	3000	Plug overview : =DCDC++ZMI+ADU.PCL-E01-J1 - =DCDC++ZMI+ADU.PCL-E11-J3
	ADU.PCL	3001	Plug overview : =DCDC++ZMI+ADU.PCL-J2 - =DCDC++ZMI+ADU.PCL-XC425.1
	ADU.PPU	1	Title page / cover sheet
	ADU.PPU	10	Layout
	ADU.PPU	11	Layout
	ADU.PPU	20	Designation terminals and connectors
	ADU.PPU	100	Interconnections - PPU/PCL
	ADU.PPU	300	Door Control Maneuver
	ADU.PPU	301	Door Control Maneuver
	ADU.PPU	302	General Fault Maneuver
	ADU.PPU	400	PCB´s - Inputs/Outputs
	ADU.PPU	1000	Bill of materials
	ADU.PPU	3000	Plug overview : =DCDC++ZMI+ADU.PPU-E12-J2 - =DCDC++ZMI+ADU.PPU-XC31.2
	ADU.PRC	1	Title page / cover sheet
	ADU.PRC	1000	Bill of materials
	ADU.PTH	1	Title page / cover sheet
	ADU.PTH	1000	Bill of materials
		1	Title page / cover sheet
		10	Layout
		11	Interconnection Areas ZMI - MOI
		12	Inductance Wiring conduction
		20	Designation terminals and connectors
		100	Interconnections - Comms RS485 Fans
		101	Interconnections - PCL / MOI (FANS)

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Superior level	Structure	Page	Page description
MOI		200	AC Inductance (Power busbar)
		201	AC Inductance (Terminals block)
		400	PCB control and measures
		1000	Bill of materials
		2000	Cable overview
		3000	Plug overview : =DCDC++MOI+-E04-J1 - =DCDC++MOI+-XC115.1
	PMV2/BESS	1	Title page / cover sheet
	PMV2/BESS	10	Layout
	PMV2/BESS	11	Layout
	PMV2/BESS	20	Designation terminals and connectors
	PMV2/BESS	200	Power module BESS - Fan 2
	PMV2/BESS	2000	Cable overview
	PMV2/BESS	3000	Plug overview : =DCDC++MOI+PMV2/BESS-X112 - =DCDC++MOI+PMV2/BESS-XC112.2
	PMV3/PV	1	Title page / cover sheet
	PMV3/PV	10	Layout
	PMV3/PV	11	Layout
	PMV3/PV	20	Designation terminals and connectors
	PMV3/PV	200	Power module PV - Fan 3
	PMV3/PV	1000	Bill of materials
	PMV3/PV	2000	Cable overview
	PMV3/PV	3000	Plug overview : =DCDC++MOI+PMV3/PV-X113 - =DCDC++MOI+PMV3/PV-XC113.2
	ETR_BESS	1	Title page / cover sheet
	ETR_BESS	10	Layout
	ETR_BESS	20	Designation terminals and connectors
	ETR_BESS	200	Capacitor bus dc-link BESS
	ETR_BESS	201	Igbt assembly BESS
	ETR_BESS	202	Igbt assembly BESS
	ETR_BESS	400	V/I conversor

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Superior level	Structure	Page	Page description
MOI	ETR_BESS	401	PCB control and measures
	ETR_BESS	1000	Bill of materials
	ETR_BESS	2000	Cable overview
	ETR_BESS	3000	Plug overview : =DCDC++MOI+ETR_BESS-E03-J1 - =DCDC++MOI+ETR_BESS-XC026.1
	ETR_BESS	3001	Plug overview : =DCDC++MOI+ETR_BESS-XC031.1 - =DCDC++MOI+ETR_BESS-XC043.1
	ETR_PV	1	Title page / cover sheet
	ETR_PV	10	Layout
	ETR_PV	20	Designation terminals and connectors
	ETR_PV	200	Capacitor bus dc-link PV
	ETR_PV	201	Igbt assembly PV
	ETR_PV	202	Igbt assembly PV
	ETR_PV	400	V/I conversor
	ETR_PV	401	PCB control and measures
	ETR_PV	1000	Bill of materials
	ETR_PV	2000	Cable overview
	ETR_PV	3000	Plug overview : =DCDC++MOI+ETR_PV-E03-J1 - =DCDC++MOI+ETR_PV-XC026.1
	ETR_PV	3001	Plug overview : =DCDC++MOI+ETR_PV-XC031.1 - =DCDC++MOI+ETR_PV-XC043.1

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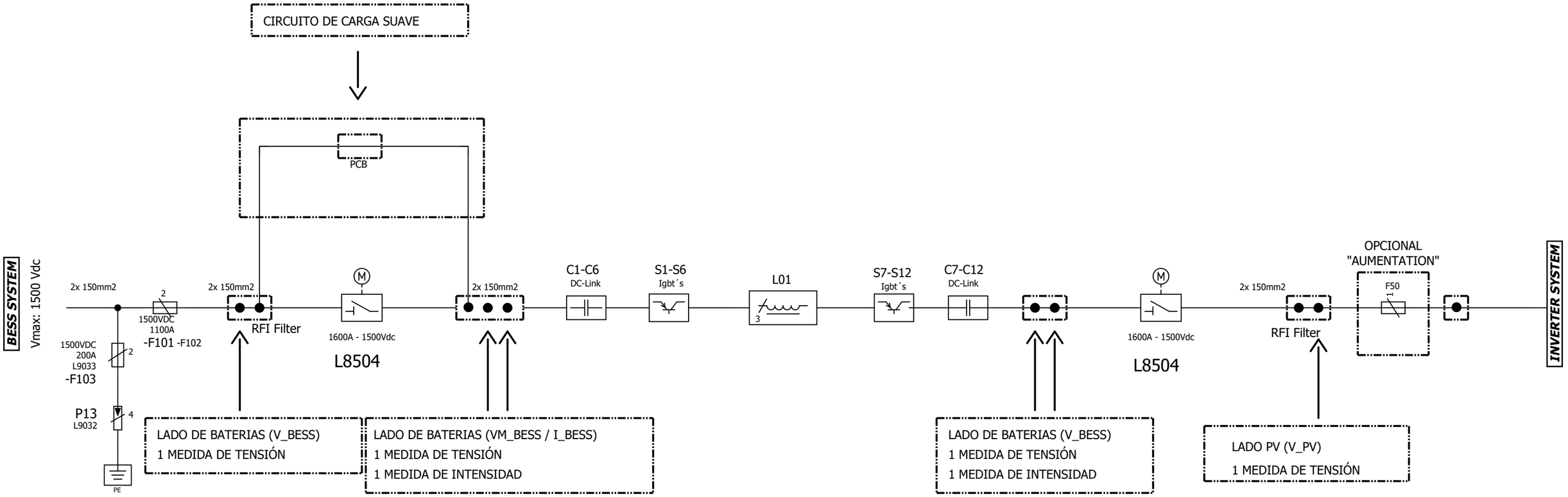
Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=+++			

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project ESQ_FD1200U000002_000	Name Plug overview : =+++ - =+++	Installation:	A3
	REVISED	REVISION 0.0.0			Mounting location:	Previous page &/11 / Next page =DCDC&DG/1
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 3000 / Total number of pages 181

Name	Date	Description	Remarks
0.1	OCT-2022	Liberación prototipo HW0.1	
0.2	MAY-2023	Liberación prototipo HW0.2	
1.0	NOV-2023	Liberación prototipo HW1.0	
1.0	ABR-2024	ACTUALIZACIÓN A M&L	

Single Line Diagram --> DC/DC 1200kW



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	Single line diagram	Mounting location:	Previous page 1 / Next page ++ZMI&/1
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 50 / Total number of pages 181

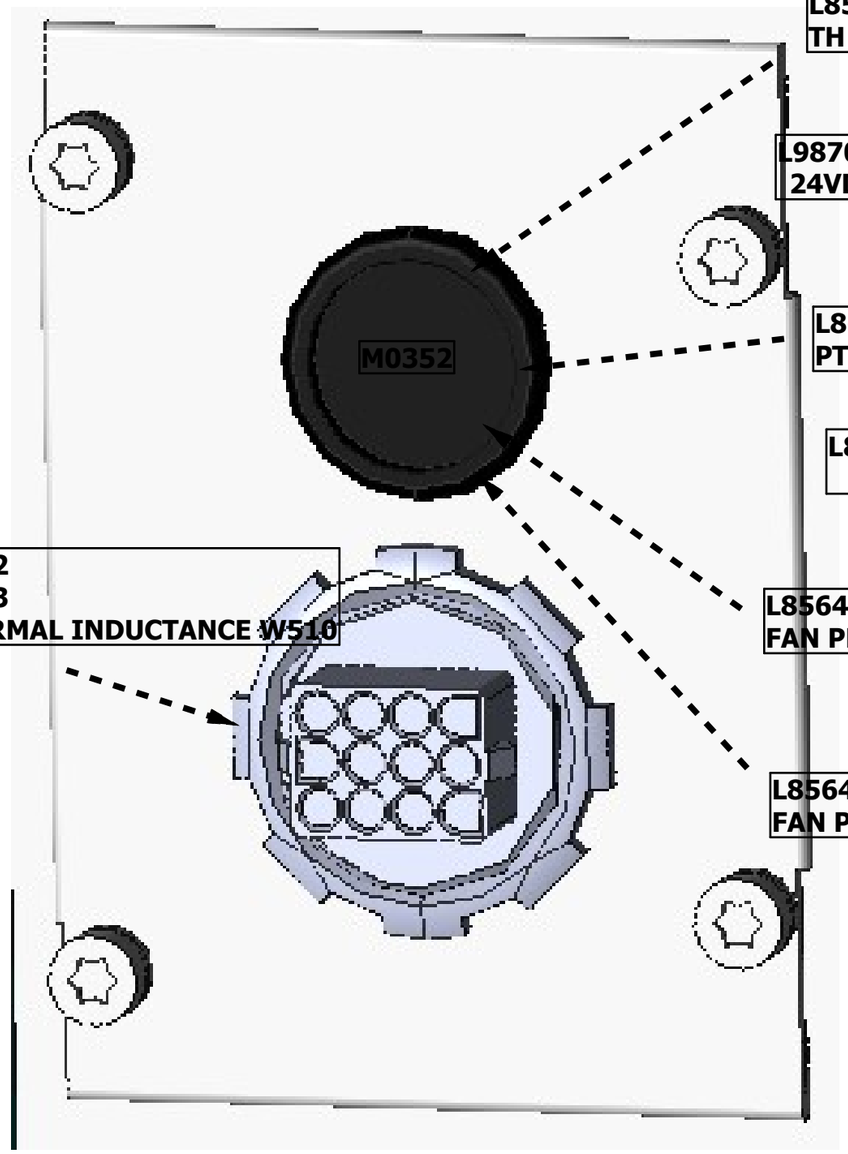
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

DCDC DCDC GEN 3 CONVERTER

ZMI INVERTER ASSEMBLY AREA

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

Interconnection Areas
ZMI - MOI



L8565 (Manguito 5-6)
TH IND (W051)

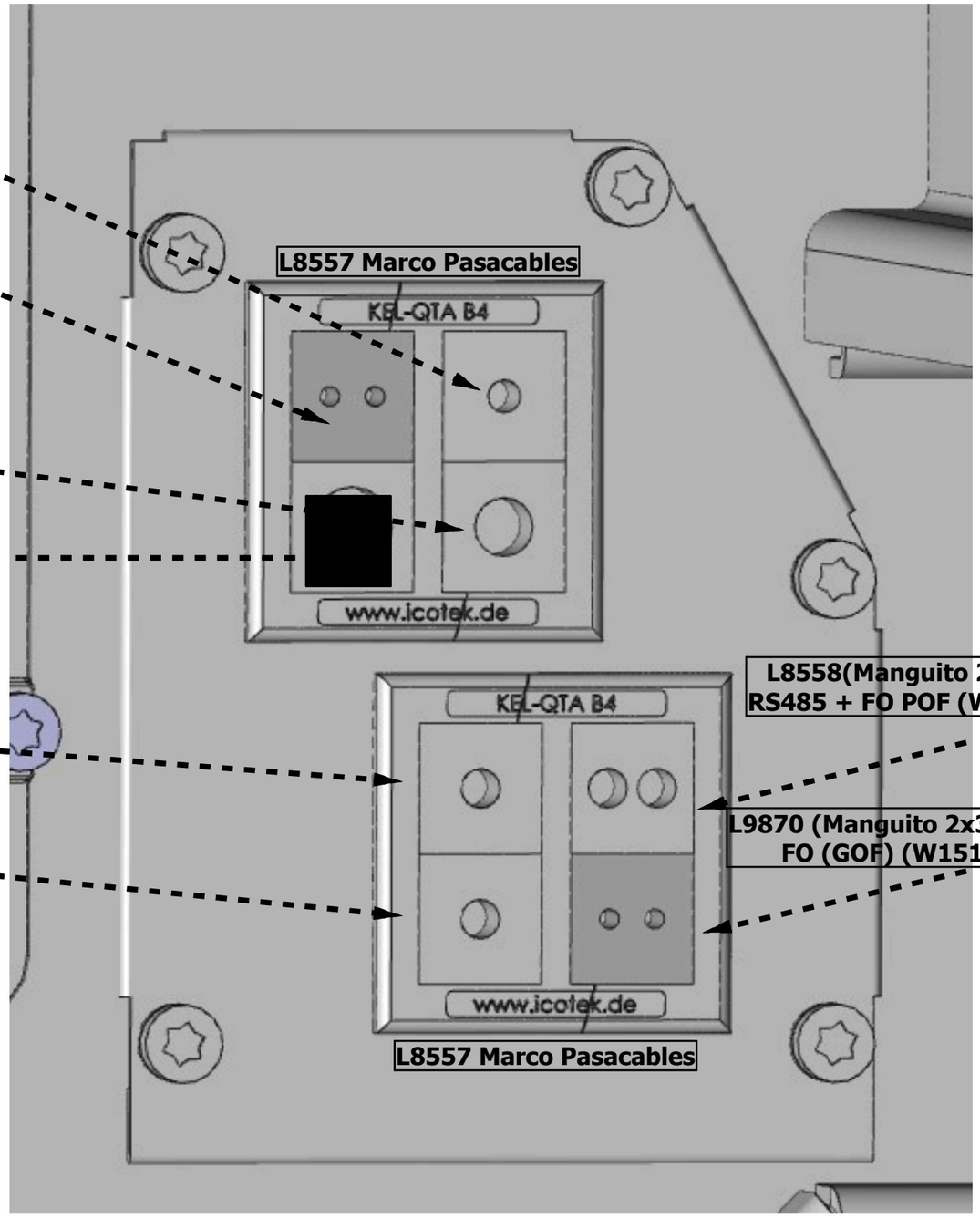
L9870 (Manguito 2x3mm)
24VDC 2kV PCB MODULO

L8625 (Manguito 9-10)
PT100 INDC.1-2-3 (W050)

L8562 (Manguito Ciego)
SPARE

L8564 (Manguito 6-7)
FAN PMV2 W112

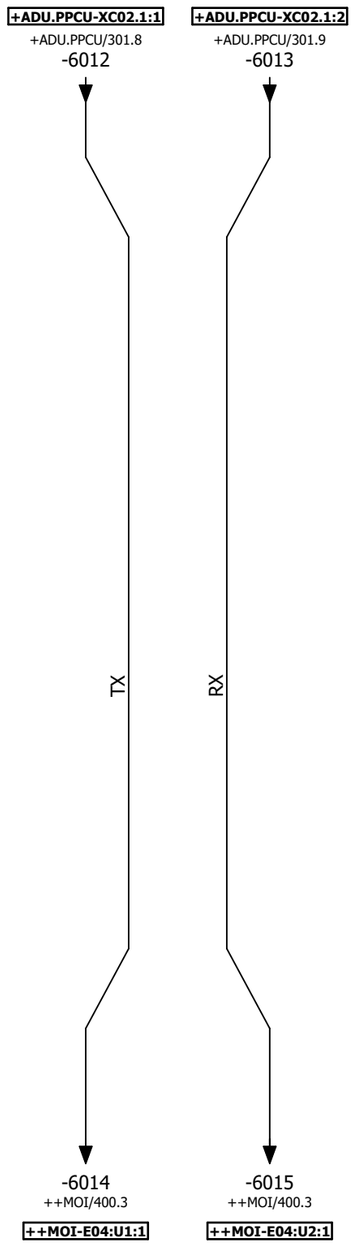
L8564 (Manguito 6-7)
FAN PMV3 W113



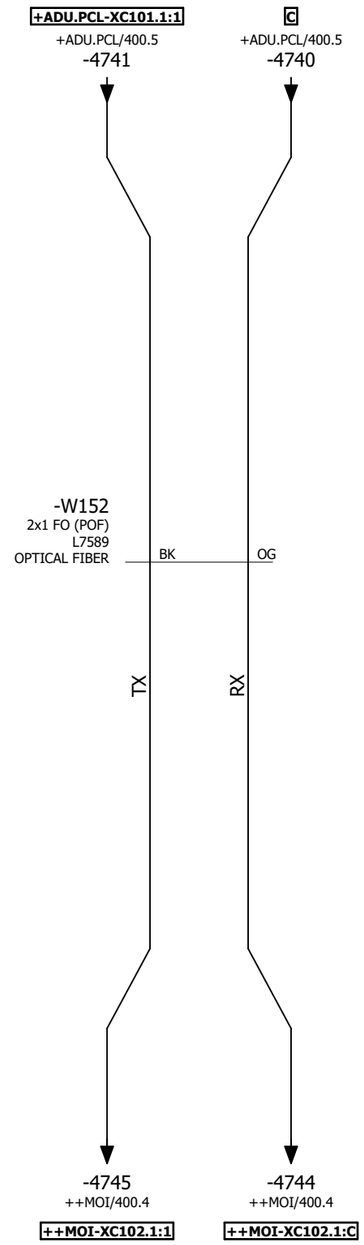
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: Interconnection Areas ZMI - MOI	Installation: DCDC GEN 3 CONVERTER — DCDC — A3
	REVISED	REVISION 0.0.0			Mounting location: — Previous page 1 — Next page 100
	APPROVED A.CABALLERO				Project Code: FD1200U000002 — Page 11 — Total number of pages 181

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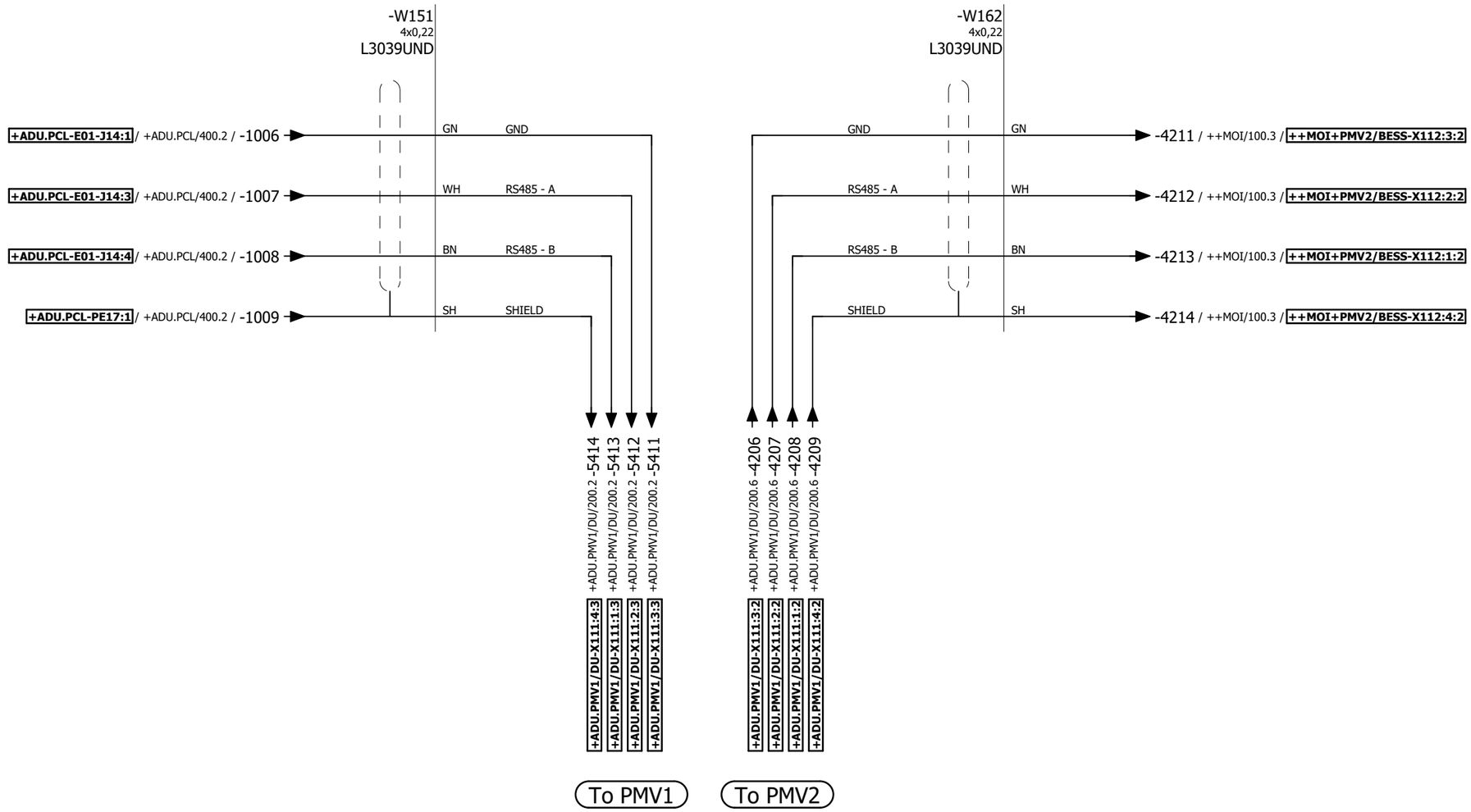
STAND ALONE



FPGA



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Interconnection - Optical Fiber	Mounting location:	Previous page 11	Next page 102
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 100	Total number of pages 181



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POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
REVISED		ESQ_FD1200U000002_000	Interconnections - Comms RS485 Fans	Mounting location:	Previous page 100 / Next page &CAB/2000
APPROVED A.CABALLERO	REVISION 0.0.0			Project Code: FD1200U000002	Page 102 / Total number of pages 181

Cable overview

Cable designation	Function text	ERP
=DCDC++ZMI+-W151		L3039UND
=DCDC++ZMI+-W152	OPTICAL FIBER	L7589
=DCDC++ZMI+-W162	SHIELD	L3039UND

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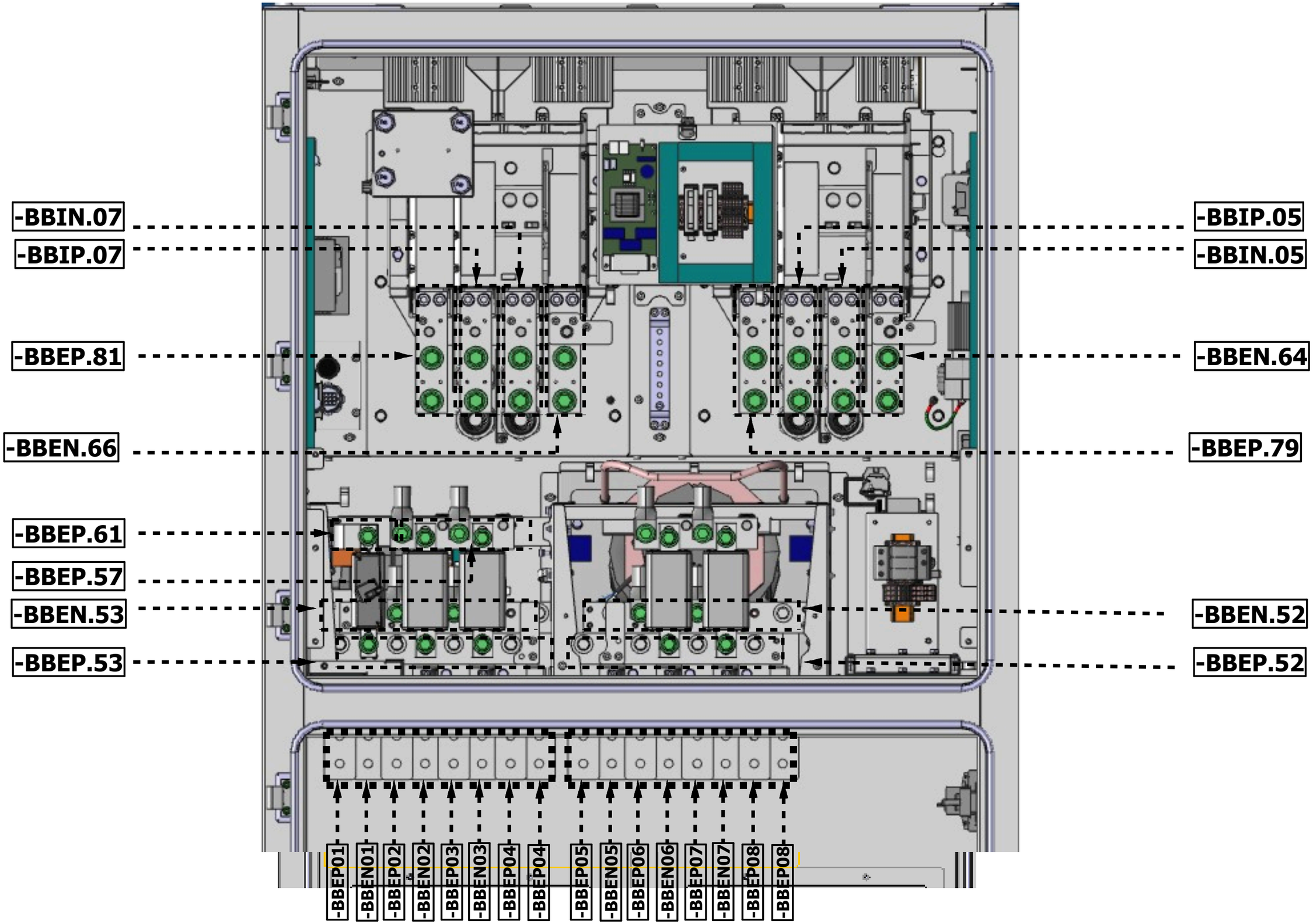
DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU	DU CABINET

ADU

DU CABINET

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

General layout



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Layout	Mounting location: DU CABINET ADU	Previous page 1 / Next page +.PADU/1
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 10 / Total number of pages 181

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CV35 SALIDA 30

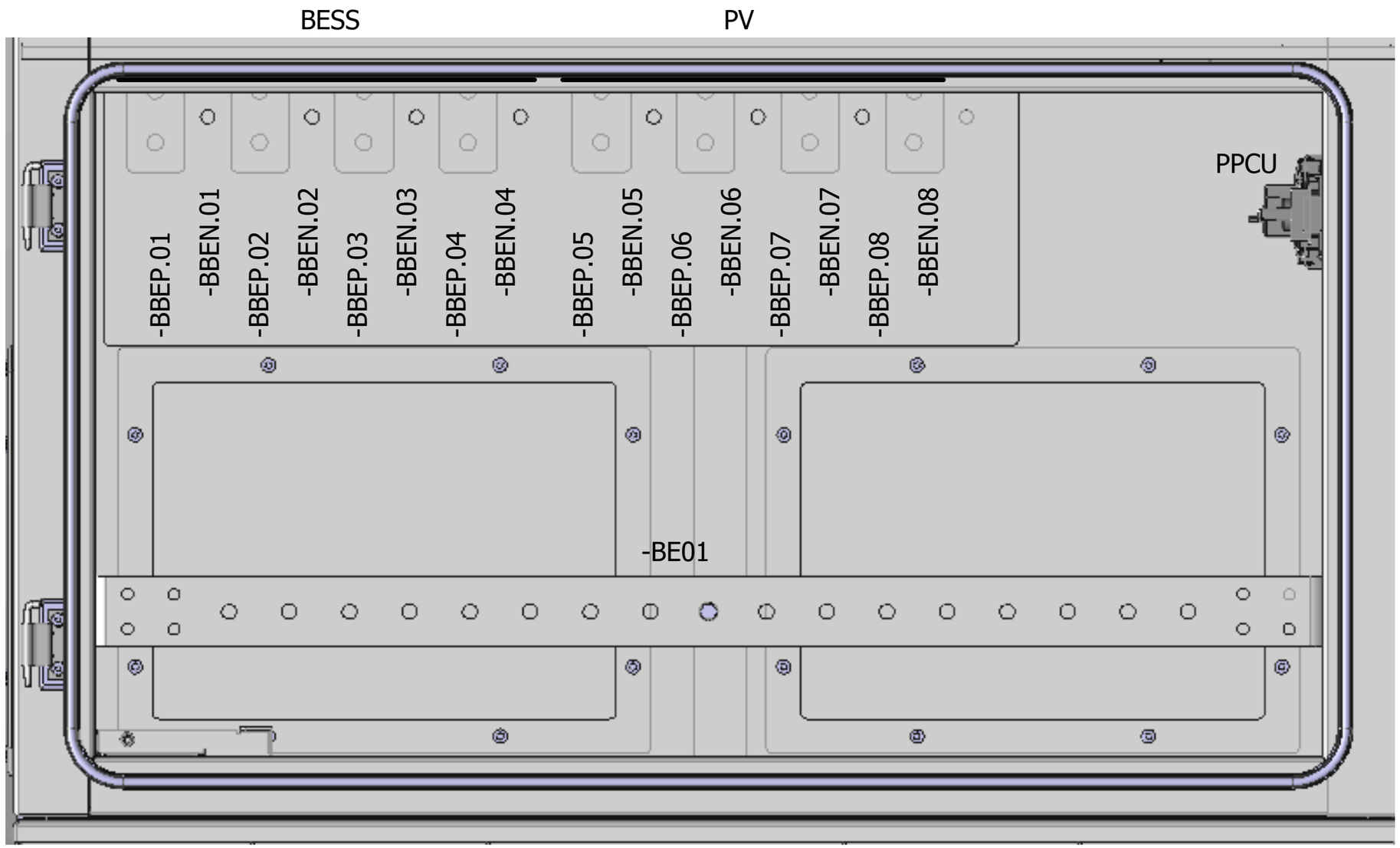
DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PADU	CUSTOMER BESS INPUT

PADU

CUSTOMER BESS INPUT

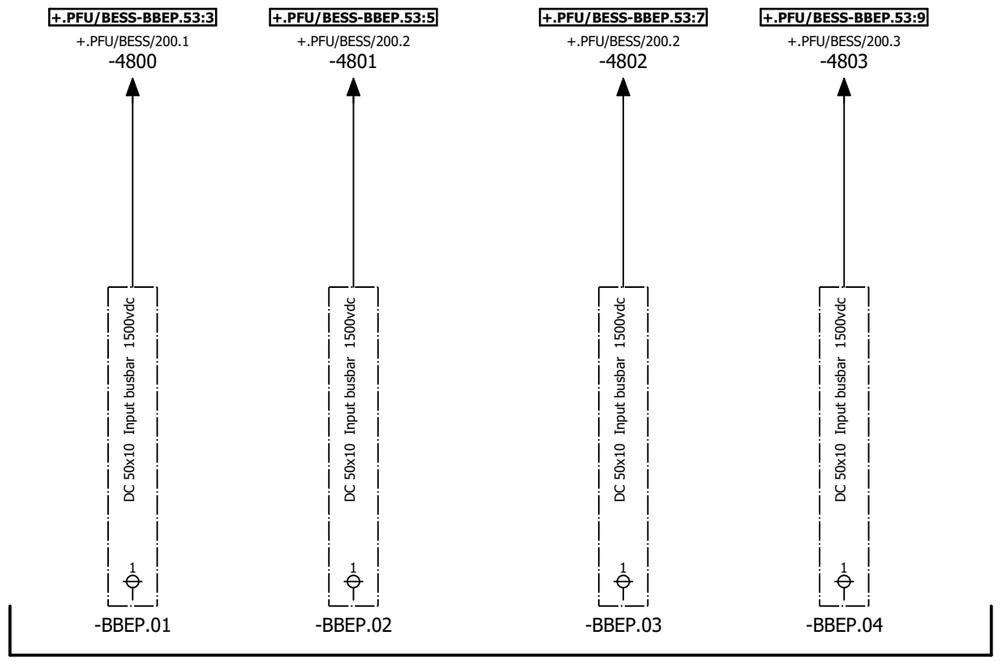
Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

Layout



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Layout	Mounting location: DU CABINET ADU.PADU	Previous page 1 / Next page 200
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 10 / Total number of pages 181

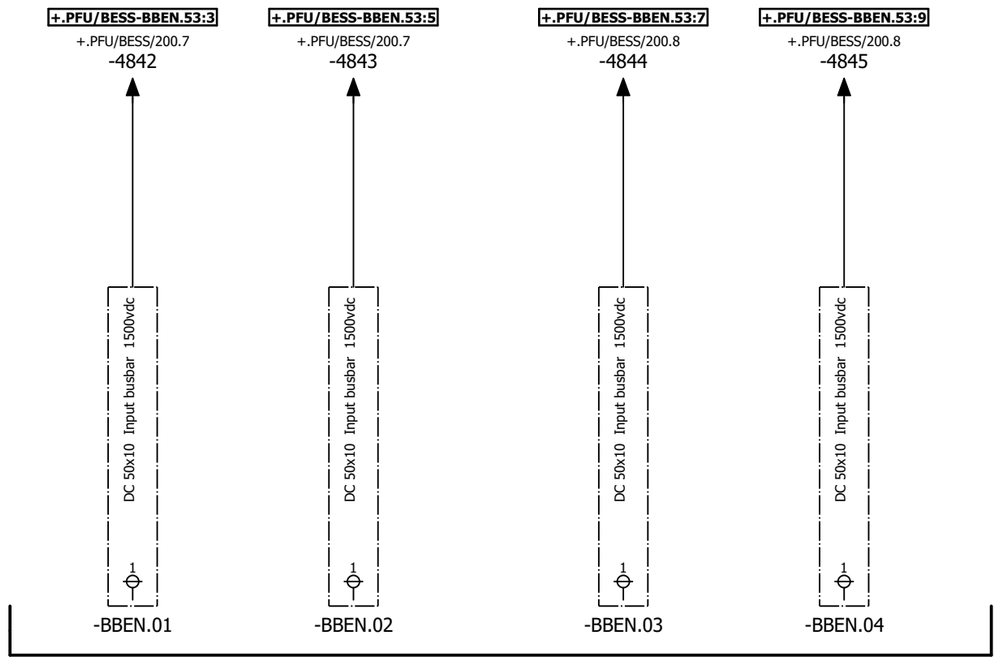
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DC CONNECTION FROM BESS SYSTEM TO DU INPUT BUSBARS 50x10 1500vdc

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	DC Input 1 (+) to PFU1	Mounting location: DU CABINET ADU.PADU	Previous page 10	Next page 201
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 200	Total number of pages 181

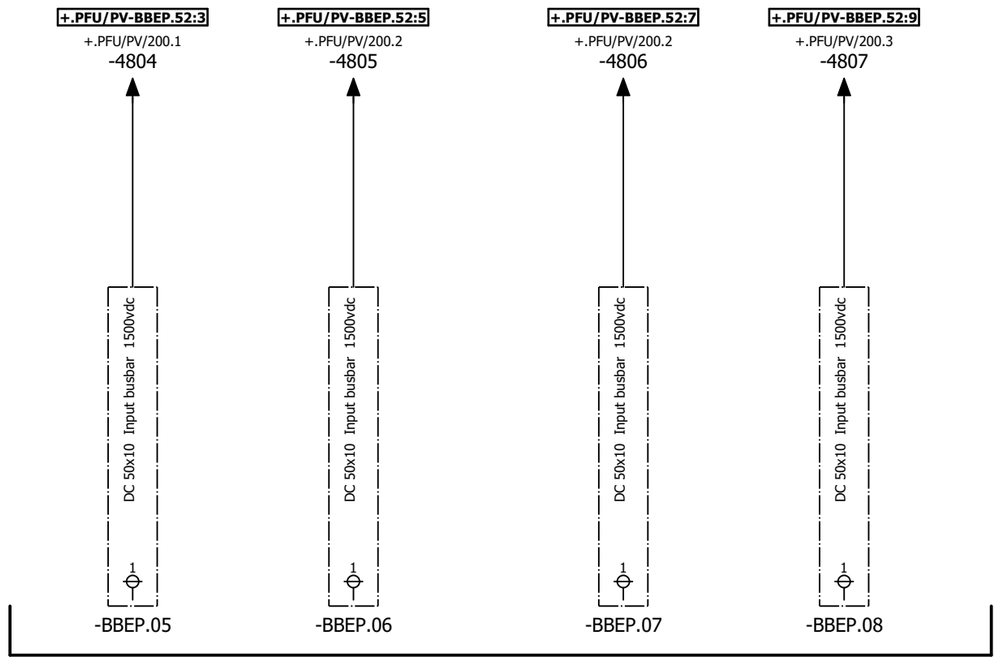
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DC CONNECTION FROM BESS SYSTEM TO DU INPUT BUSBARS 50x10 1500vdc

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION	ESQ_FD1200U000002_000	DC Input 1 (-) to PFU1	Mounting location: DU CABINET ADU.PADU	Previous page 200 / Next page 202
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 201 / Total number of pages 181

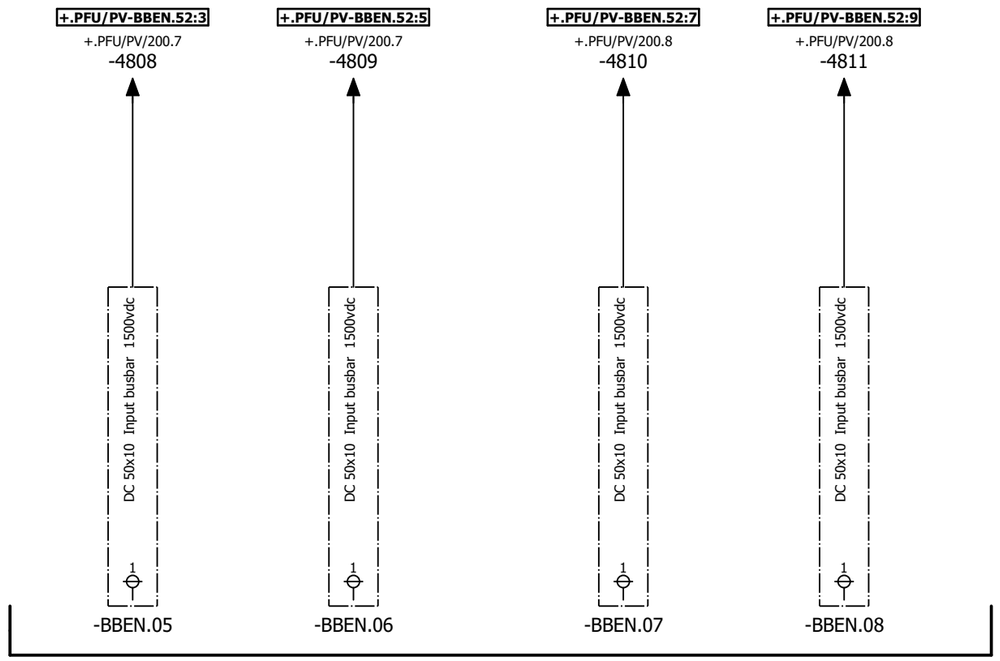
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DC CONNECTION FROM PV SYSTEM TO DU INPUT BUSBARS 50x10 1500vdc

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	DC Input 2 (+) to PFU2	Mounting location: DU CABINET ADU.PADU	Previous page 201 / Next page 203
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 202 / Total number of pages 181

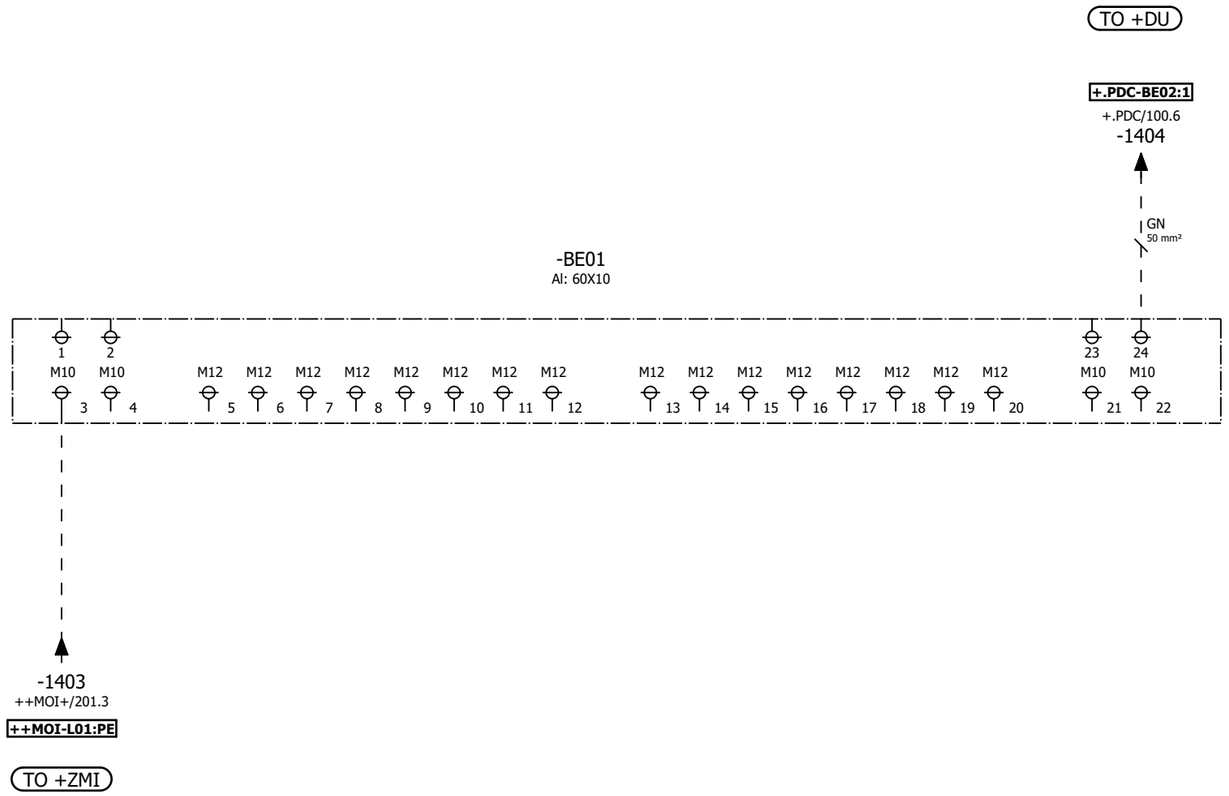
0	1	2	3	4	5	6	7	8	9



DC CONNECTION FROM PV SYSTEM TO DU INPUT BUSBARS 50x10 1500vdc

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project ESQ_FD1200U000002_000	Name DC Input 2 (-) to PFU2	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PADU	Previous page 202 / Next page 204
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 203 / Total number of pages 181

Earth connection



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project ESQ_FD1200U000002_000	Name Earth connection	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PADU	Previous page 203 / Next page +.PPCU/1
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 204 / Total number of pages 181

POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

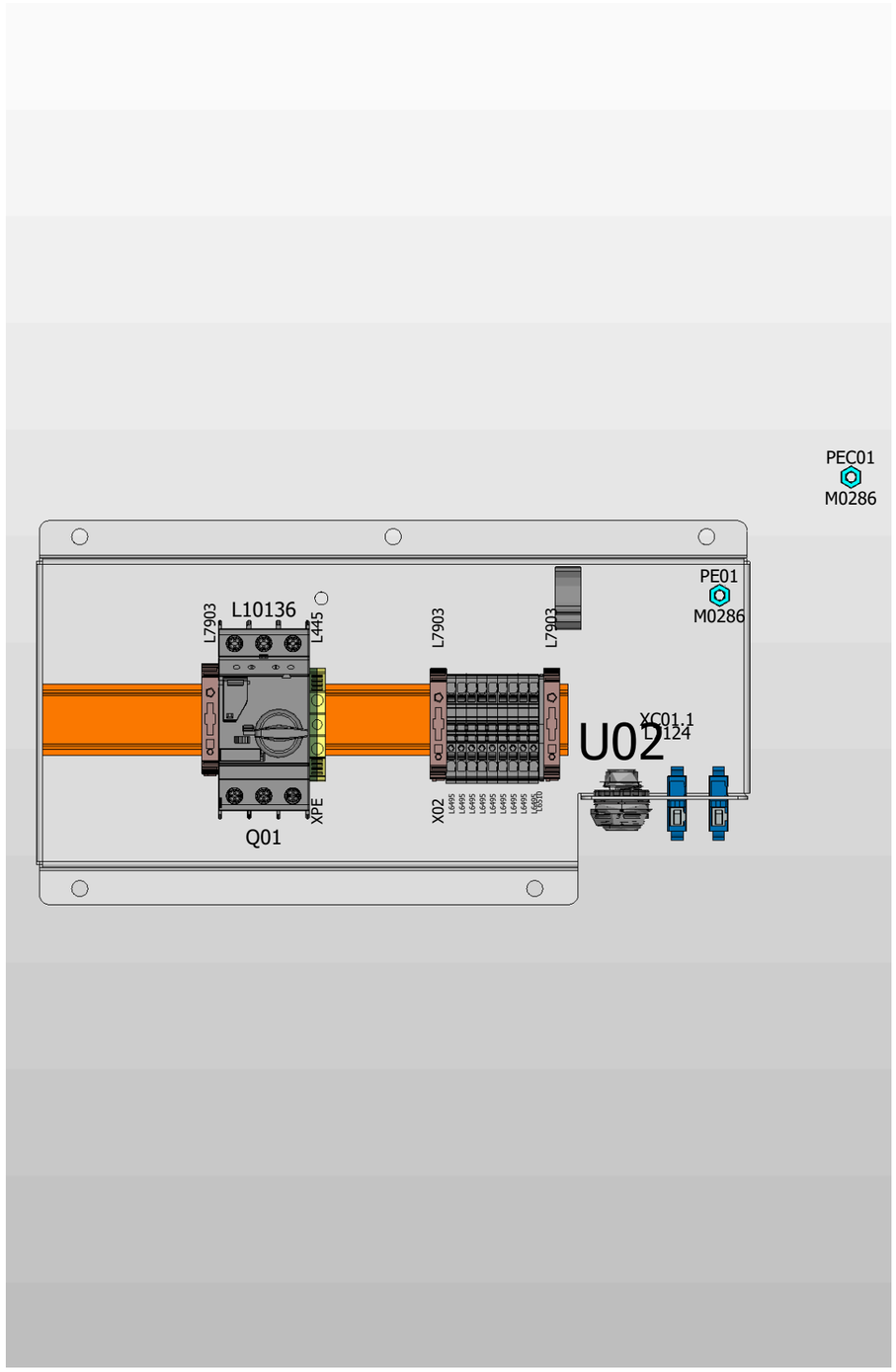
DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PPCU	MAIN CUSTOMER PANEL

PPCU

MAIN CUSTOMER PANEL

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

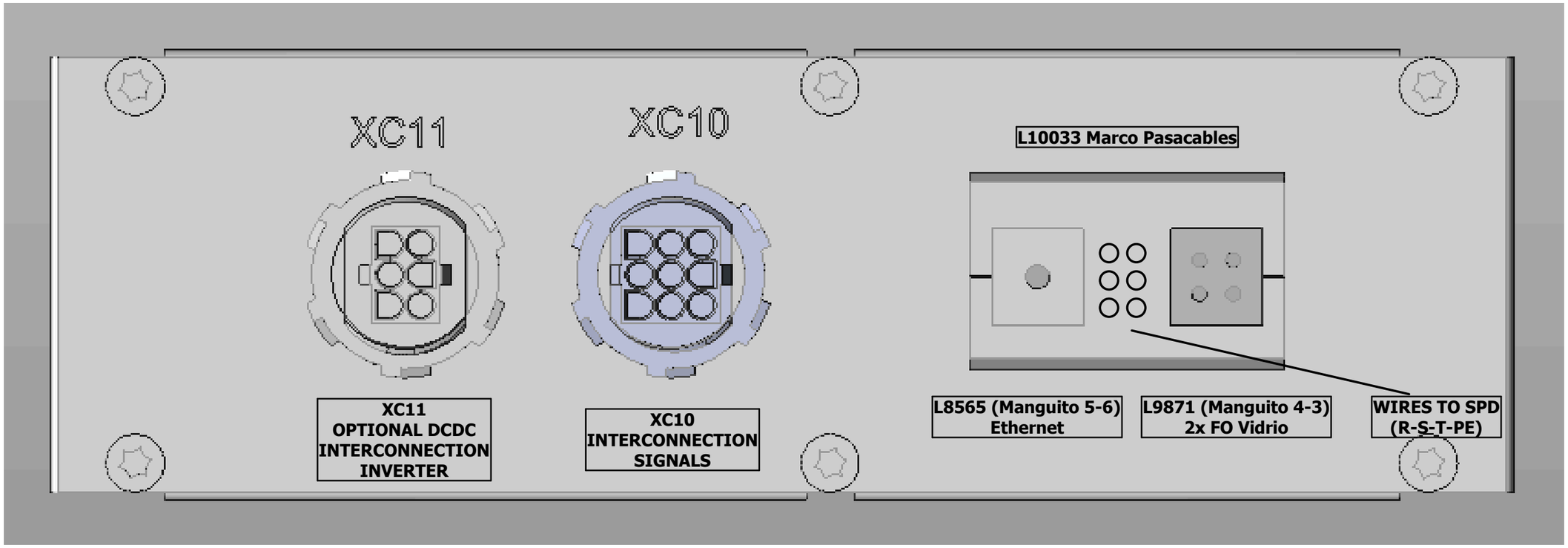
LAYOUT PANEL



Enclosure legend

IME	ERP	Description
Q01	L10136	GUARDAMOTOR MMS-32H-22A 3P 14-22A UL
U64	L1062	DIN 35x15mm
U02	L4811	INSTALLATION SOCKET RJ45 CAT5E IP65

Interconnection Areas
User zone PPU - DU



0	1	2	3	4	5	6	7	8	9

-X02 = Customer Signals Interconnections

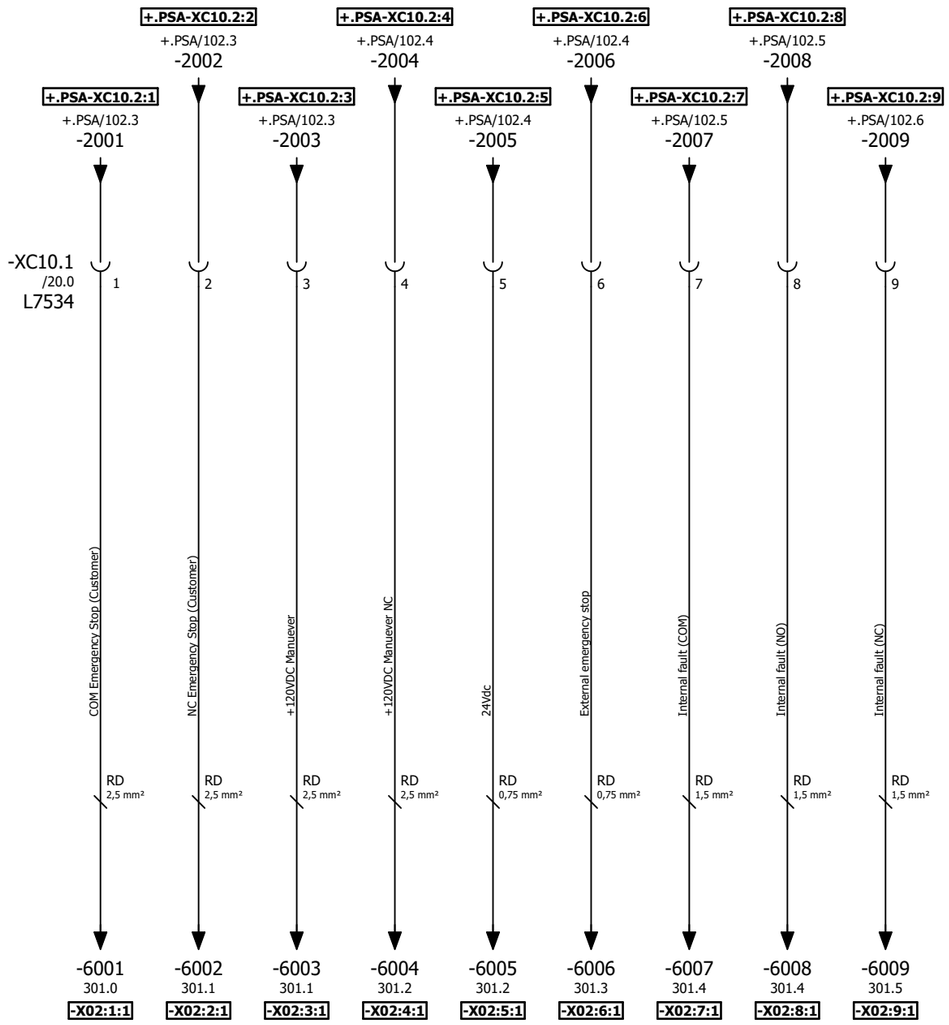
-XC10.1 = M&L 9P AEREAL INTERCONNECTION PPCU

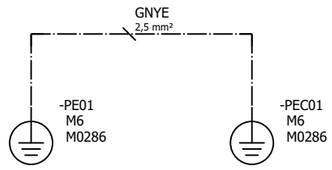
XC01.1 = CONECTOR CONEXION F.O. A SWITCH
XC02.1 = CONECTOR CONEXION STANDALONE

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Designation terminals and connectors	Mounting location:	DU CABINET ADU.PPCU	Previous page 11	Next page 100
APPROVED A.CABALLERO	0.0.0			Project Code:	FD1200U000002	Page 20	Total number of pages 181

Interconnections PPCU-PSA

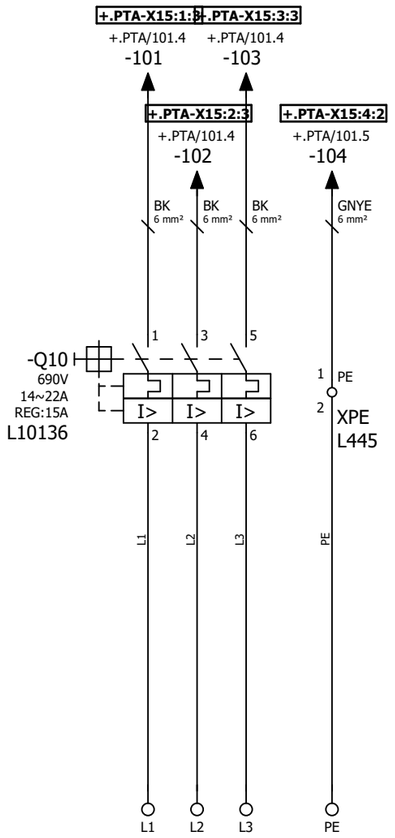




POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Interconnections PPCU/PSA	Mounting location: DU CABINET ADU.PPCU	Previous page 20 / Next page 300
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 100 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

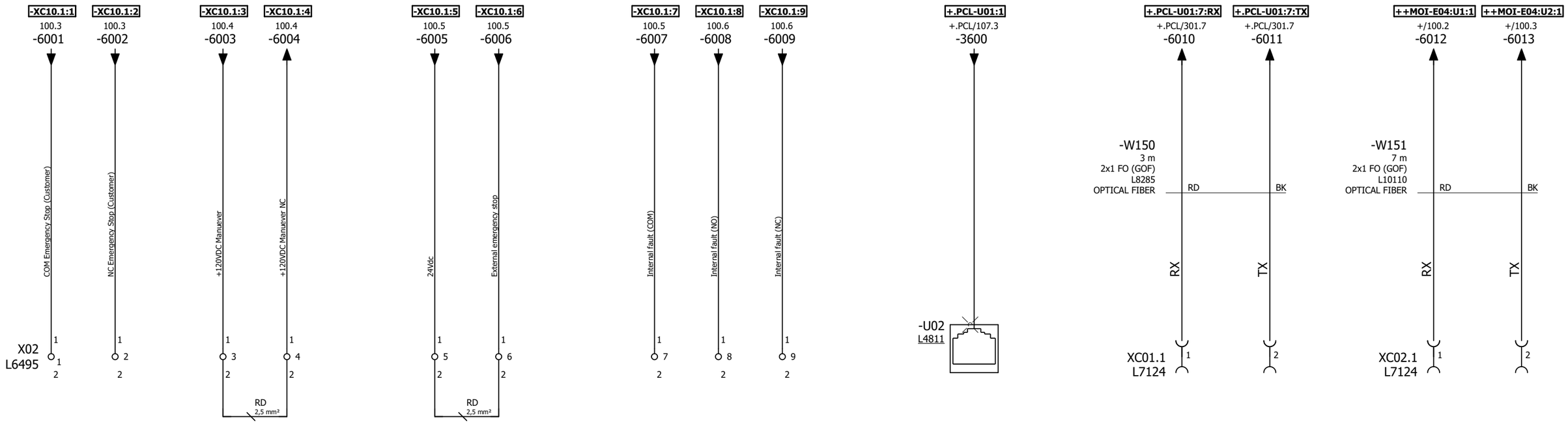
Auxiliary power Supply (PTA)



AUXILIARY
POWER SUPPLY
UL: 3x208VAC+PE
IEC:3x400VAC+PE

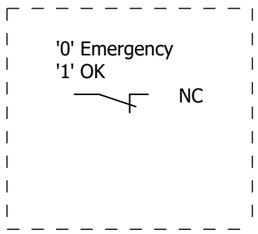
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project ESQ_FD1200U000002_000	Name Interconnection Auxiliary Power Supply	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PPCU	Previous page 100 / Next page 301
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 300 / Total number of pages 181

Customer Signals Interconnections

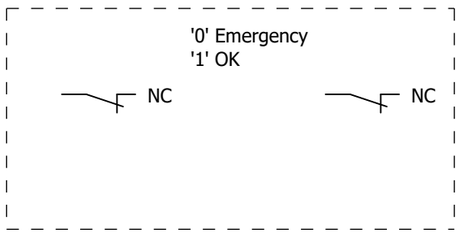


**NOTE:
If used, retire bridges between:
X02:3:2 - X02:4:2
X02:5:2 - X02:6:2

EMERGENCY STOP (TO CUSTOMER)

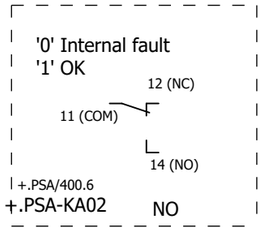


EMERGENCY STOP (FROM CUSTOMER)



EMERGENCY STOP (FROM CUSTOMER)

DCDC INTERNAL FAULT (TO CUSTOMER)



CUSTOMER COMMUNICATION ETHERNET CONNECTION

CUSTOMER COMMUNICATION OPTICAL FIBER COMMUNICATION SC/SC OM3 GOF 50/125um

F.O. CONNECTION STANDALONE SC/SC OM3 GOF 50/125um (PEE.E21101304Ax)

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Interconnection Customer Signals	Mounting location: DU CABINET ADU.PPCU	Previous page 300 / Next page &MAT/1000
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 301 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-Q01	L7903	8630740000	End bracket	1
-Q01	L10136	0705001400	GUARDAMOTOR MMS-32H-22A 3P 14-22A UL	1
-Q10	L10136	0705001400	GUARDAMOTOR MMS-32H-22A 3P 14-22A UL	1
-Q10	L7903	8630740000	End bracket	2
-U02	L4811	4000-73000-0010000	INSTALLATION SOCKET RJ45 CAT5E IP65	1
-U64	L1062	0236500000	DIN 35x15mm	1
-XPE	L445	1010200000	T.STRIP GROUND SCREW 1-TIER 6mm GREEN	1

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED		ESQ_FD1200U000002_000	Bill of materials	Mounting location:	DU CABINET ADU.PPCU	Previous page &/301	Next page &CAB/2000
APPROVED A.CABALLERO	REVISION 0.0.0			Project Code:	FD1200U000002	Page 1000	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

Cable overview

Cable designation	Function text	ERP
=DCDC++ZMI+ADU.PPCU-W150	OPTICAL FIBER	L8285
=DCDC++ZMI+ADU.PPCU-W151	OPTICAL FIBER	L10110

0	1	2	3	4	5	6	7	8	9

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PPCU-XC01.1	CONECTOR CONEXION F.O. A SWITCH	L7124	1
=DCDC++ZMI+ADU.PPCU-XC02.1	CONECTOR CONEXION STANDALONE	L7124	1
=DCDC++ZMI+ADU.PPCU-XC10.1	M&L 9P AEREAL INTERCONNECTION PPCU	L7534	1
		L5709	7
		E1203	2

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED		ESQ_FD1200U000002_000	Plug overview : =DCDC++ZMI+ADU.PPCU-XC01.1 - =DCDC++ZMI+ADU.PPCU-XC10.1	Mounting location:	DU CABINET ADU.PPCU	Previous page &CAB/2000	Next page +.PFU/BESS&/1
APPROVED A.CABALLERO	REVISION 0.0.0			Project Code:	FD1200U000002	Page 3000	Total number of pages 181

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CV35 SALIDA 30

DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PFU/BESS	FUSES PRE-ASSEMBLY BESS

PFU/BESS

FUSES PRE-ASSEMBLY BESS

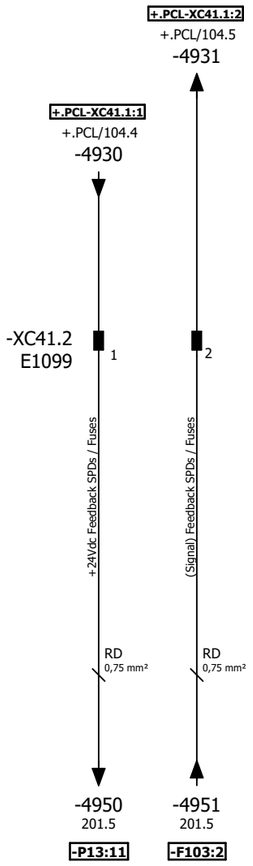
Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

0	1	2	3	4	5	6	7	8	9

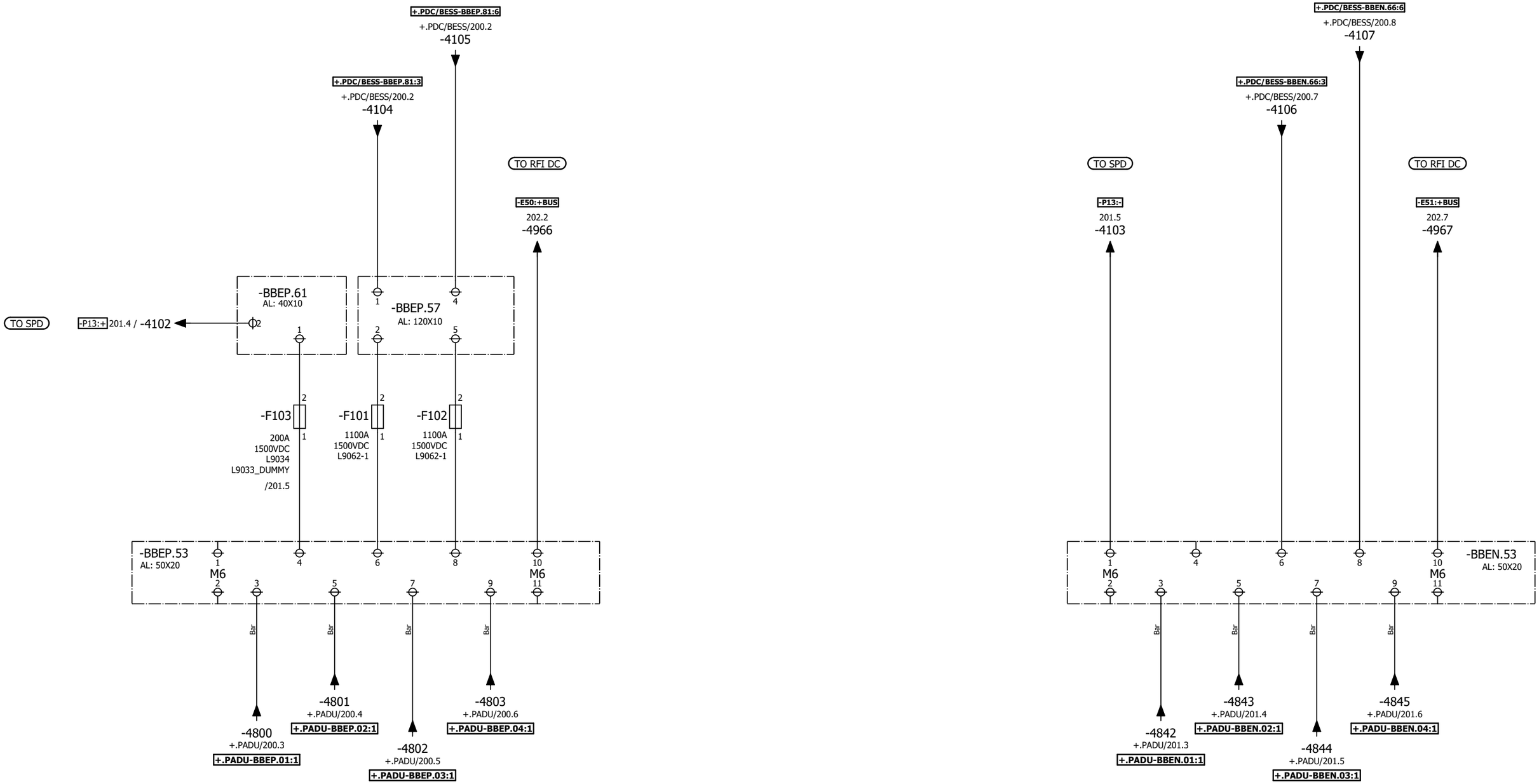
-XC41.2 = CONECTOR PANEL 2P SPD

Interconnections PFU-BESS/PCL

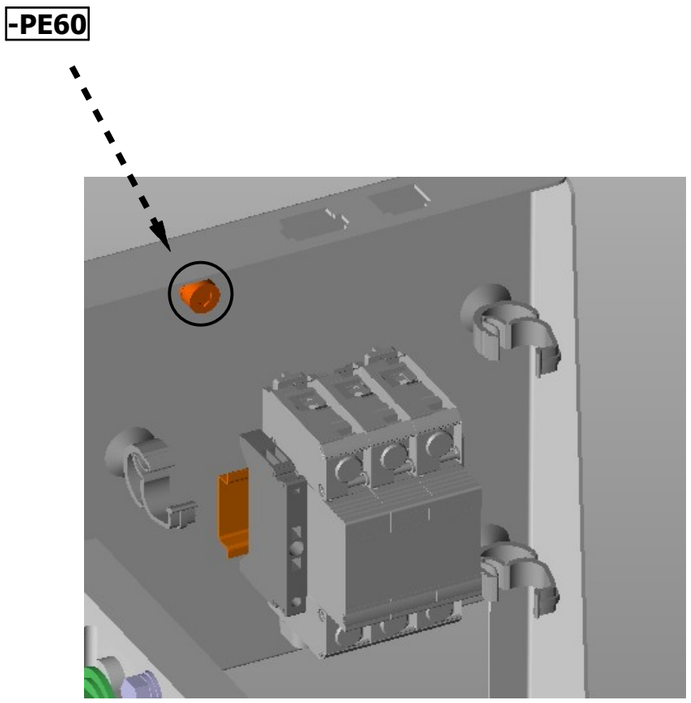
Feedback SPD & Fuses

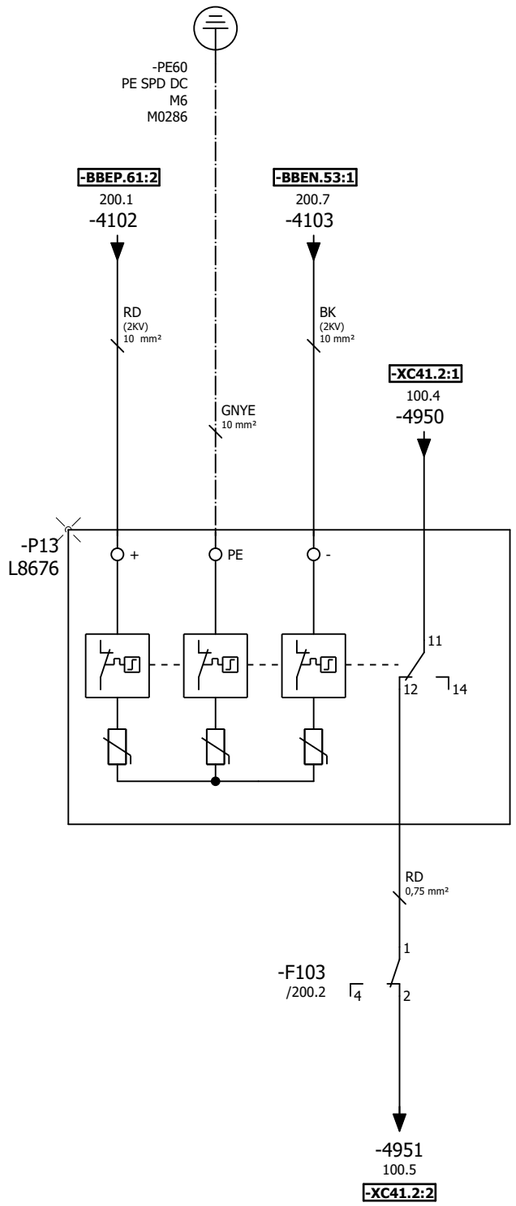


POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project ESQ_FD1200U000002_000	Name Interconnections SPD / Fuses	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PFU/BESS	Previous page 20	Next page 200
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 100	Total number of pages 181



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	DC Input 1 (+)	Mounting location: DU CABINET ADU.PFU/BESS	Previous page 100	Next page 201
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 200	Total number of pages 181





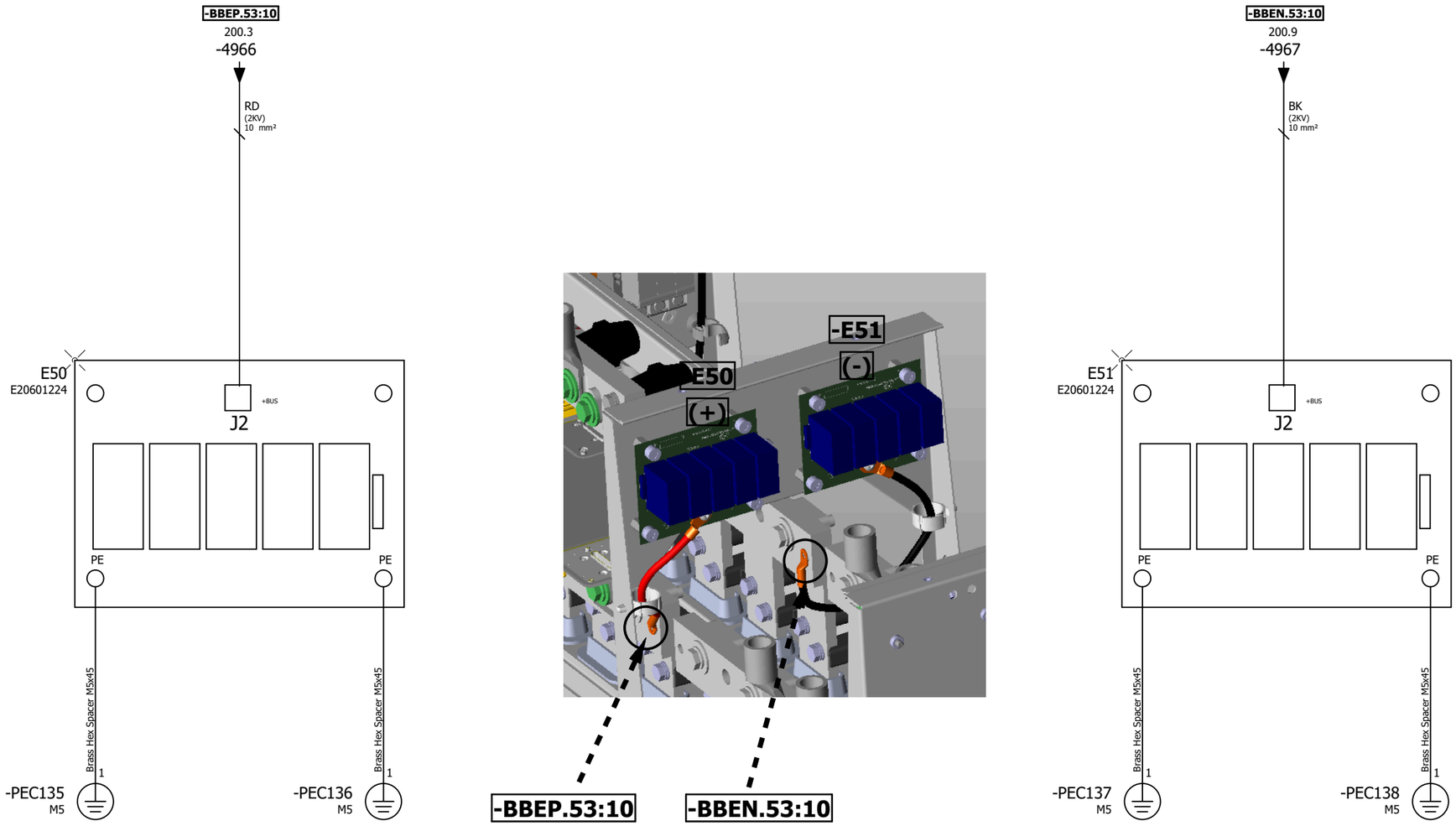
SPD - Type I+II

0	1	2	3	4	5	6	7	8	9

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC / A3
REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	Surge protector / Feedback fuse	Mounting location: DU CABINET ADU.PFU/BESS	Previous page 200 / Next page 202
APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 201 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION	ESQ_FD1200U000002_000	RFI DC	Mounting location: DU CABINET ADU.PFU/BESS	Previous page 201 / Next page &MAT/1000
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 202 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-E50	E20601224AA	E20601224AA	RFI DC HEM GENIII PCB BOARD	1
-E51	E20601224AA	E20601224AA	RFI DC HEM GENIII PCB BOARD	1
-F101	L9062-1	PSX3XLUB1100	FUSE 1100A - 1500VDC - aR BOLTED	1
-F101	L3714UND	1724151687	ADHESIVE LABEL LM MT300 17x9 GE	1
-F102	L9062-1	PSX3XLUB1100	FUSE 1100A - 1500VDC - aR BOLTED	1
-F102	L3714UND	1724151687	ADHESIVE LABEL LM MT300 17x9 GE	1
-F103	L9033_DUMMY	PSX1XLUB0200	FUSE 200A - 1500VDC - aR BOLTED	1
-F103	L9034	MSXL2000C	FUSE MICROSWITCH PSX - LITTEL	1
-F103	L3714UND	1724151687	ADHESIVE LABEL LM MT300 17x9 GE	1
-P13	L8676	2530590000	VPU PV I+II 3 R 1500 SURGE PROTECTOR DEV.	1

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Bill of materials	Mounting location:	DU CABINET ADU.PFU/BESS	Previous page &/202	Next page &CON/3000
APPROVED A.CABALLERO	0.0.0			Project Code:	FD1200U000002	Page 1000	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PFU/BESS-XC41.2	CONECTOR PANEL 2P SPD	E1099	1
		E1204	2

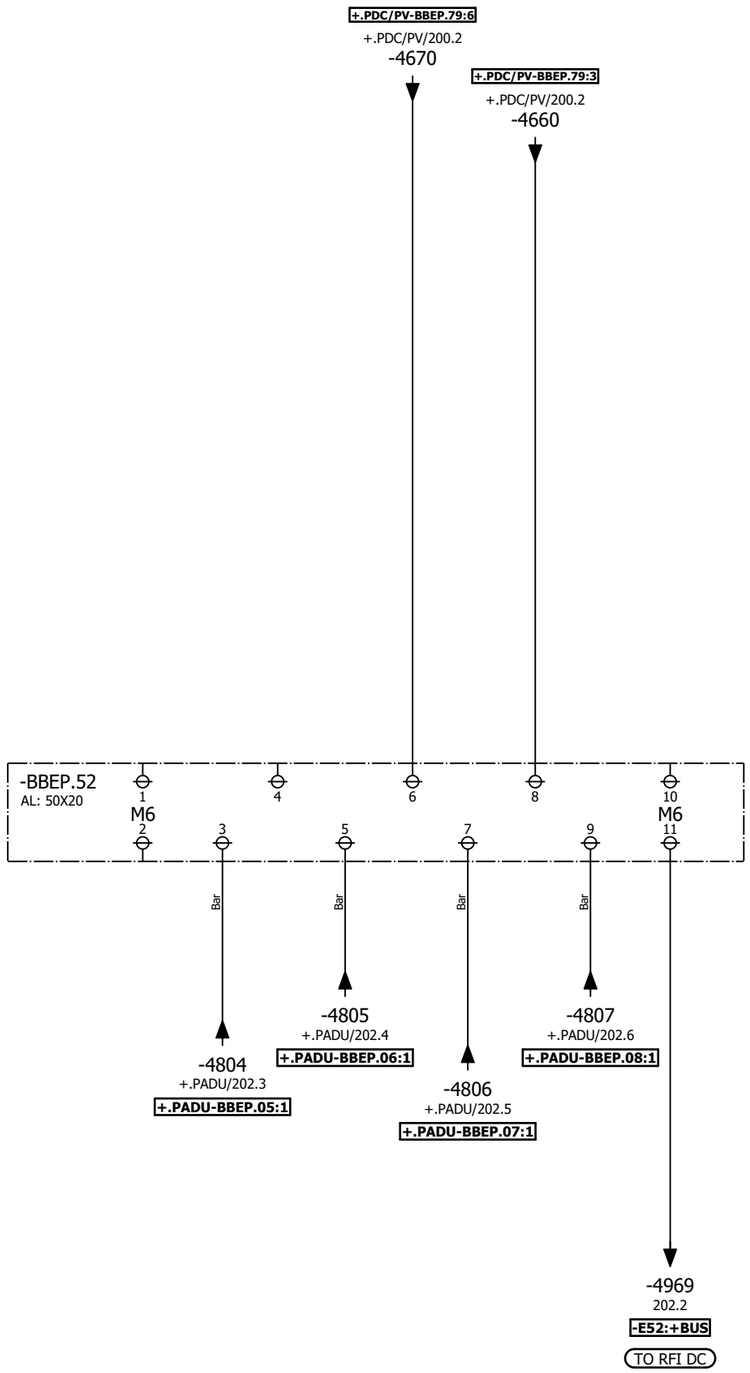
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

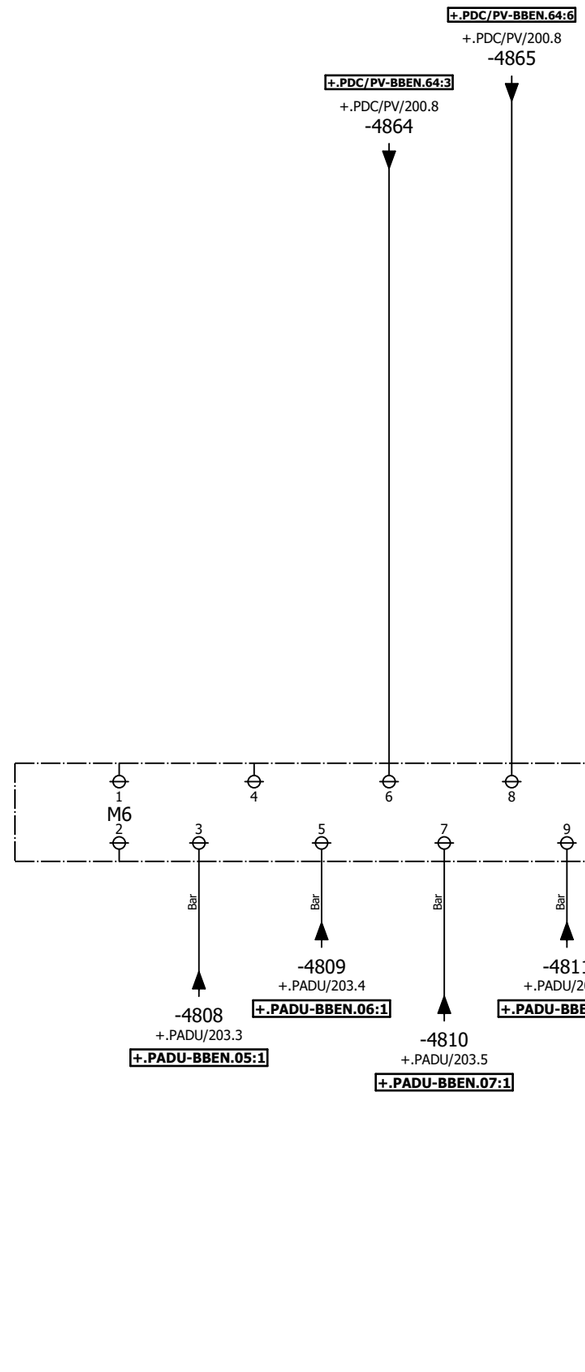
DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PFU/PV	FUSES PRE-ASSEMBLY PV OPTIONAL

PFU/PV

FUSES PRE-ASSEMBLY PV OPTIONAL

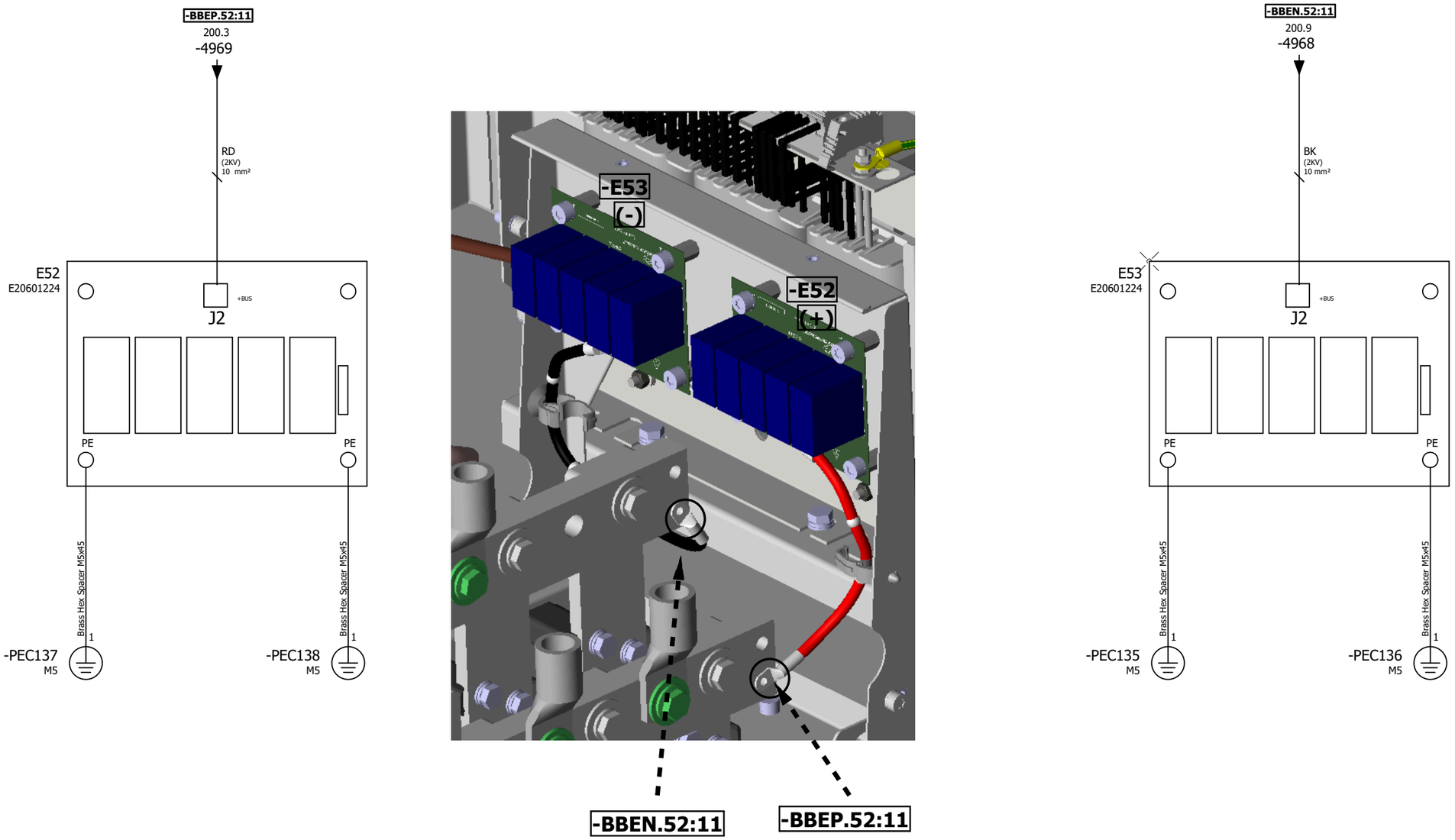
Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		





POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	DC Input 1 (+)	Mounting location: DU CABINET ADU.PFU/PV	Previous page 1	Next page 202
APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 200	Total number of pages 181



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC A3
	REVISED	REVISION	ESQ_FD1200U000002_000	RFI DC	Mounting location: DU CABINET ADU.PFU/PV Previous page 200 Next page &MAT/1000
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002 Page 202 Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-E52	E20601224AA	E20601224AA	RFI DC HEM GENIII PCB BOARD	1
-E53	E20601224AA	E20601224AA	RFI DC HEM GENIII PCB BOARD	1

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED		ESQ_FD1200U000002_000	Bill of materials	Mounting location:	DU CABINET ADU.PFU/PV	Previous page &/202	Next page +.PDC/BESS&/1
APPROVED A.CABALLERO	REVISION 0.0.0			Project Code:	FD1200U000002	Page 1000	Total number of pages 181

POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PDC/BESS	DC DISCONNECTOR PRE-ASSEMBLY BESS

PDC/BESS

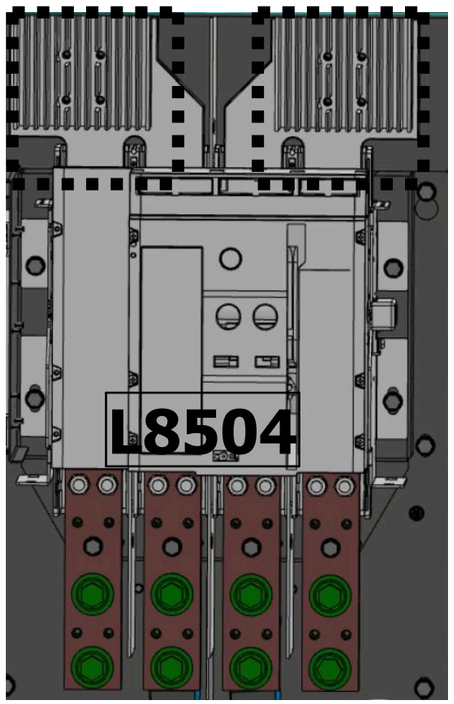
DC DISCONNECTOR PRE-ASSEMBLY BESS

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

0	1	2	3	4	5	6	7	8	9

-XC470.1 = CONNECTOR DISCONNECTOR SWITCH 1 - OPERATING

-XC471.1 = CONNECTOR DISCONNECTOR SWITCH 1 - AUX.



L7753

BBEP.81

BBIP.07

BBIN.07

BBEN.66

LOAD

Busbars + Heatsinks

-SI01
/300.1
4x1600A 1500Vdc

1 2 3 4 5 6 7 8

+.PCL-F11:2
+.PCL/102.3
-4034

-BBEP.81
Cu: 50x20

RD
(2KV)
2,5 mm²

1 2 M6 M6
3 M12
4 5 M6 M6
6 M12

(+) PV STRINGS
BN
2KV
150 mm²

-4104
+.PFU/BESS/200.2
+.PFU/BESS-BBEP.57:1

(+) PV STRINGS
BN
2KV
150 mm²

-4105
+.PFU/BESS/200.3
+.PFU/BESS-BBEP.57:4

+ PV STRINGS

+.PDC-E60-J4:5
+.PDC/400.8
-4007

-BBIP.07
Cu: 50x20

To soft load board
RD
(2KV)
2,5 mm²

1 2 M6 M6
3 M12
4 5 M6 M6
6 M12

(+) DC/AC MODULE - Rear Side
BN
2KV
150 mm²

-4017
++MOI+ETR_BESS/200.2
++MOI+ETR_BESS-BBIP.20:2

(+) DC/AC MODULE - Rear Side
BN
2KV
150 mm²

-4018
++MOI+ETR_BESS/200.2
++MOI+ETR_BESS-BBIP.20:1

+ DC/AC MODULE

Busbars + Heatsinks

+.PDC-E60-J4:1
+.PDC/400.8
-4008

-BBIN.07
Cu: 50x20

To soft load board
BK
(2KV)
2,5 mm²

1 2 M6 M6
3 M12
4 5 M6 M6
6 M12

(-) DC/AC MODULE - Rear Side
BN
2KV
150 mm²

-4026
++MOI+ETR_BESS/200.7
++MOI+ETR_BESS-BBIN.20:1

(-) DC/AC MODULE - Rear Side
BN
2KV
150 mm²

-4027
++MOI+ETR_BESS/200.7
++MOI+ETR_BESS-BBIN.20:2

- DC/AC MODULE

+.PCL-F12:2
+.PCL/102.4
-4035

-BBEN.66
Cu: 50x20

BK
(2KV)
2,5 mm²

1 2 M6 M6
3 M12
4 5 M6 M6
6 M12

(-) PV STRINGS
BN
2KV
150 mm²

-4106
+.PFU/BESS/200.8
+.PFU/BESS-BBEN.53:6

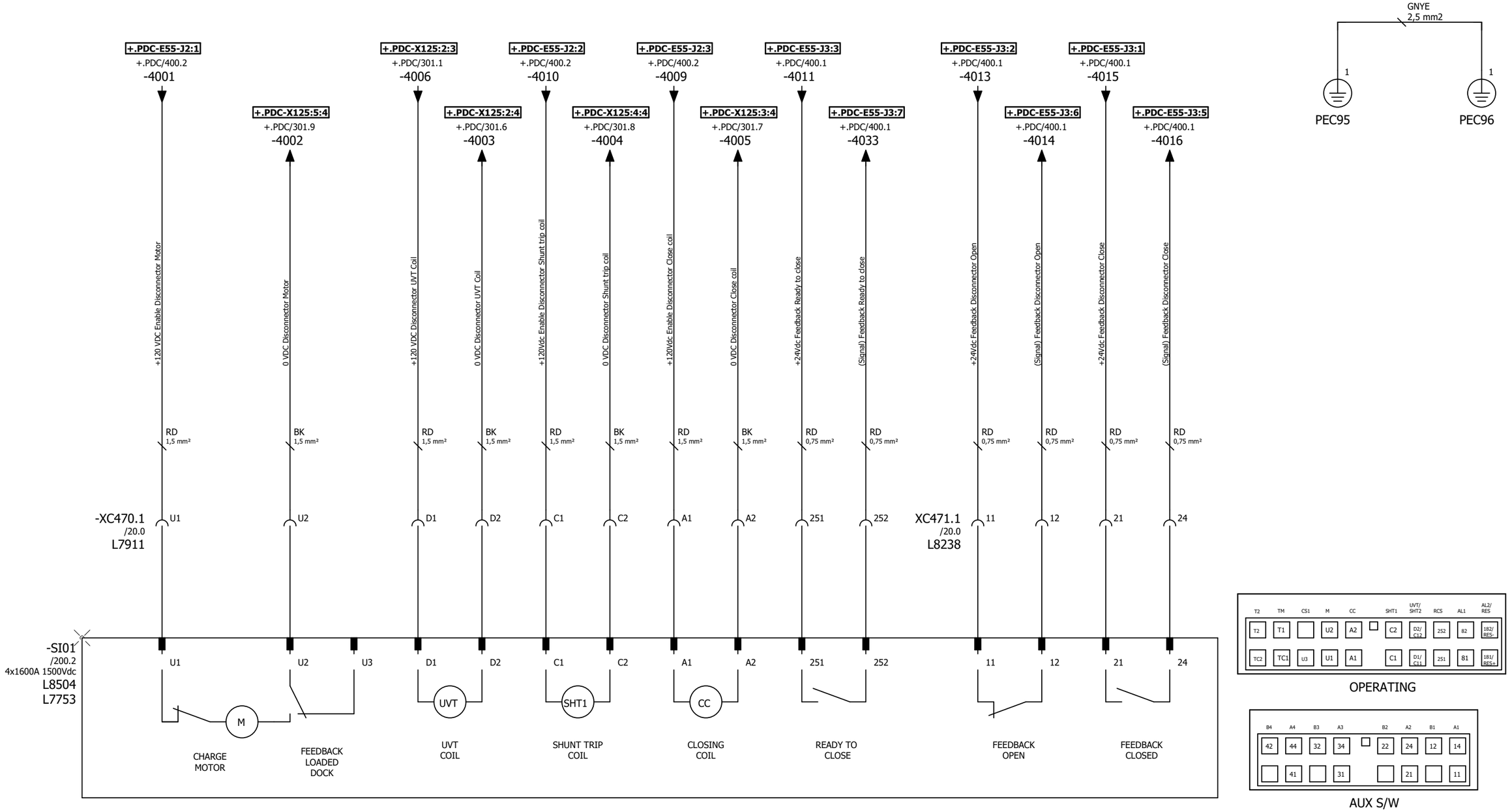
(-) PV STRINGS
BN
2KV
150 mm²

-4107
+.PFU/BESS/200.8
+.PFU/BESS-BBEN.53:8

- PV STRINGS

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Power DC Disconnector (DC+/-)	Mounting location: DU CABINET ADU.PDC/BESS	Previous page 20	Next page 300
APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 200	Total number of pages 181



This diagram is based on "CONNECTED"position of a circuit breaker and Opening, Motor charging, Releasing of locking plate should be normal condition.

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Disconnector switch DC 1	Mounting location: DU CABINET ADU.PDC/BESS	Previous page 200	Next page &MAT/1000
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 300	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-SI01	L8504	2950009300	4x1600A-C-P IEC DISCONNECTOR SWITCH	1
-SI01	L7753	70223472602	Short bars + heatsink for DDV-16C4-00P	1

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Bill of materials	Mounting location:	DU CABINET ADU.PDC/BESS	Previous page &/300	Next page &CON/3000
APPROVED A.CABALLERO	0.0.0			Project Code:	FD1200U000002	Page 1000	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PDC/BESS-XC470.1	CONNECTOR DISCONNECTOR SWITCH 1 - OPERATING	L7911	1
		L7912	10
=DCDC++ZMI+ADU.PDC/BESS-XC471.1	CONNECTOR DISCONNECTOR SWITCH 1 - AUX.	L8238	1
		L7912	4

POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PDC/PV	DC DISCONNECTOR PRE-ASSEMBLY PV

PDC/PV

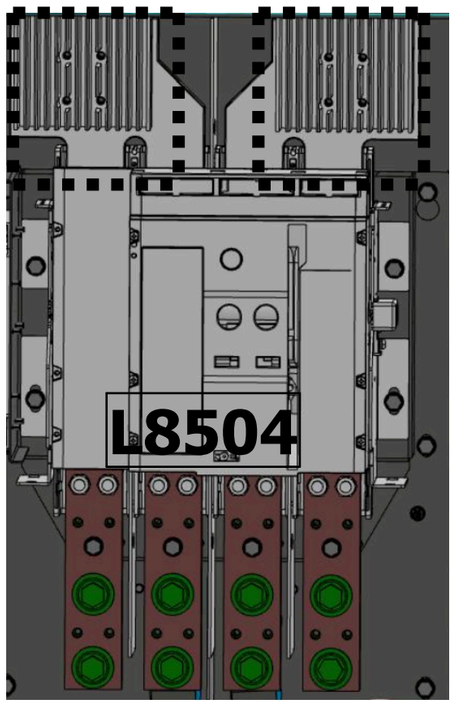
DC DISCONNECTOR PRE-ASSEMBLY PV

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

0	1	2	3	4	5	6	7	8	9

-XC475.1 = CONNECTOR DISCONNECTOR SWITCH 2 - OPERATING

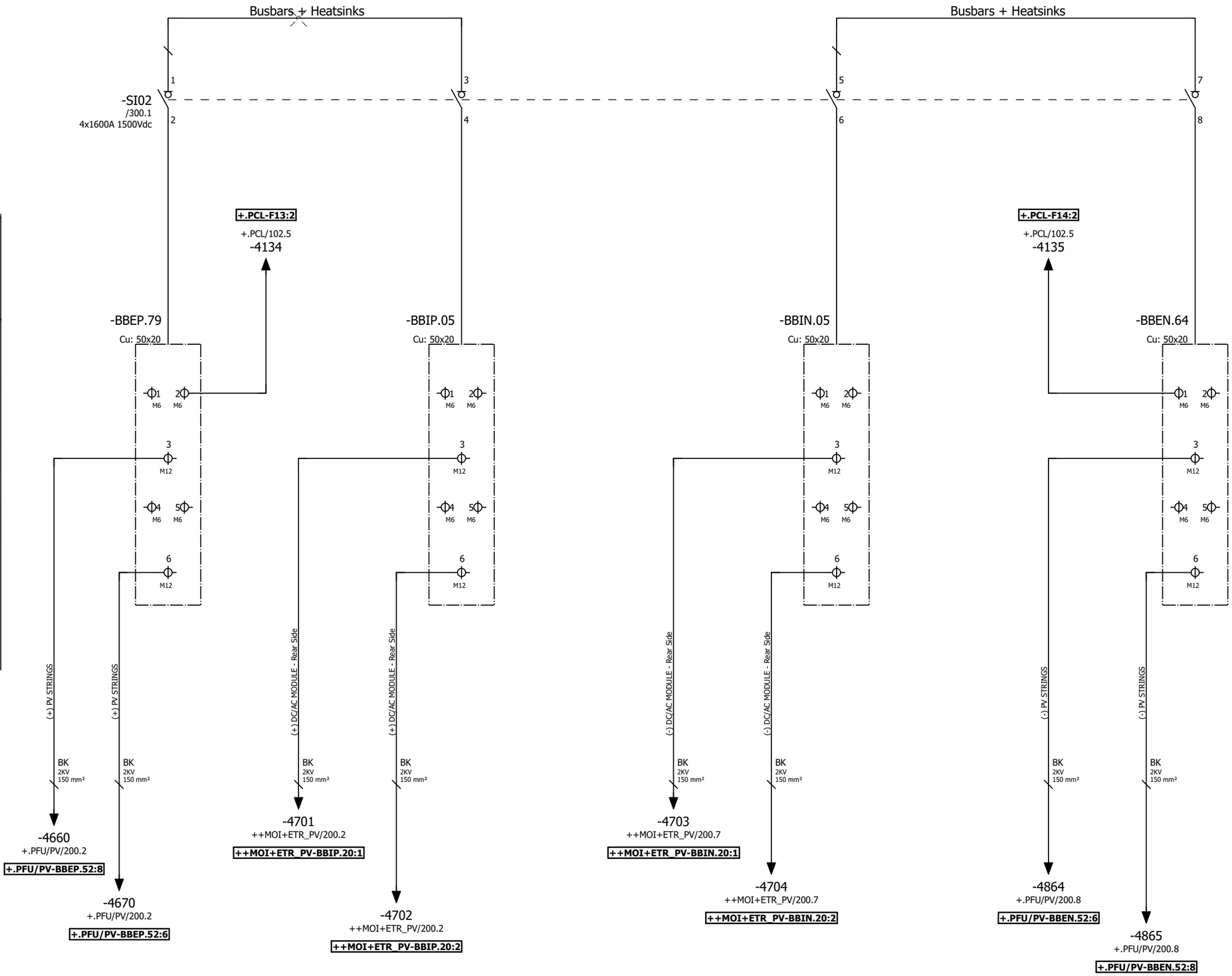
-XC476.1 = CONNECTOR DISCONNECTOR SWITCH 2 - AUX.



L7753

BBEP.79 BBIP.05 BBIN.05 BBEN.64

LOAD

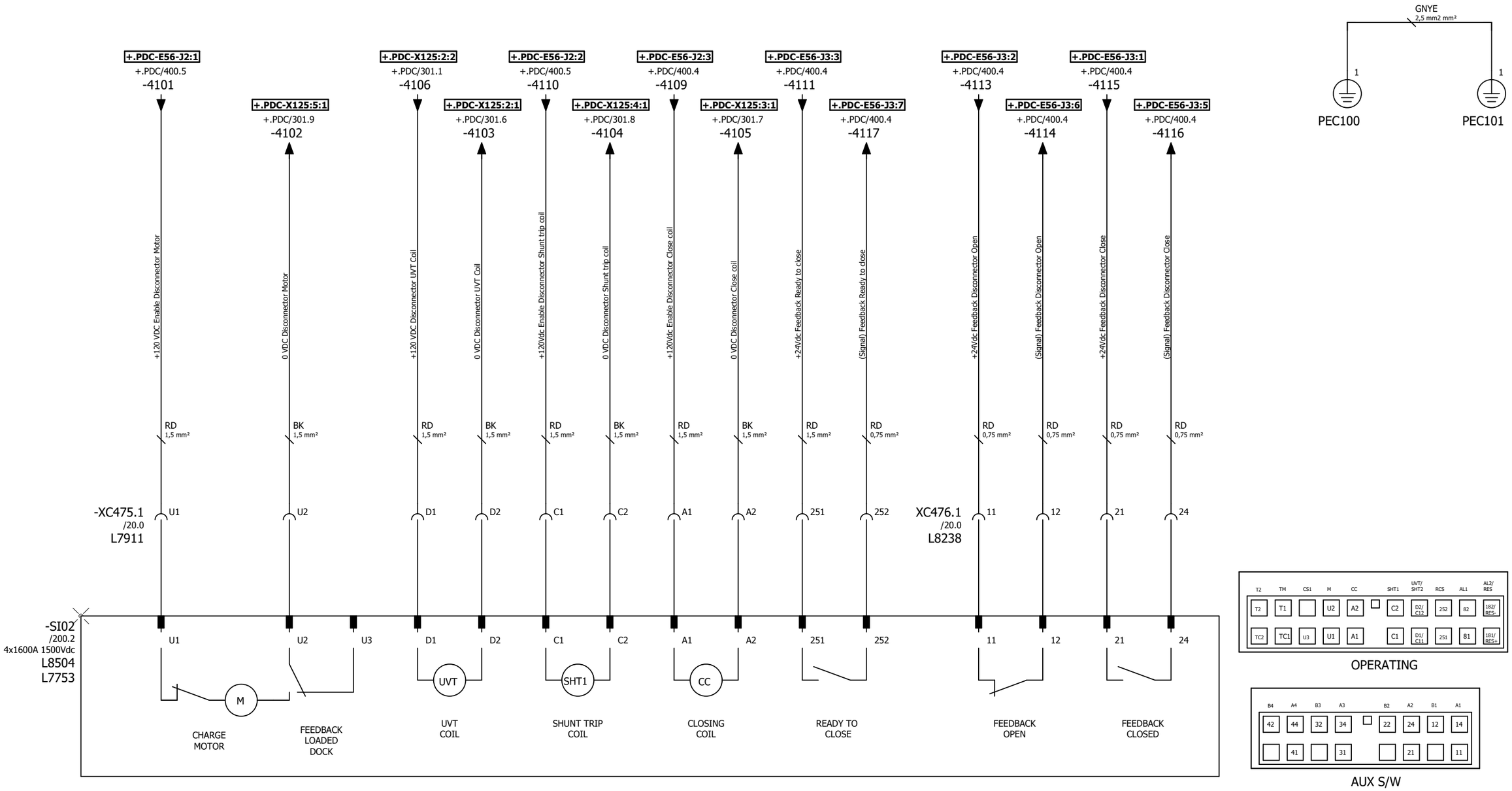


+ PV STRINGS **+ DC/AC MODULE** **- DC/AC MODULE** **- PV STRINGS**

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name
REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	Power DC Disconnector (DC+/-)
APPROVED A.CABALLERO			

Installation:	DCDC GEN 3 CONVERTER		DCDC	A3
Mounting location:	DU CABINET ADU.PDC/PV	Previous page 20	Next page 300	
Project Code:	FD1200U000002	Page 200	Total number of pages 181	



This diagram is based on "CONNECTED"position of a circuit breaker and Opening, Motor charging, Releasing of locking plate should be normal condition.

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Disconnector switch DC 1	Mounting location: DU CABINET ADU.PDC/PV	Previous page 200	Next page &MAT/1000
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 300	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-SI02	L8504	2950009300	4x1600A-C-P IEC DISCONNECTOR SWITCH	1
-SI02	L7753	70223472602	Short bars + heatsink for DDV-16C4-00P	1

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DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Bill of materials	Mounting location: DU CABINET ADU.PDC/PV		Previous page &/300	Next page &CON/3000
APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002		Page 1000	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PDC/PV-XC475.1	CONNECTOR DISCONNECTOR SWITCH 2 - OPERATING	L7911	1
		L7912	10
=DCDC++ZMI+ADU.PDC/PV-XC476.1	CONNECTOR DISCONNECTOR SWITCH 2 - AUX.	L8238	1
		L7912	4

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CV35 SALIDA 30

DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PDC	DC DISCONNECTORS PANEL MANUEVER

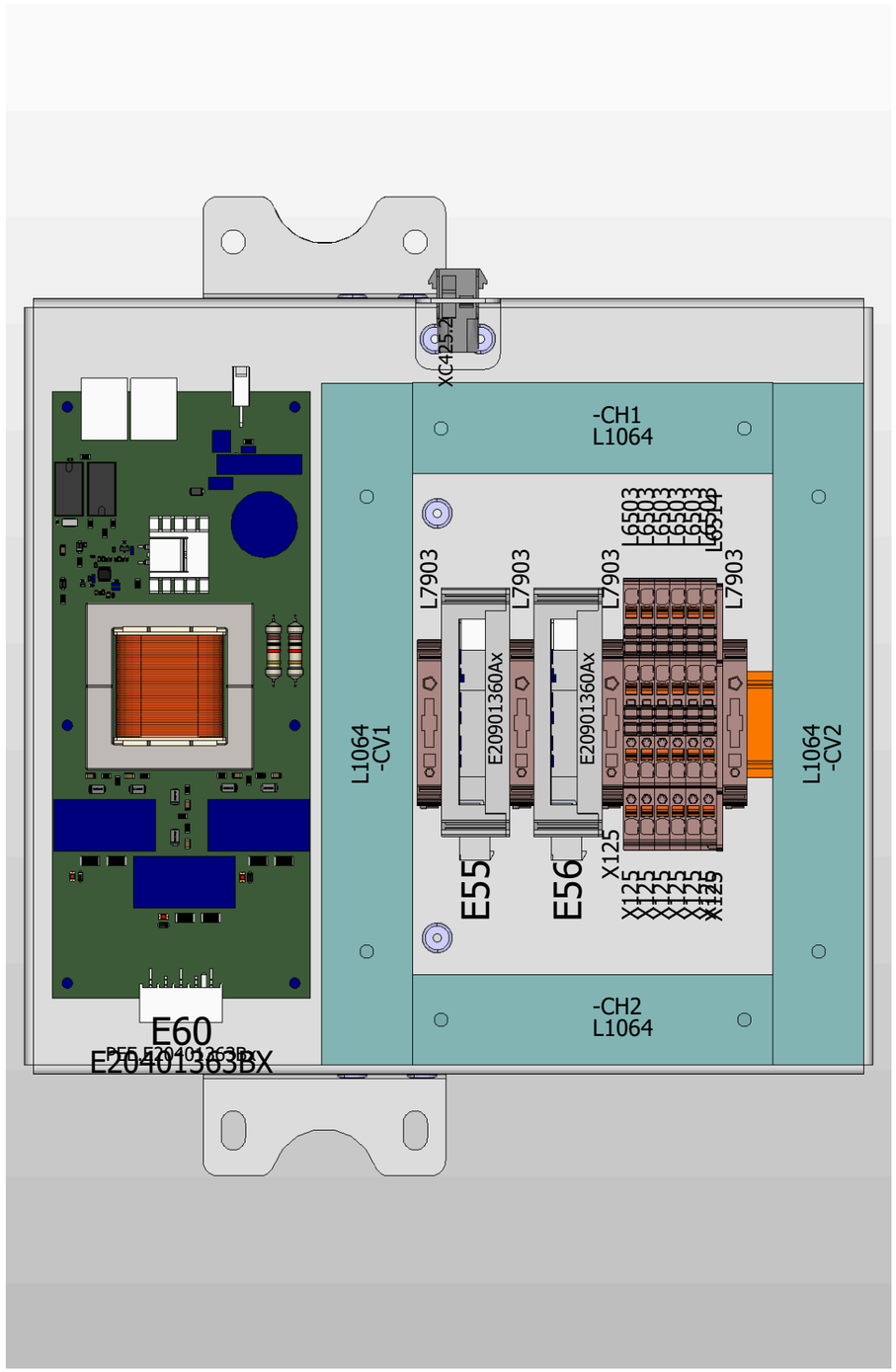
PDC

DC DISCONNECTORS PANEL MANUEVER

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

0	1	2	3	4	5	6	7	8	9

LAYOUT PANEL



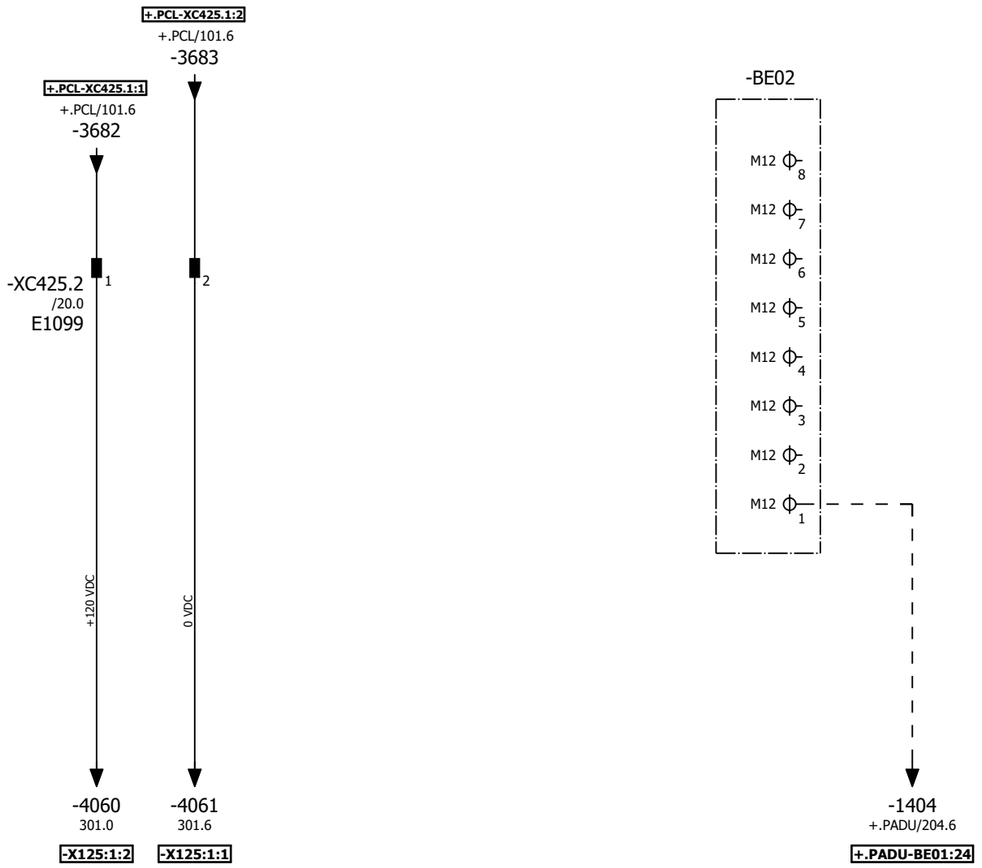
Enclosure legend

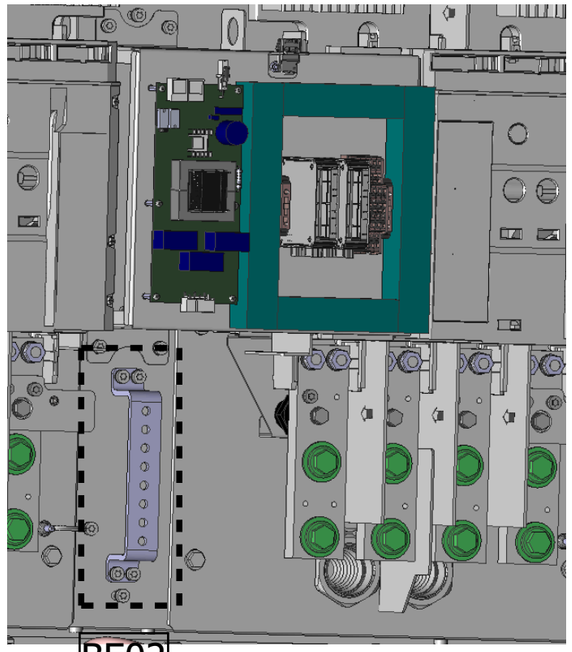
IME	ERP	Description
E60	E20401363BX	SOFT LOAD BOARD
E55;E56	PCB E20901360Ax	TARJETA DE ENTRADAS/SALIDAS DIGITALES (DCDC)
CA1	L1061	DIN RAIL 35x7,5mm
CH1;CH2;CV1;CV2	L1064	CABLE DUCT UNEX 42X30 EN U23X

0	1	2	3	4	5	6	7	8	9

-X125 = Distribution 120 Vdc
-XC425.2 = CONECTOR PANEL 2P PCL

Interconnections PDC/PCL

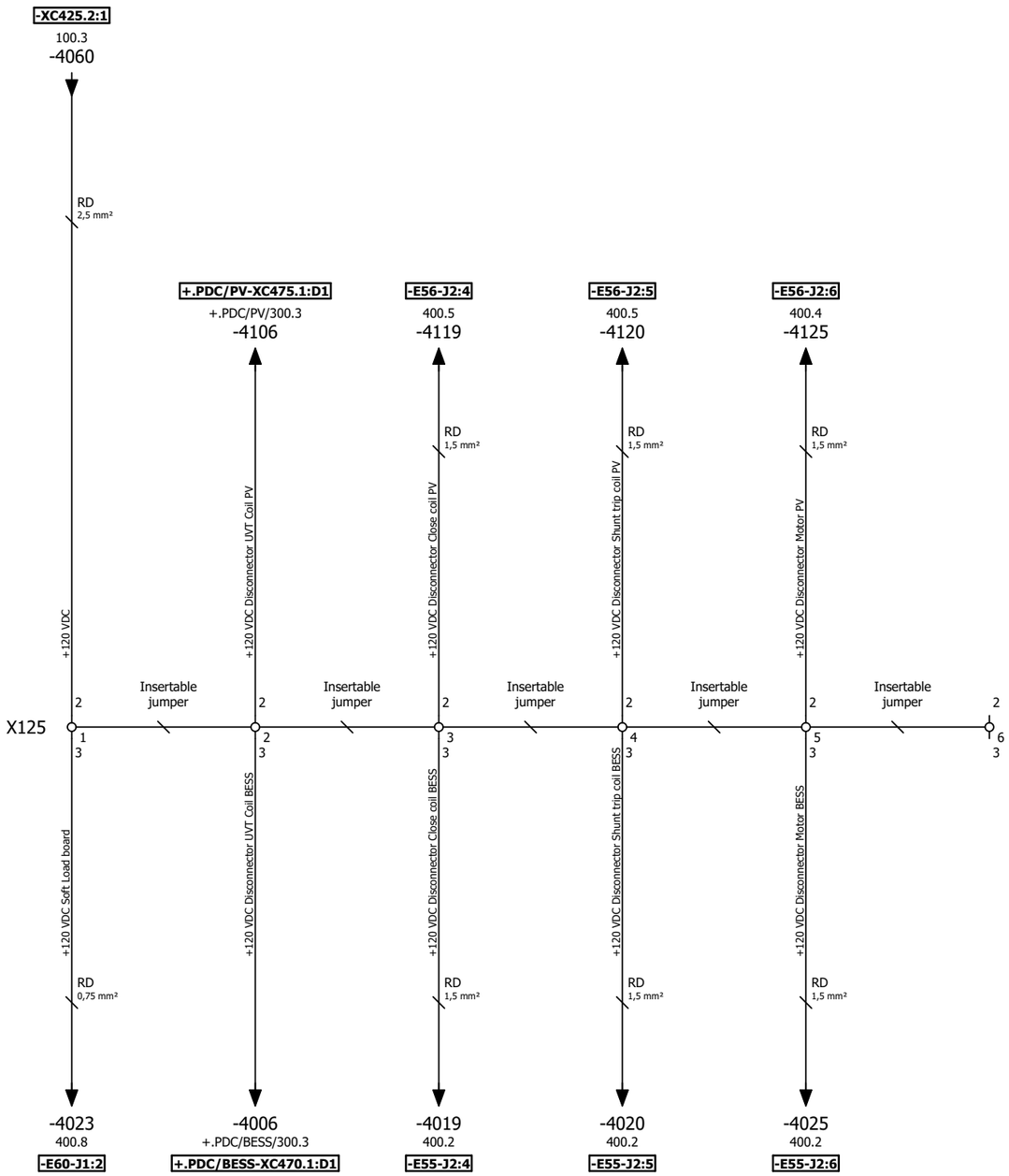


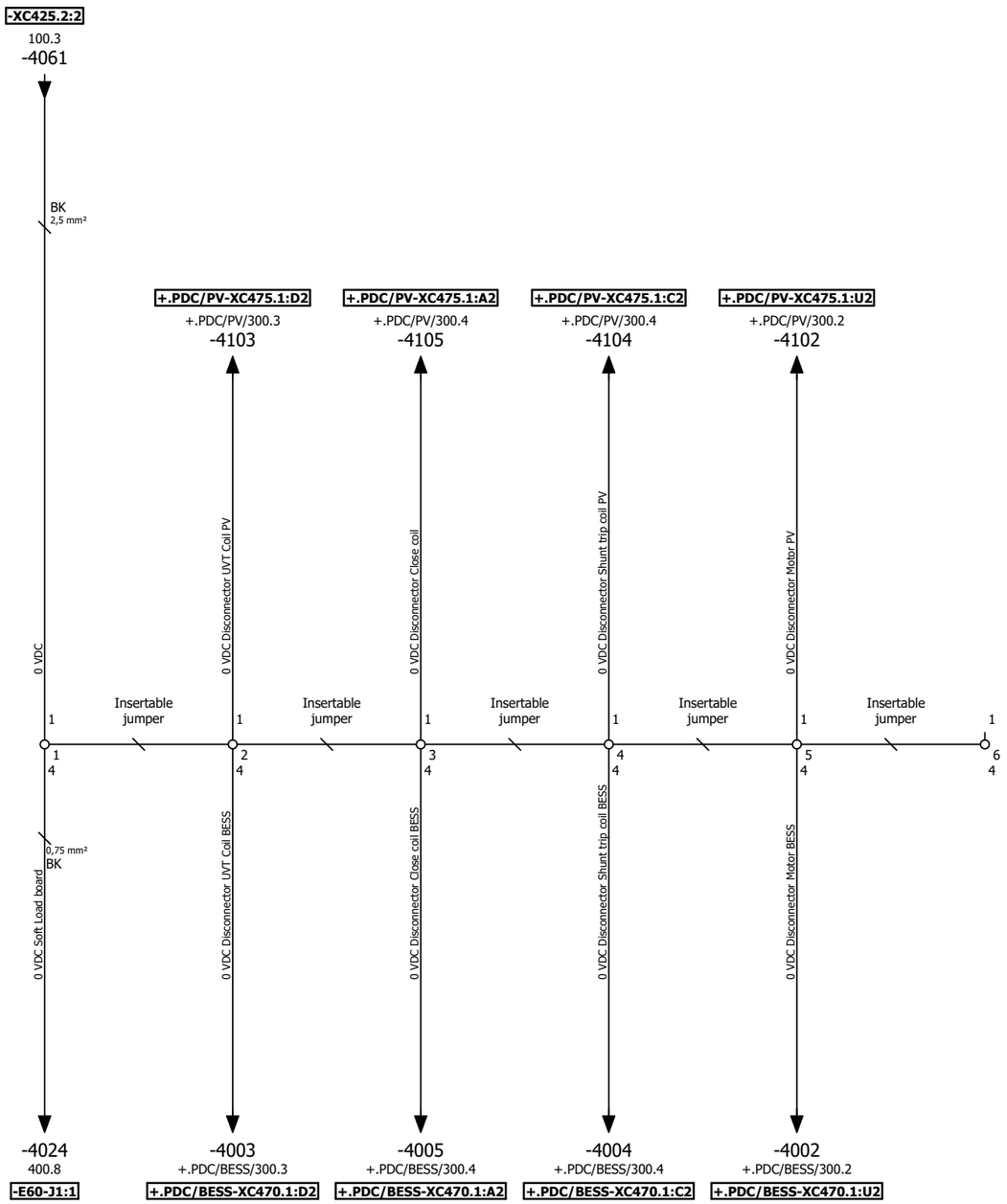


POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: Interconnection	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PDC	Previous page 20 / Next page 301
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 100 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

Distribution 120Vdc





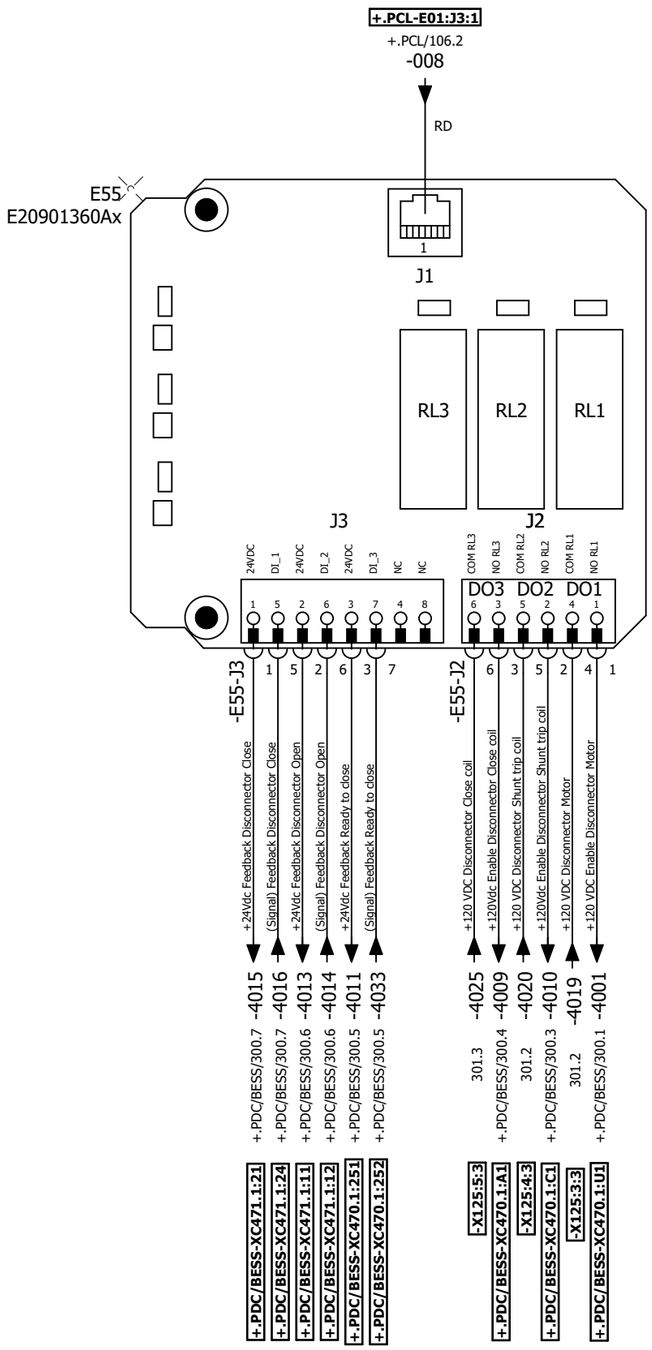
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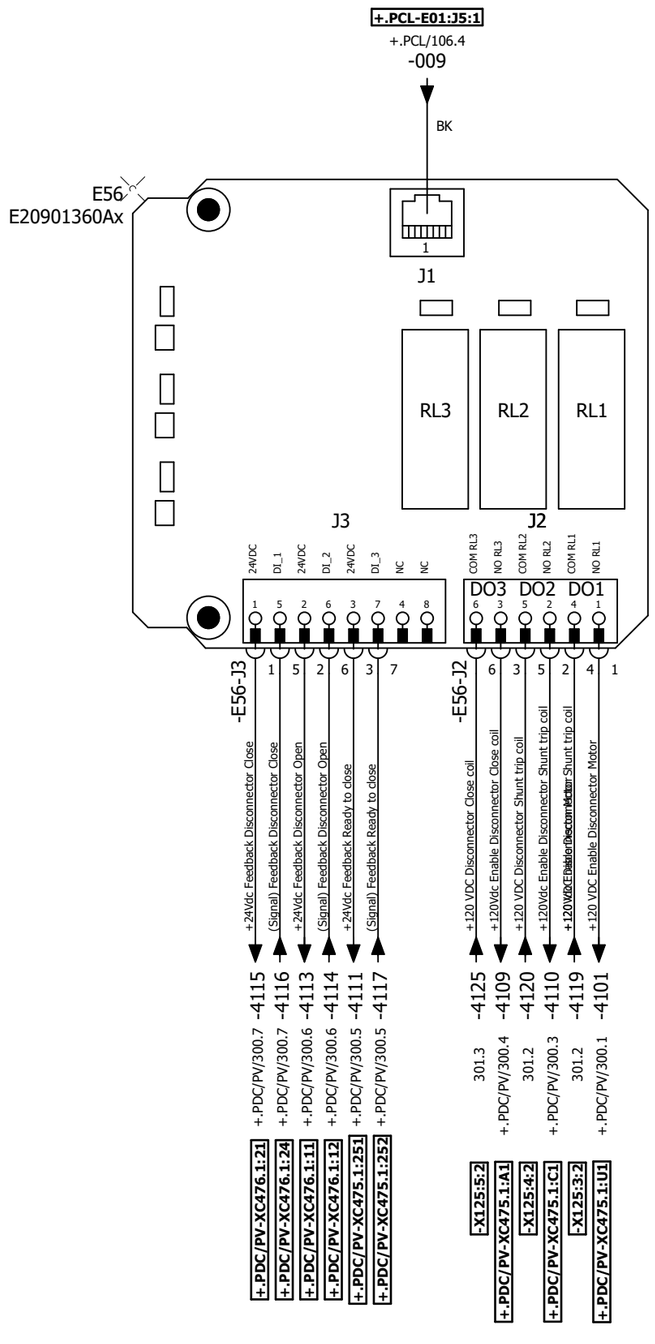
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Distribution 120 VDC	Mounting location: DU CABINET ADU.PDC	Previous page 100	Next page 400
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 301	Total number of pages 181

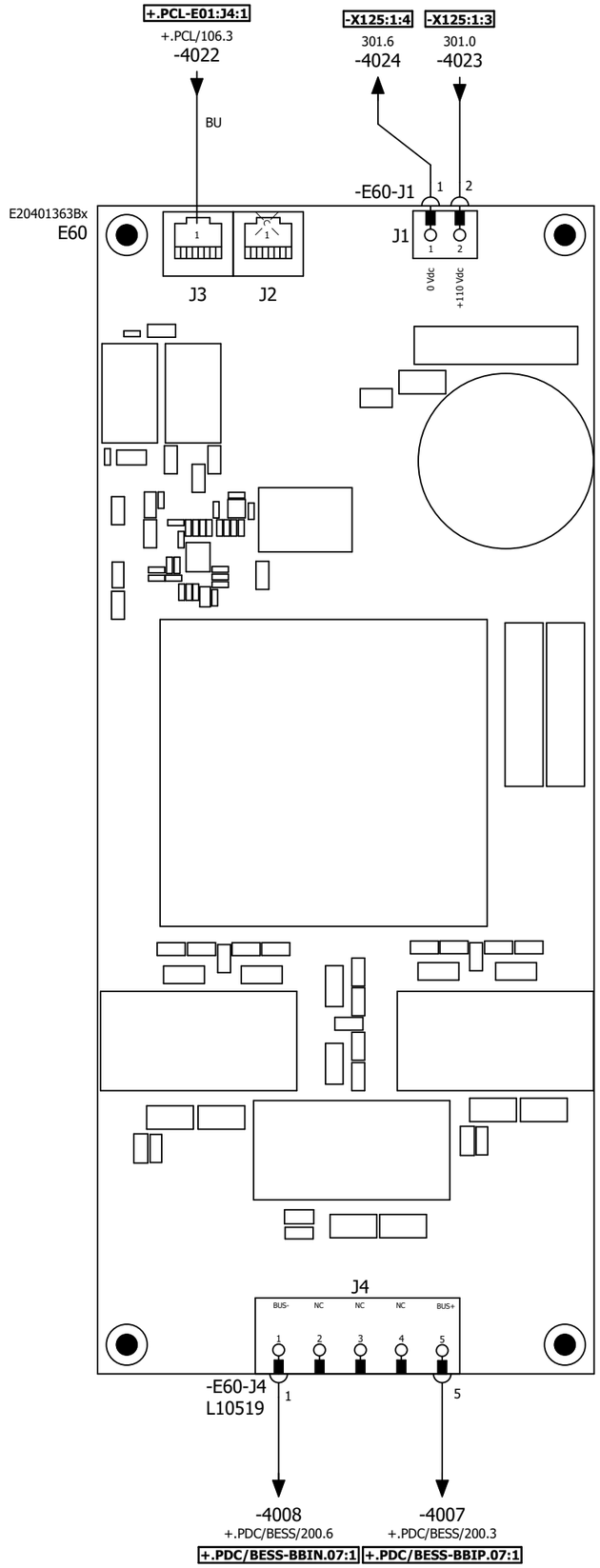
PCB - I/O DC DISCONNECTOR

BESS



PV





0	1	2	3	4	5	6	7	8	9

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER		A3
REVISED	REVISION	ESQ_FD1200U000002_000	PCB´s - Disconnector - Module	Mounting location: DU CABINET ADU.PDC	Previous page 301	Next page &MAT/1000
APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 400	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	4
-E55	PCB E20901360Ax	PCB E20901360Ax	TARJETA DE ENTRADAS/SALIDAS DIGITALES (DCDC)	1
-E55	L7903	8630740000	End bracket	2
-E56	PCB E20901360Ax	PCB E20901360Ax	TARJETA DE ENTRADAS/SALIDAS DIGITALES (DCDC)	1
-E56	L7903	8630740000	End bracket	2
-E60	E20401363BX	E20401363BA	SOFT LOAD BOARD	1
-E60-J4	L10519	2-2178029-5	CONECTOR RAST5 AÉREO 5 VIAS PASO 5	1
-E60-J4	L10521	964202-1	TERMINAL HEMBRA RAST5 AWG13-17 TE	2

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PDC-E55-J2	+120 VDC Enable Disconnector Motor		
=DCDC++ZMI+ADU.PDC-E55-J3	+24Vdc Feedback Disconnector Close		
=DCDC++ZMI+ADU.PDC-E56-J2	+120 VDC Enable Disconnector Motor		
=DCDC++ZMI+ADU.PDC-E56-J3	+24Vdc Feedback Disconnector Close		
=DCDC++ZMI+ADU.PDC-E60-J1	+120 VDC Enable Disconnector Motor		
=DCDC++ZMI+ADU.PDC-E60-J4	+120 VDC Enable Disconnector Motor	L10519	1
		L10521	2
=DCDC++ZMI+ADU.PDC-J2			
=DCDC++ZMI+ADU.PDC-J3			
=DCDC++ZMI+ADU.PDC-XC425.2	CONECTOR PANEL 2P PCL	E1099	1
		E1204	2
		E1099	1

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CV35 SALIDA 30

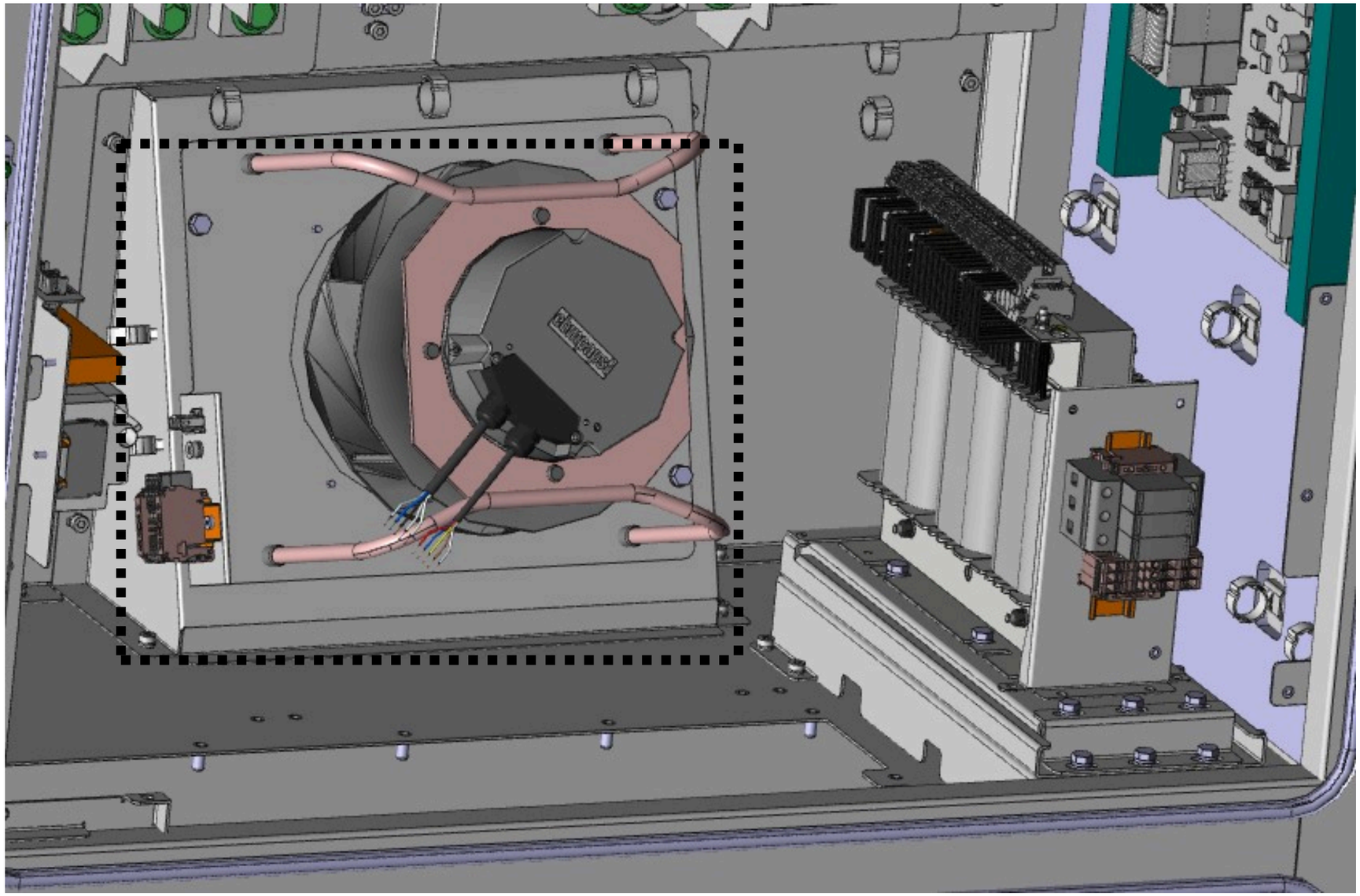
DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PMV1/DU	FAN PRE-ASSEMBLY 1 DU

PMV1/DU

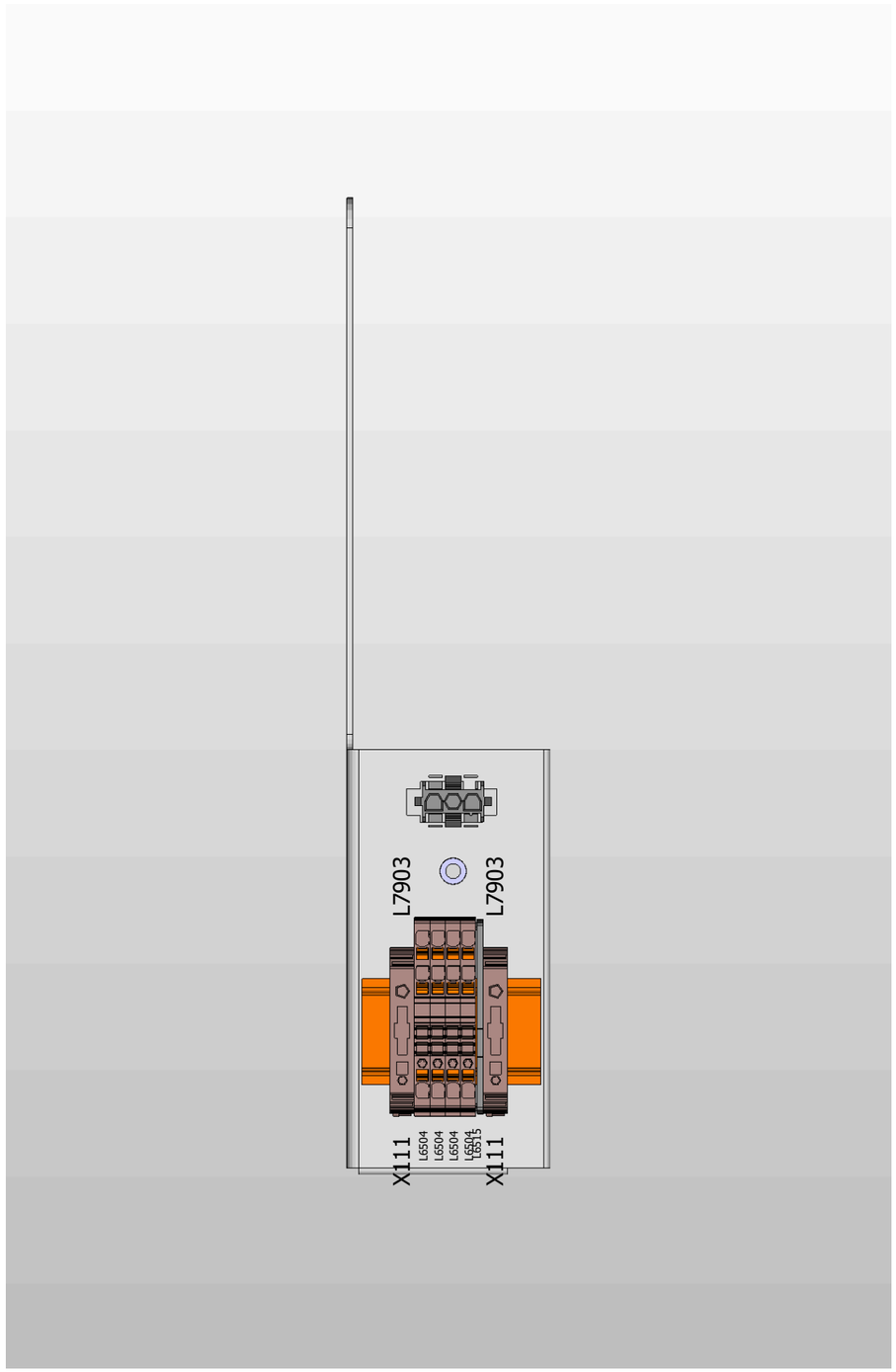
FAN PRE-ASSEMBLY 1 DU

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

PRE-ASSEMBLY LAYOUT PMV1



LAYOUT PANEL



Enclosure legend

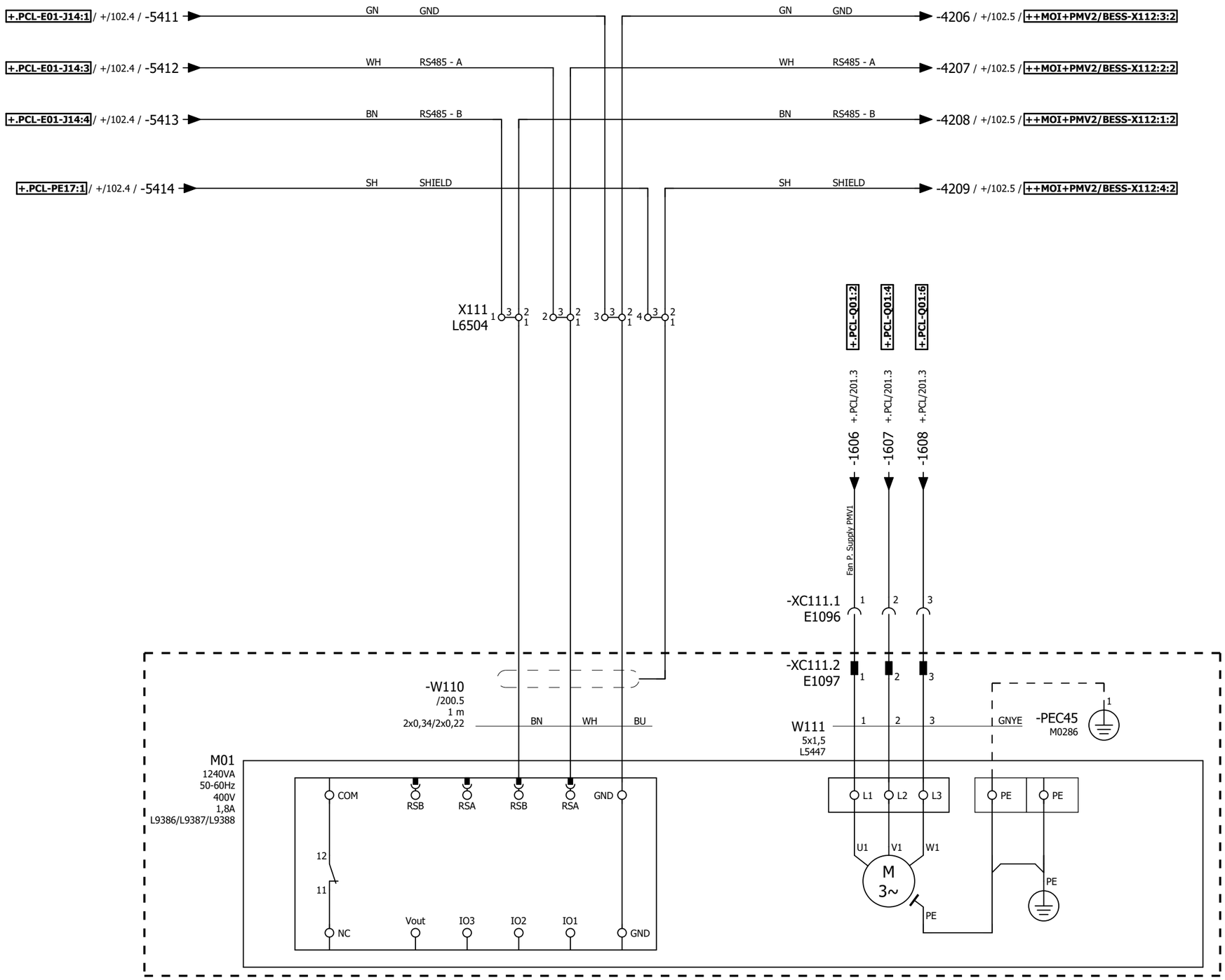
IME	ERP	Description
	L1061	DIN RAIL 35x7,5mm

0	1	2	3	4	5	6	7	8	9

-X111 = RS485 TERMINAL PMV1
-XC111.1 = CONECTOR AEREO POWER SUPPLY DU FAN
-XC111.2 = CONECTOR PANEL POWER SUPPLY DU FAN

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Designation terminals and connectors	Mounting location: DU CABINET ADU.PMV1/DU	Previous page 11	Next page 200
APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 20	Total number of pages 181



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	DU - Fan 1	Mounting location: DU CABINET ADU.PMV1/DU	Previous page 20 / Next page &CAB/2000
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 200 / Total number of pages 181

Cable overview

Cable designation	Function text	ERP
=DCDC++ZMI+ADU.PMV1/DU-W110	SHIELD	L8023
=DCDC++ZMI+ADU.PMV1/DU-W111	Fan P. Supply PMV1	L5447

0	1	2	3	4	5	6	7	8	9

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PMV1/DU-X111	RS485 TERMINAL PMV1		
=DCDC++ZMI+ADU.PMV1/DU-XC111.1	CONECTOR AEREO POWER SUPPLY DU FAN	E1096	1
		E1203	3
=DCDC++ZMI+ADU.PMV1/DU-XC111.2	CONECTOR PANEL POWER SUPPLY DU FAN	E1097	1
		E1204	3

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER			A3
REVISED		ESQ_FD1200U000002_000	Plug overview : =DCDC++ZMI+ADU.PMV1/DU-X111 -	Mounting location: DU CABINET ADU.PMV1/DU	Previous page &CAB/2000	Next page +.PTA&/1	
APPROVED A.CABALLERO	REVISION 0.0.0			Project Code: FD1200U000002	Page 3000	Total number of pages 181	

POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

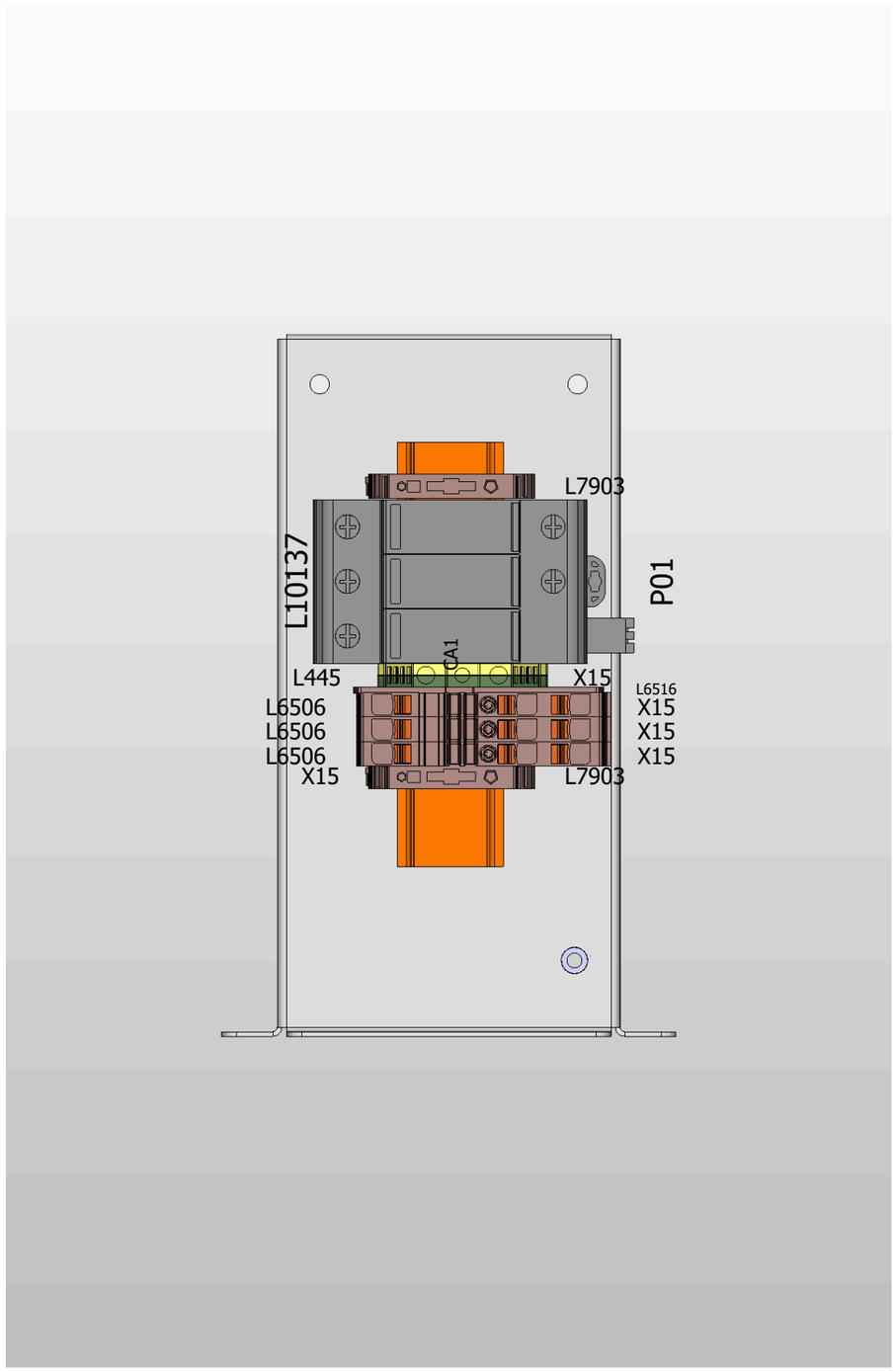
DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PTA	SERVICES AUXILIARY TRANSFORMER ASSEMBLY

PTA

SERVICES AUXILIARY TRANSFORMER ASSEMBLY

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

LAYOUT PANEL



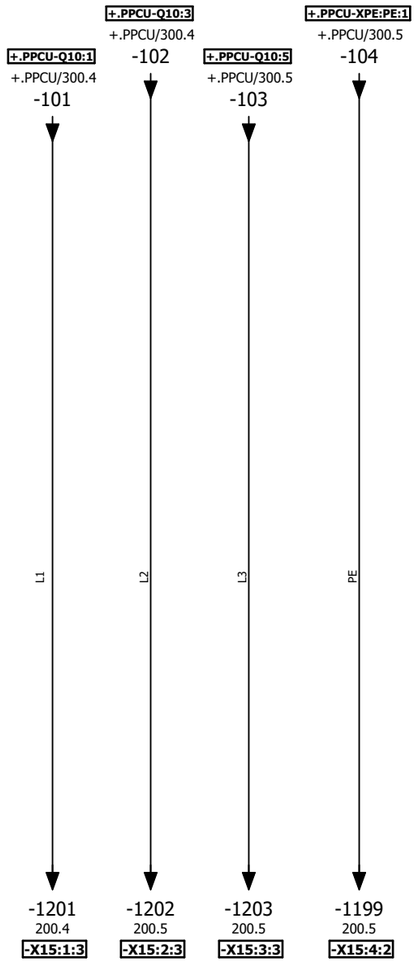
Enclosure legend

IME	ERP	Description
P01	L10137	P.SOBRETENSION 3P 480 T2
CA1	L1061	DIN RAIL 35x7,5mm

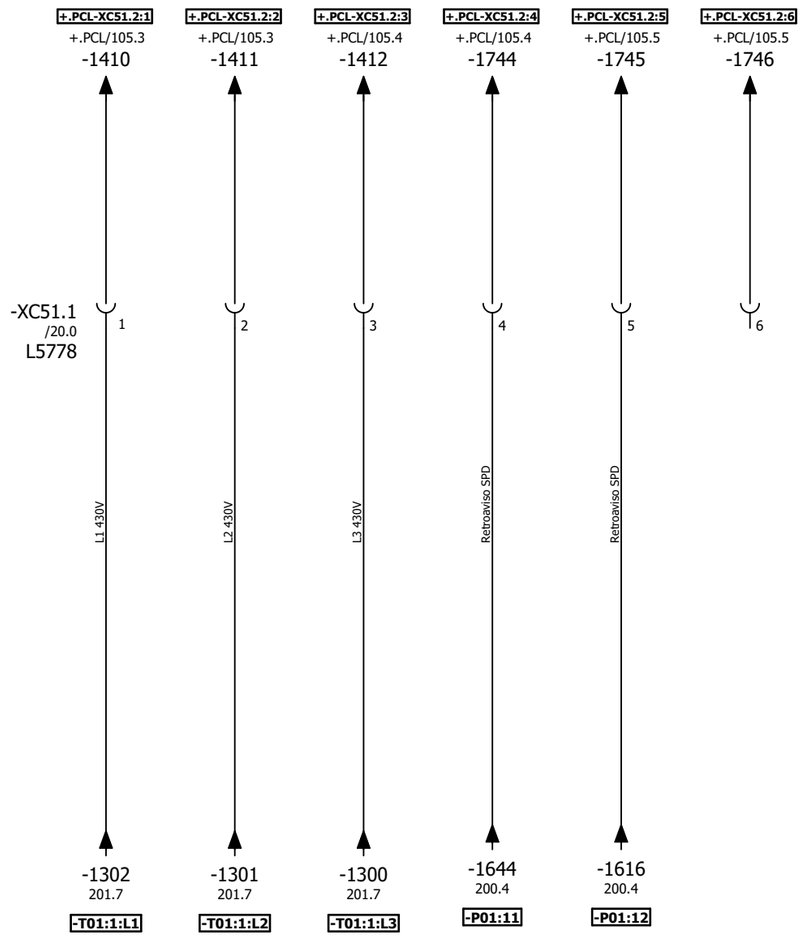
0	1	2	3	4	5	6	7	8	9

-XC51.1 = CONECTOR AEREO 6P PTA - PCL

Interconnection PTA/PPCU



Interconnection SSAA - PCL

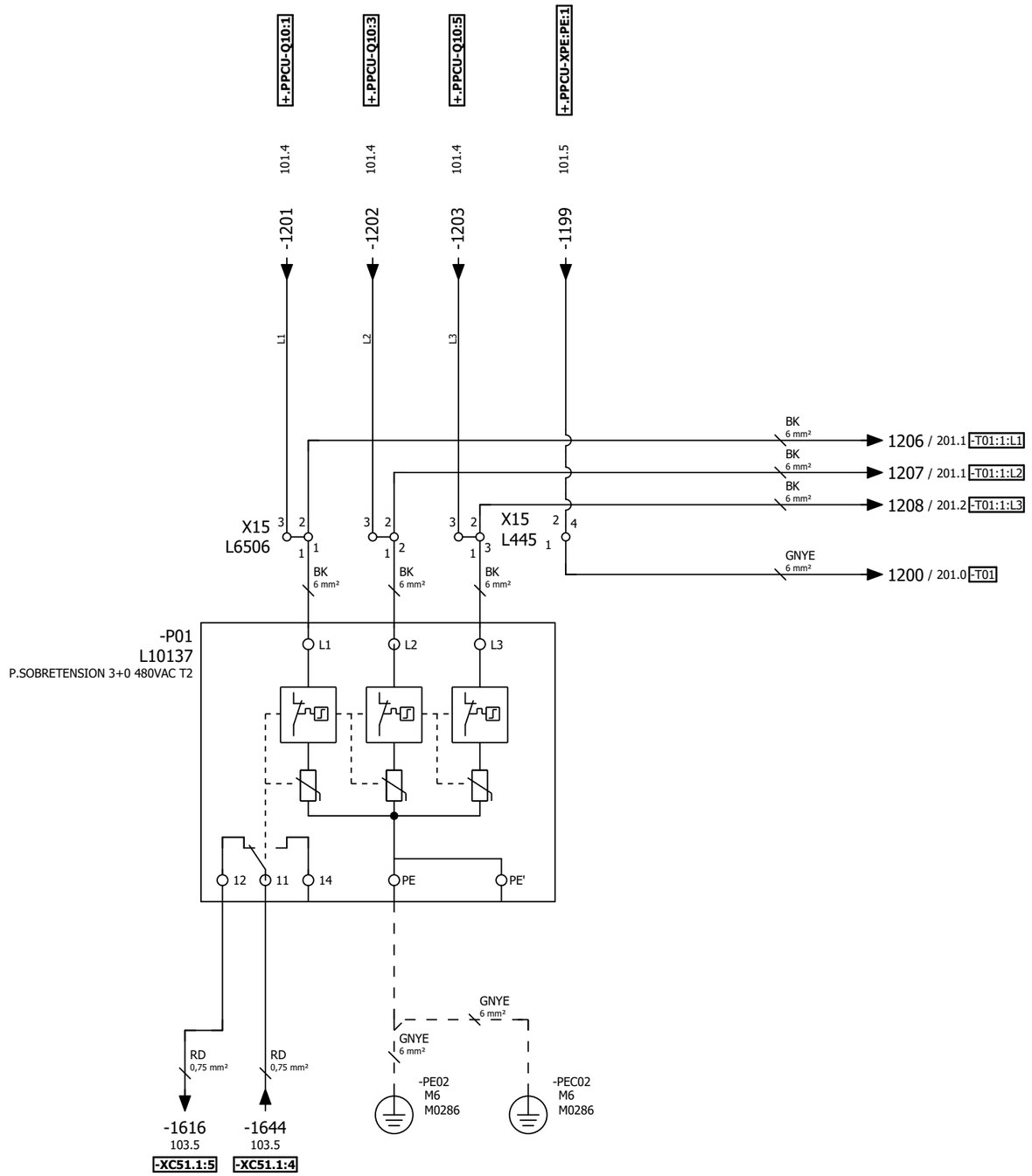


Tap 1 3X430VAC (SSAA)

SPD

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project ESQ_FD1200U000002_000	Name Interconnection PTA - PCL	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PTA	Previous page 101	Next page 200
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 103	Total number of pages 181

From SSAA Transformer



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: SPD	Installation: DCDC GEN 3 CONVERTER — DCDC — A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PTA — Previous page 103 — Next page 201
	APPROVED A.CABALLERO				Project Code: FD1200U000002 — Page 200 — Total number of pages 181

AUXILIARY
POWER SUPPLY
UL: 3x208VAC+PE
IEC:3x400VAC+PE

-X15:4:1 200.7 / -1200

-1206 200.7 -X15:1:2
-1207 200.7 -X15:2:2
-1208 200.7 -X15:3:2

T01
6KVA
L9466C
/201

PE
L1 208V
L2 208V
L3 208V
N 208V

3x208V 60HZ

L1 L1 400V 400V
L2 L2 400V 400V
L3 L3 400V 400V
N N N N 400V 400V 400V 400V

3x400V 50HZ

RD 0,75 mm² → 1310 / +.PSA/400.1 +.PSA-E10-J3:1
RD 0,75 mm² → 1311 / +.PSA/400.1 +.PSA-E10-J3:5

+PSA

C NC N N

Thermal switches

Tap 1 3X430VAC (5KVA) Tap 2 3X93VAC (1KVA)

N 430V
L1 430V
L2 430V
L3 430V

L1 93V
L2 93V
L3 93V

BK 2,5 mm² L2 93V → 1305 / +.PSA/201.1 +.PSA-X01:2:3
RD 2,5 mm² L1 93V → 1304 / +.PSA/201.0 +.PSA-X01:1:3

+PSA

GY 2.5 mm L3 430V → -1300 / 103.4 -XC51.1:3
BK 2,5 mm² L2 430V → -1301 / 103.4 -XC51.1:2
BN 2.5 mm L1 430V → -1302 / 103.3 -XC51.1:1

+PCL

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: SSAA Auto transformer	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PTA	Previous page 200 / Next page &MAT/1000
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 201 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-P01	L10137	590098	P.SOBRETENSION 3+0 480VAC T2	2
-P01	L7903	8630740000	End bracket	1
-T01	L9466C	L9466C	TRAFO DCDC/GEN3 6KVA IEC-UL	1
-T01	L9466D	L9466D	TRAFO DCDC/GEN3 6KVA IEC-UL	1

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED		ESQ_FD1200U000002_000	Bill of materials	Mounting location:	DU CABINET ADU.PTA	Previous page &/201	Next page &CON/3000
APPROVED A.CABALLERO	REVISION 0.0.0			Project Code:	FD1200U000002	Page 1000	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PTA-XC51.1	CONECTOR AEREO 6P PTA - PCL	L5778	1
		L5709	3
		E1203	2

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CV35 SALIDA 30

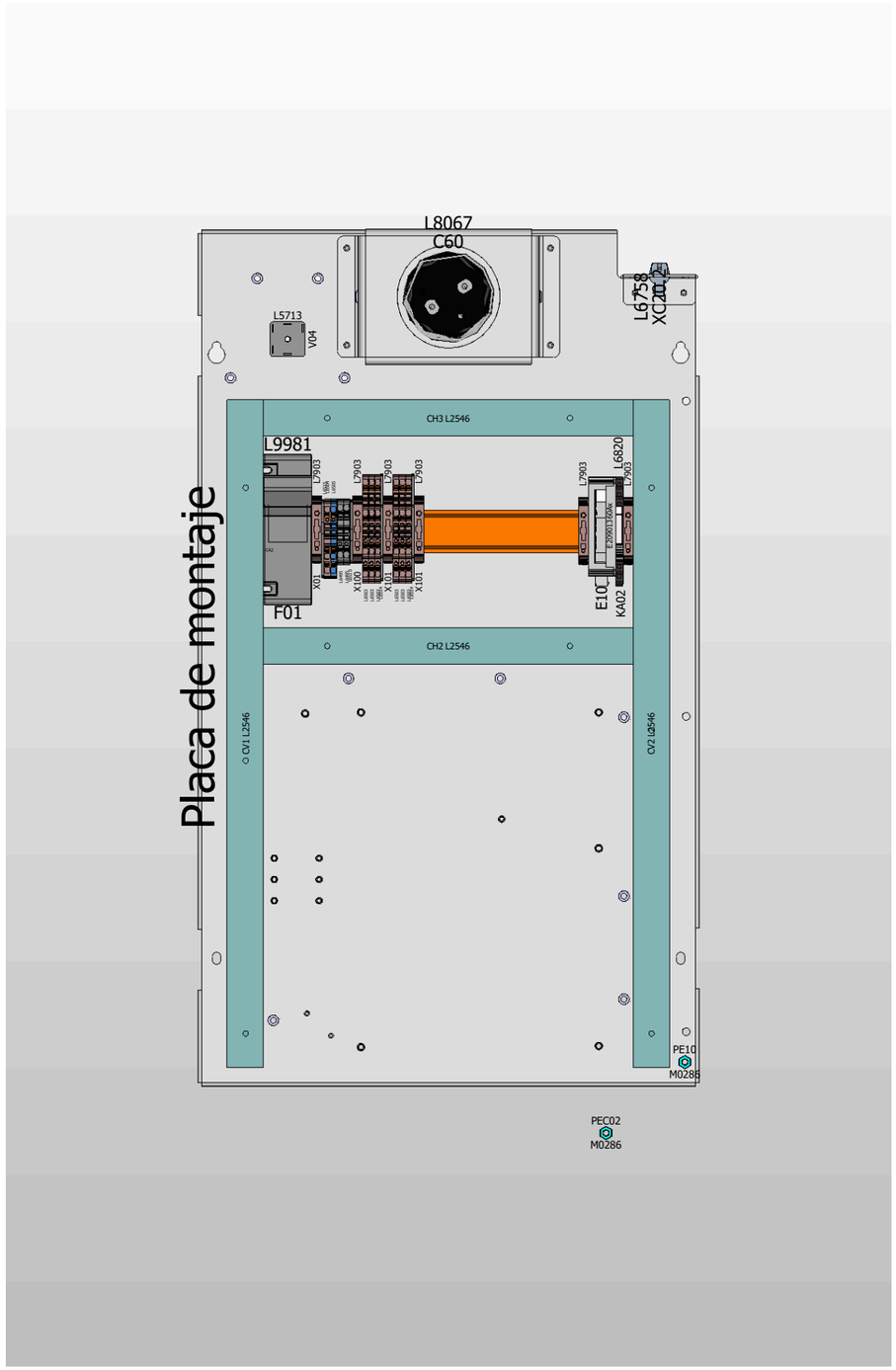
DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PSA	POWER SUPPLY PANEL

PSA

POWER SUPPLY PANEL

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

LAYOUT PANEL



Enclosure legend

IME	ERP	Description
E10	PCB E20901360Ax	TARJETA DE ENTRADAS/SALIDAS DIGITALES (DCDC)
F01	L9981	BASE PORTAFUSIBLE CH J 1P 60A
CA2	L1061	DIN RAIL 35x7,5mm
C60	L8067	CAPACITOR 22.000uF/200VDC
V04	L5713	RECTIFIER BRIDGE 3pH PUENTE 1600V 35A
CH2;CH3;CV1;CV2	L2546	CANAL UNEX 60X30 EN U43X
KA02	L6820	RELAY, 24…230 V UC; 6 A;

0	1	2	3	4	5	6	7	8	9

-X01 = Input SSAA/ECON
-X100 = 120 Vdc Capacitor
-X101 = 120 Vdc distribution

-XC10.2 = PANEL CONNECTOR - 9P SEALED CUSTOMER CONNECTION

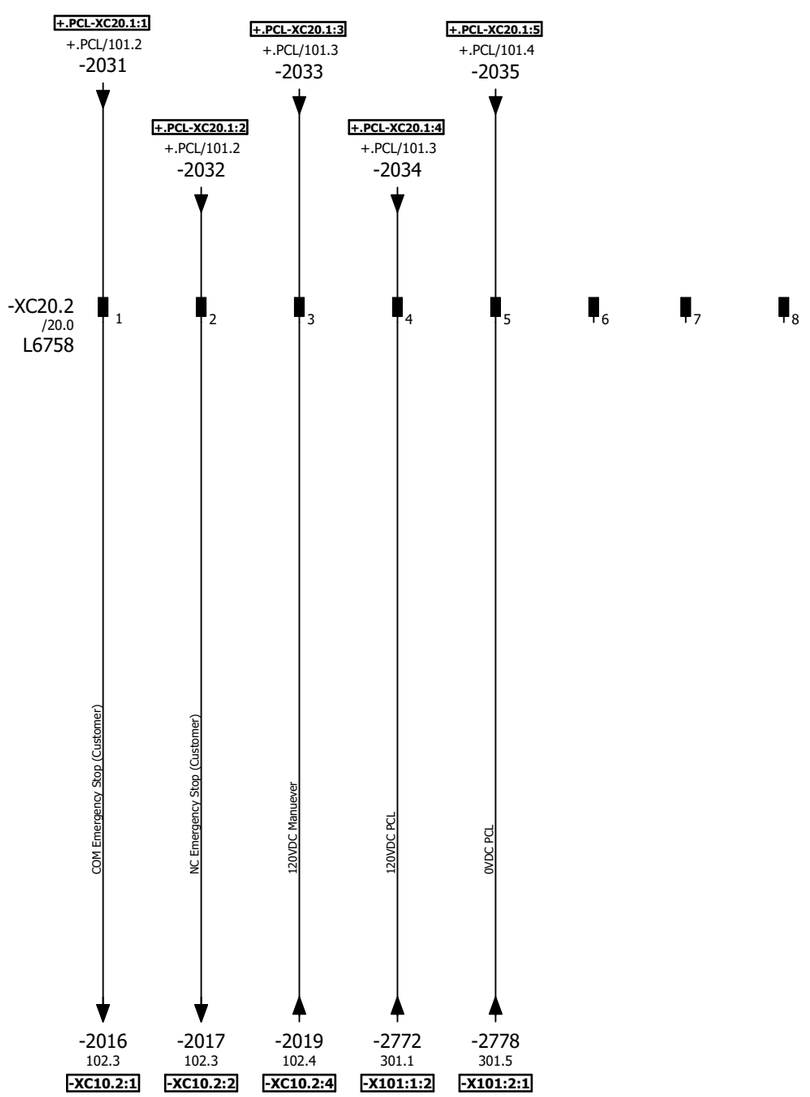
-XC20.2 = CONECTOR PANEL 8P PSA PANEL - PCL

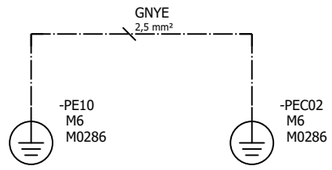
POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Designation terminals and connectors	Mounting location: DU CABINET ADU.PSA	Previous page 10	Next page 101
APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 20	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

Interconnections - PSA-PCL

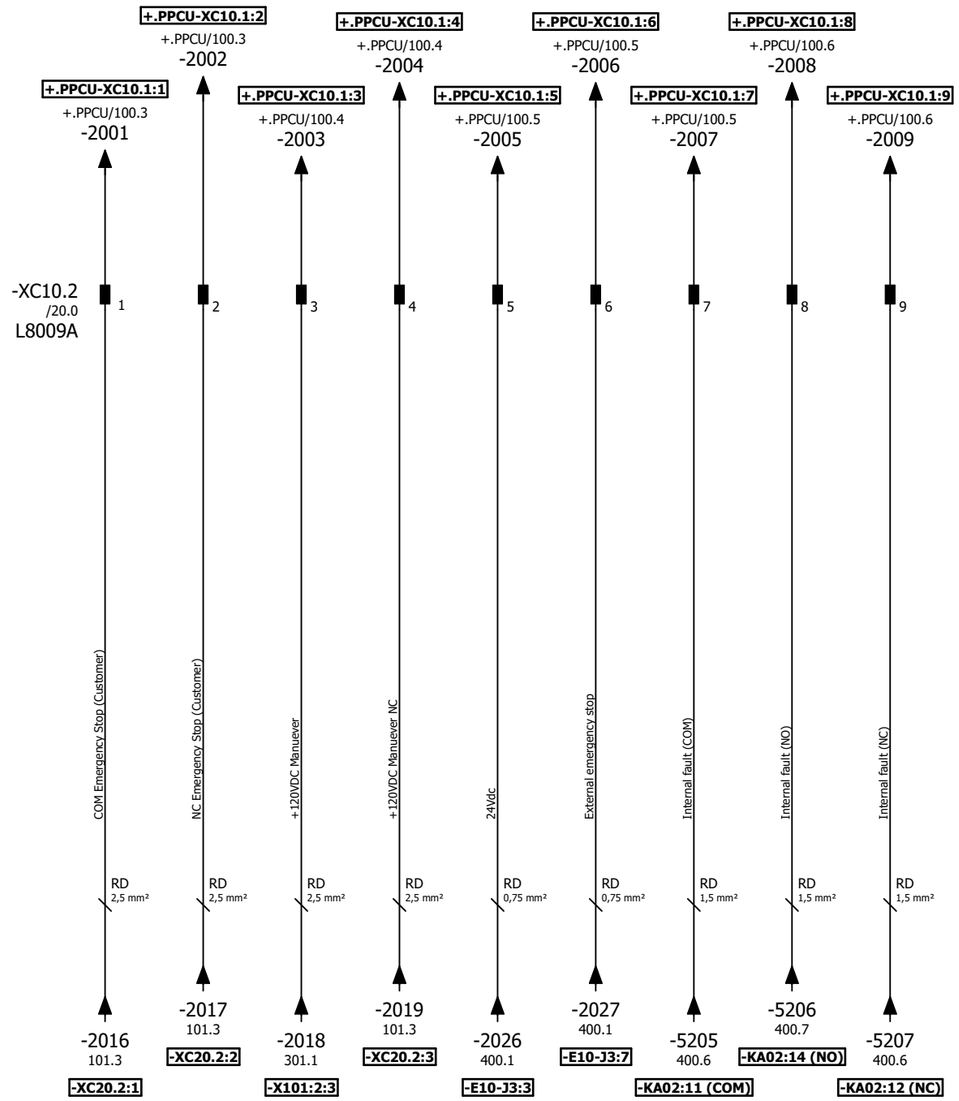




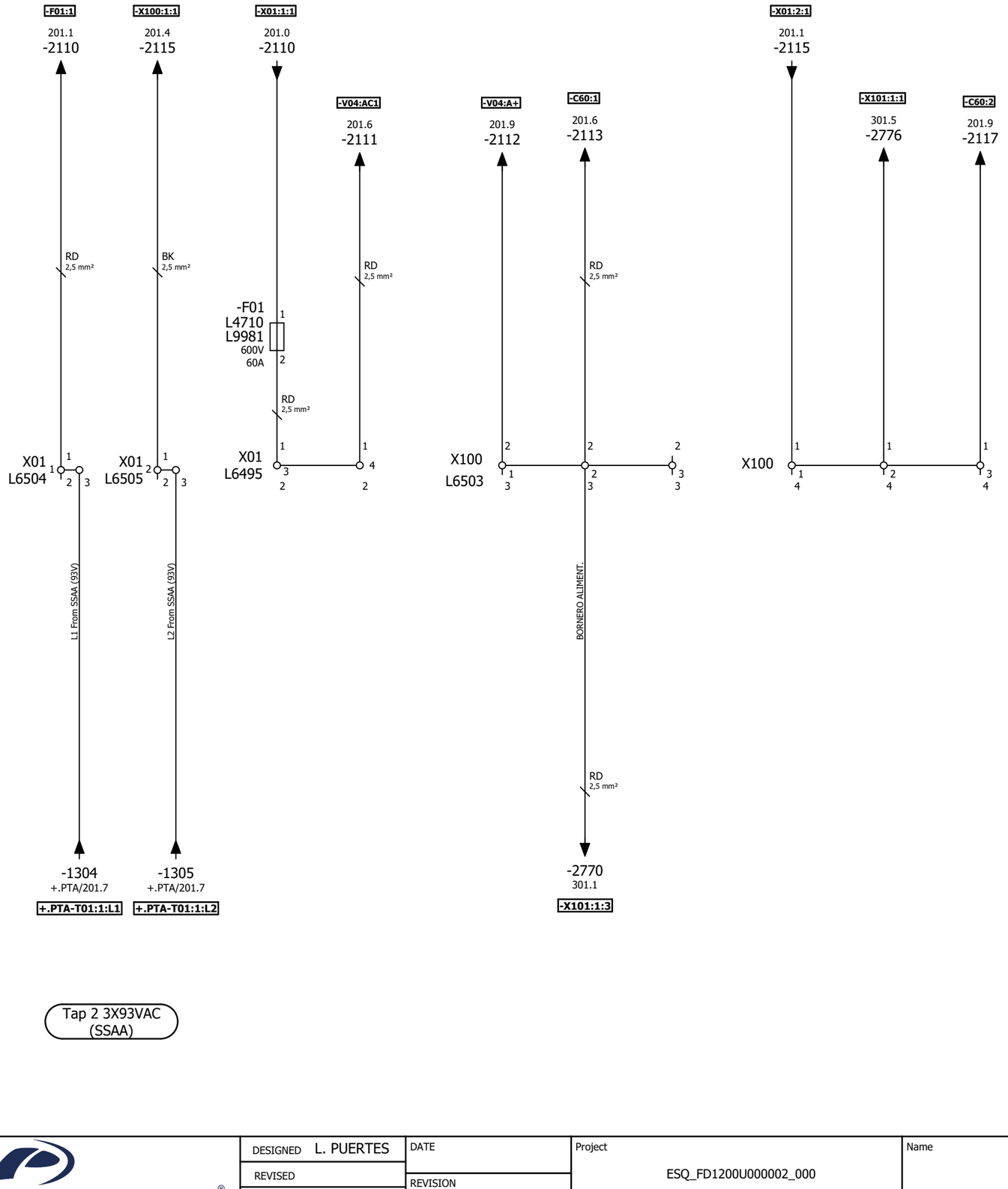
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Interconnections PSA/PCL	Mounting location: DU CABINET ADU.PSA	Previous page 20 / Next page 102
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 101 / Total number of pages 181

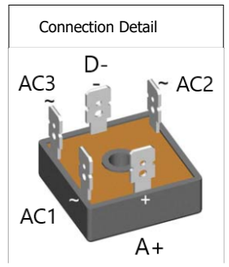
0	1	2	3	4	5	6	7	8	9

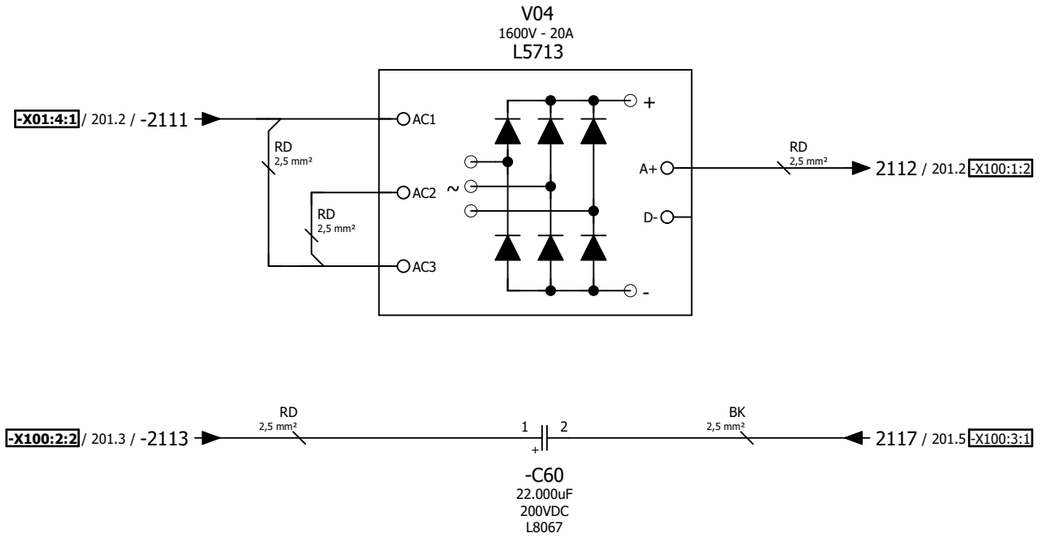
Interconnections PSA/PPCU



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	Interconnections PSA/PPCU	Mounting location: DU CABINET ADU.PSA	Previous page 101 / Next page 201
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 102 / Total number of pages 181







0	1	2	3	4	5	6	7	8	9

POWER ELECTRONICS®

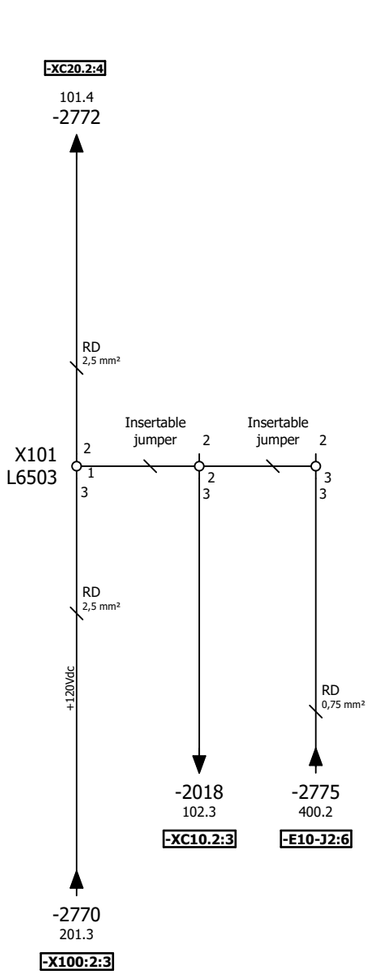
DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
REVISED		ESQ_FD1200U000002_000	Bridge Rectifier	Mounting location: DU CABINET ADU.PSA	Previous page 102	Next page 301
APPROVED A.CABALLERO	REVISION 0.0.0			Project Code: FD1200U000002	Page 201	Total number of pages 181

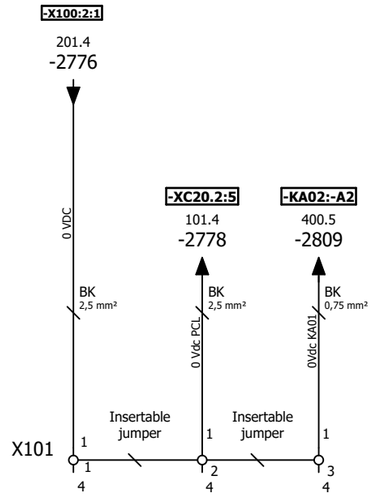
0	1	2	3	4	5	6	7	8	9

Power supply 120Vdc PCL

Power supply 0vdc PCL

DC-



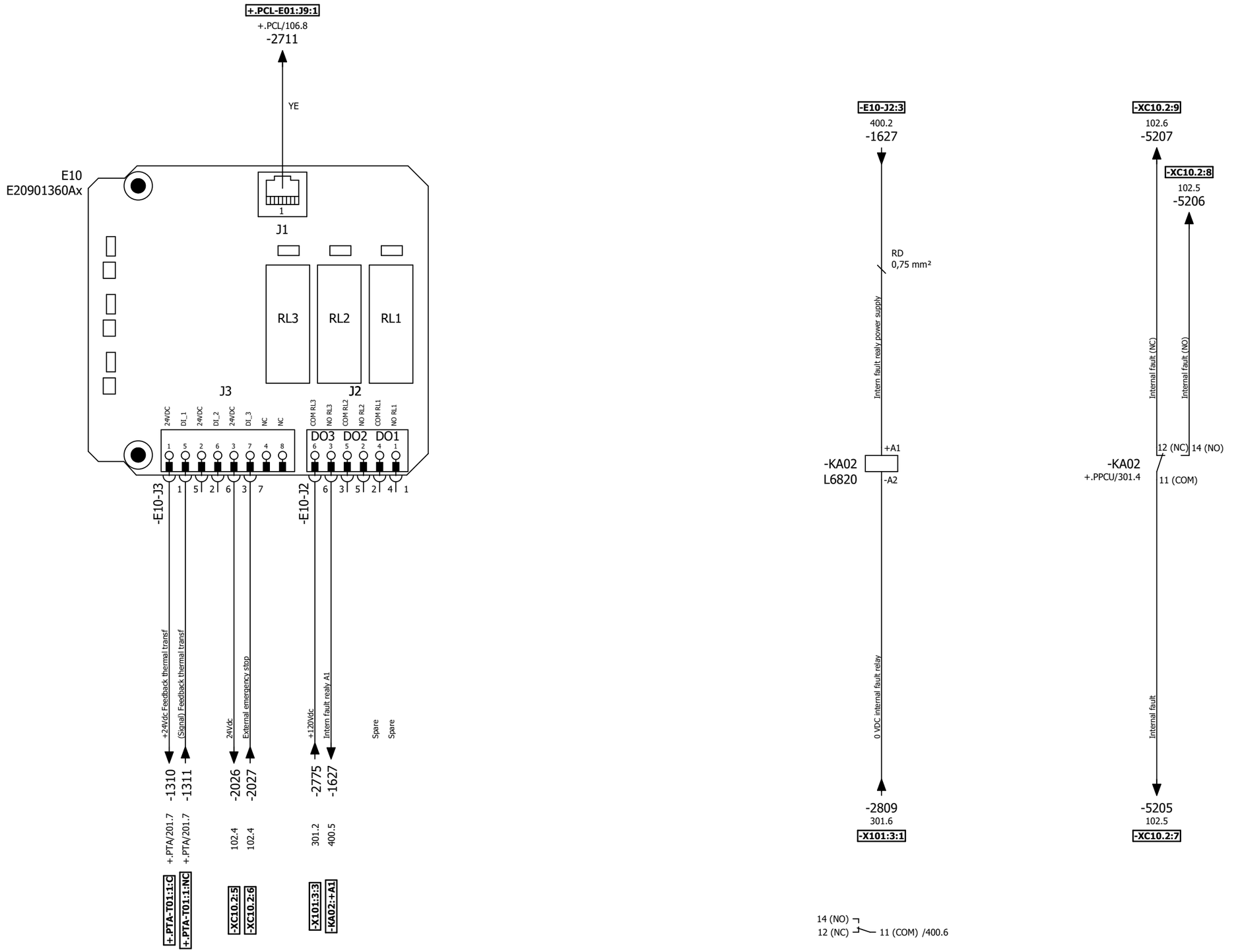


DC+

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project ESQ_FD1200U000002_000	Name 120Vdc Distribution	Installation: DCDC GEN 3 CONVERTER DCDC		A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PSA	Previous page 201	Next page 400
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 301	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

PCB - DC/DC (PSA)



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	PCB - Inputs/Outputs	Mounting location: DU CABINET ADU.PSA · Previous page 301 · Next page &MAT/1000
	APPROVED A.CABALLERO				Project Code: FD1200U000002 · Page 400 · Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-C60	L8067	ALS70A1005NP	CAPACITOR 22.000uF/200VDC	2
-CA2	L1061	0514500000	DIN RAIL 35x7,5mm	2
-E10	PCB E20901360A x	PCB E20901360Ax	TARJETA DE ENTRADAS/SALIDAS DIGITALES (DCDC)	1
-E10	L7903	8630740000	End bracket	4
-F01	L9981	CH60J1	BASE PORTAFUSIBLE CH J 1P 60A	2
-F01	L4710	LPJ-60SP	@@@@@@@FALTA TRADUCIR@@@@@@@	1
-KA02	L6820	1122850000	RELAY, 24…230 V UC; 6 A;	2
-V04	L5713	VS-36MT-160	RECTIFIER BRIDGE 3pH PUENTE 1600V 35A	2

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Bill of materials	Mounting location:	DU CABINET ADU.PSA	Previous page &/400	Next page &CON/3000
APPROVED A.CABALLERO	0.0.0			Project Code:	FD1200U000002	Page 1000	Total number of pages 181

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PSA-E10-J2	Spare		
=DCDC++ZMI+ADU.PSA-E10-J3	+24Vdc Feedback thermal transf		
=DCDC++ZMI+ADU.PSA-J2			
=DCDC++ZMI+ADU.PSA-J3			
=DCDC++ZMI+ADU.PSA-XC10.2	PANEL CONNECTOR - 9P SEALED CUSTOMER CONNECTION	L8009A	1
		L5710	9
=DCDC++ZMI+ADU.PSA-XC20.2	CONECTOR PANEL 8P PSA PANEL - PCL	L6758	1
		L5710	5
		L6758	1

POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PCL	CONTROL PANEL

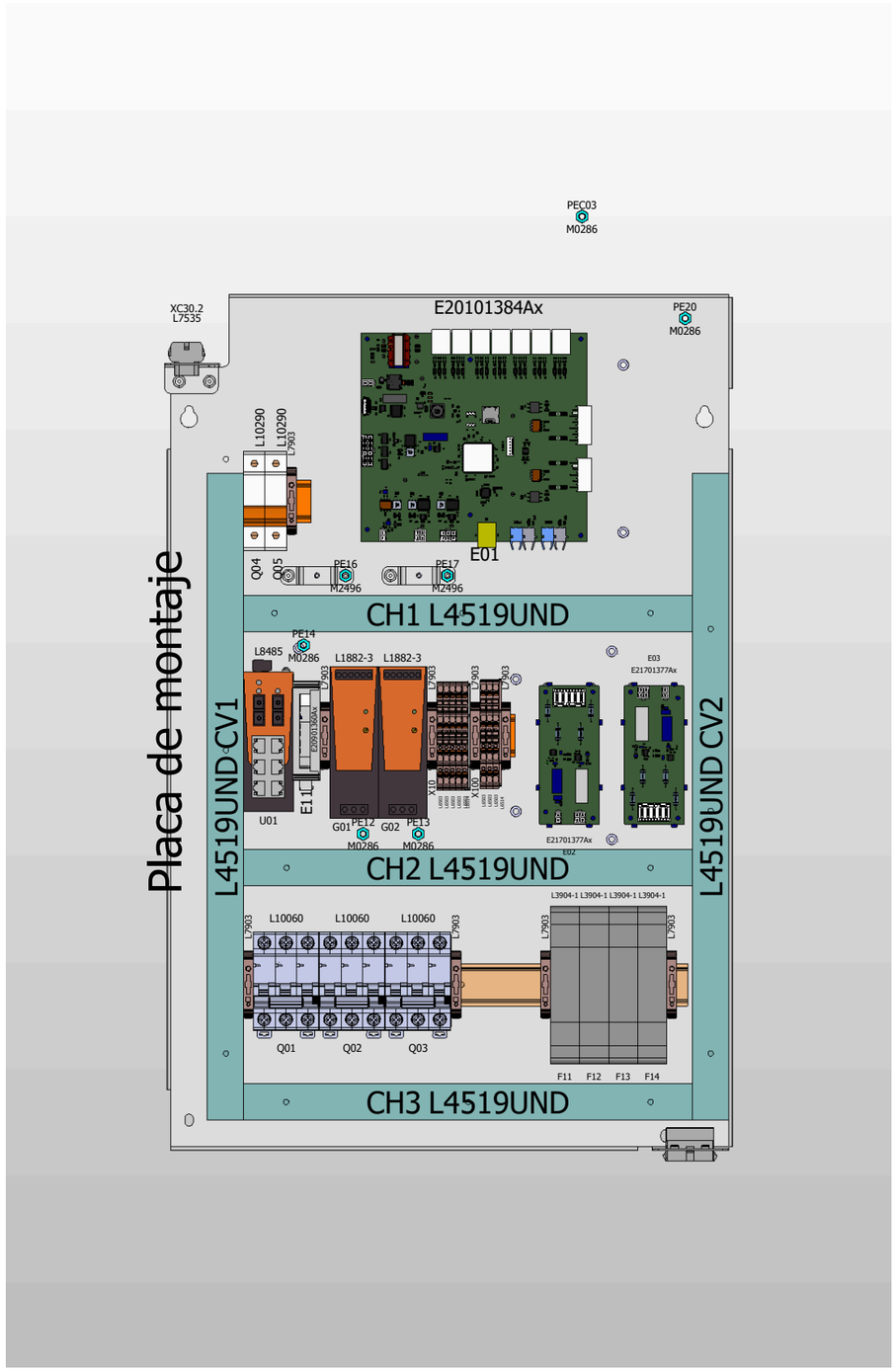
PCL

CONTROL PANEL

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

0	1	2	3	4	5	6	7	8	9

LAYOUT PANEL



Enclosure legend

IME	ERP	Description
E02;E03	PCB E21701377Ax	TARJETA DE SEÑALIZACIÓN DE BUS DCDC GEN3
E01	E20101384Ax	TARJETA CONTROL CENTRAL DCDC GEN 3
E11	PCB E20901360Ax	TARJETA DE ENTRADAS/SALIDAS DIGITALES (DCDC)
U01	L8485	SWITCH FO-ETH 2xSC + 6 ETH UNMANAGED
CH1-CH3;CV1;CV2	L4519UND	Unex slotted trunking 60X30 in U43X
F11-F14	L3904-1	FUSE HOLDER 32A 1500V 1P PML 10x85 DF
C1-C3	L1061	DIN RAIL 35x7,5mm
Q01-Q03	L10060	BK63H 3P C6A 6kA 480Y/277VAC UL1077
G01;G02	L1882-3	PRO ECO 120W 24V 5A
Q04;Q05	L10290	BK63HU 1P C4A 10kA 277VAC UL1077
F11-F14	L4635	FUSE 4A 1500VDC 10KA 10x85 UL DF

0	1	2	3	4	5	6	7	8	9

-X10 = Terminal distribution 24Vdc

-X100 = Terminal distribution 120VDC

-XC20.1 = CONECTOR AEREO 8P PSA-PCL

-XC30.1 = CONECTOR AEREO 9P PCL-PSA+PDU
-XC30.2 = CONECTOR PANEL 9P PCL

-XC031.1 = CONECTOR AEREO 8P INTERCONECTION PCL-PPU

-XC41.1 = CONECTOR AEREO 2P PCL-SPD

-XC51.2 = CONECTOR PANEL 6P PCL - PTA BASE

-XC101.1 = CONNECTOR OPTICAL FIBER DUPLEX (CONTROL DCDC PCB)

-XC111.1 = CONECTOR AEREO POWER SUPPLY DU FAN
-XC115.2 = M&L SEALED AERIAL POWER SUPPLY DU FAN

-XC425.1 = CONECTOR AEREO 2P PDC

-XC307.1 = NTC CONNECTOR - FAN ADMISSION TEMP.

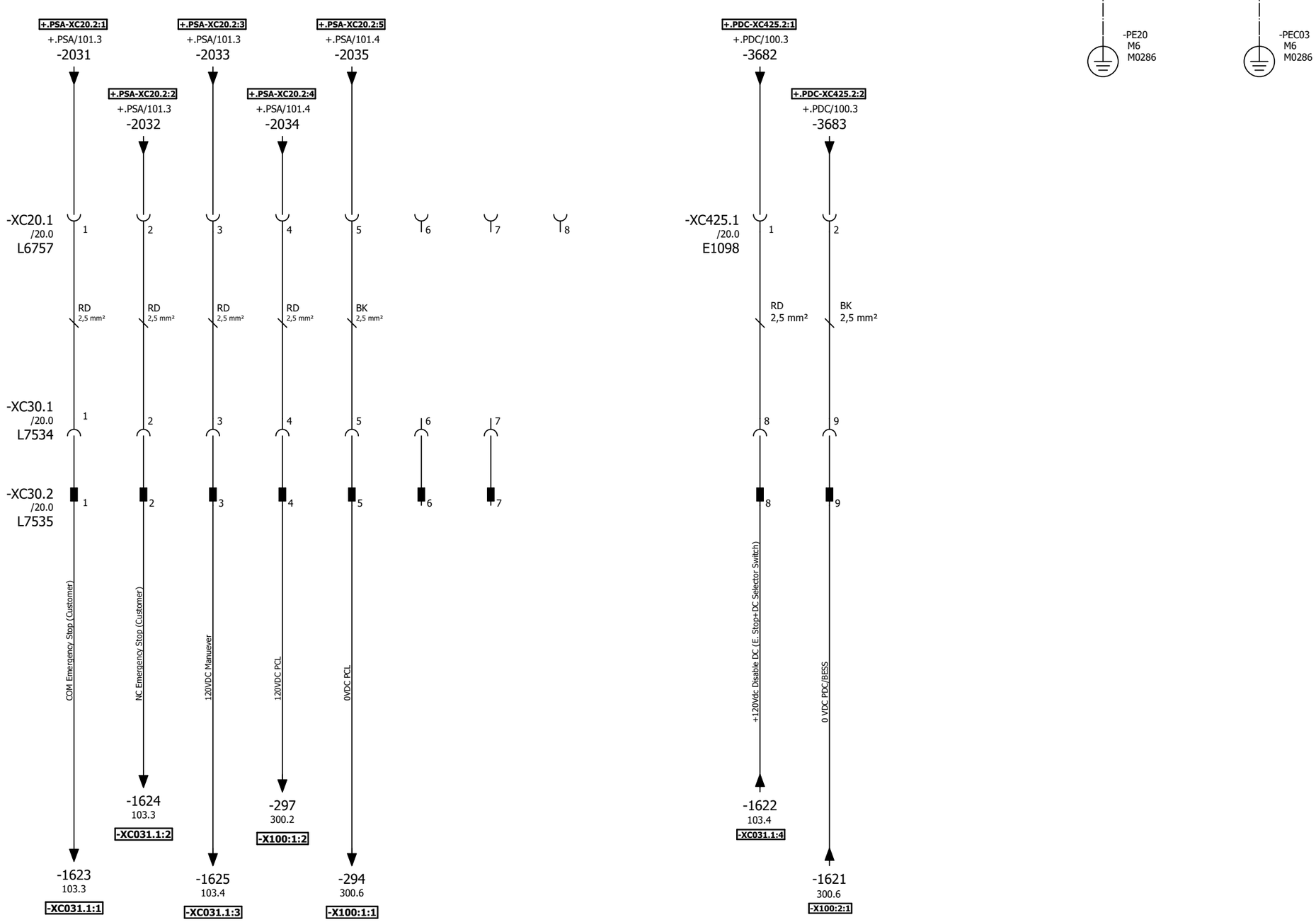
POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Designation terminals and connectors	Mounting location:	DU CABINET ADU.PCL	Previous page 10	Next page 101
APPROVED A.CABALLERO	0.0.0			Project Code:	FD1200U000002	Page 20	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

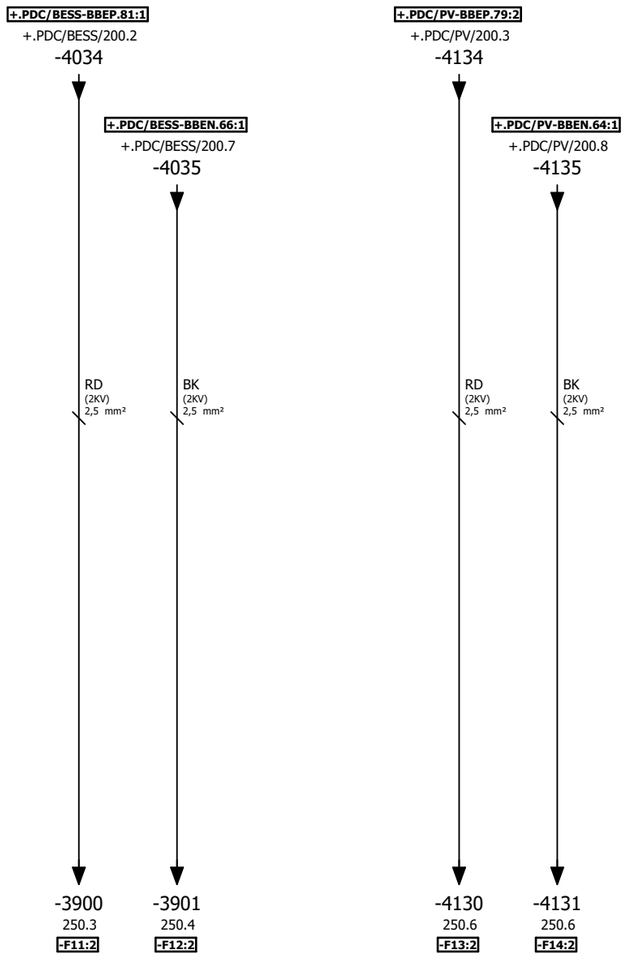
Interconnections PCL/PSA

Interconnections PCL/PDC



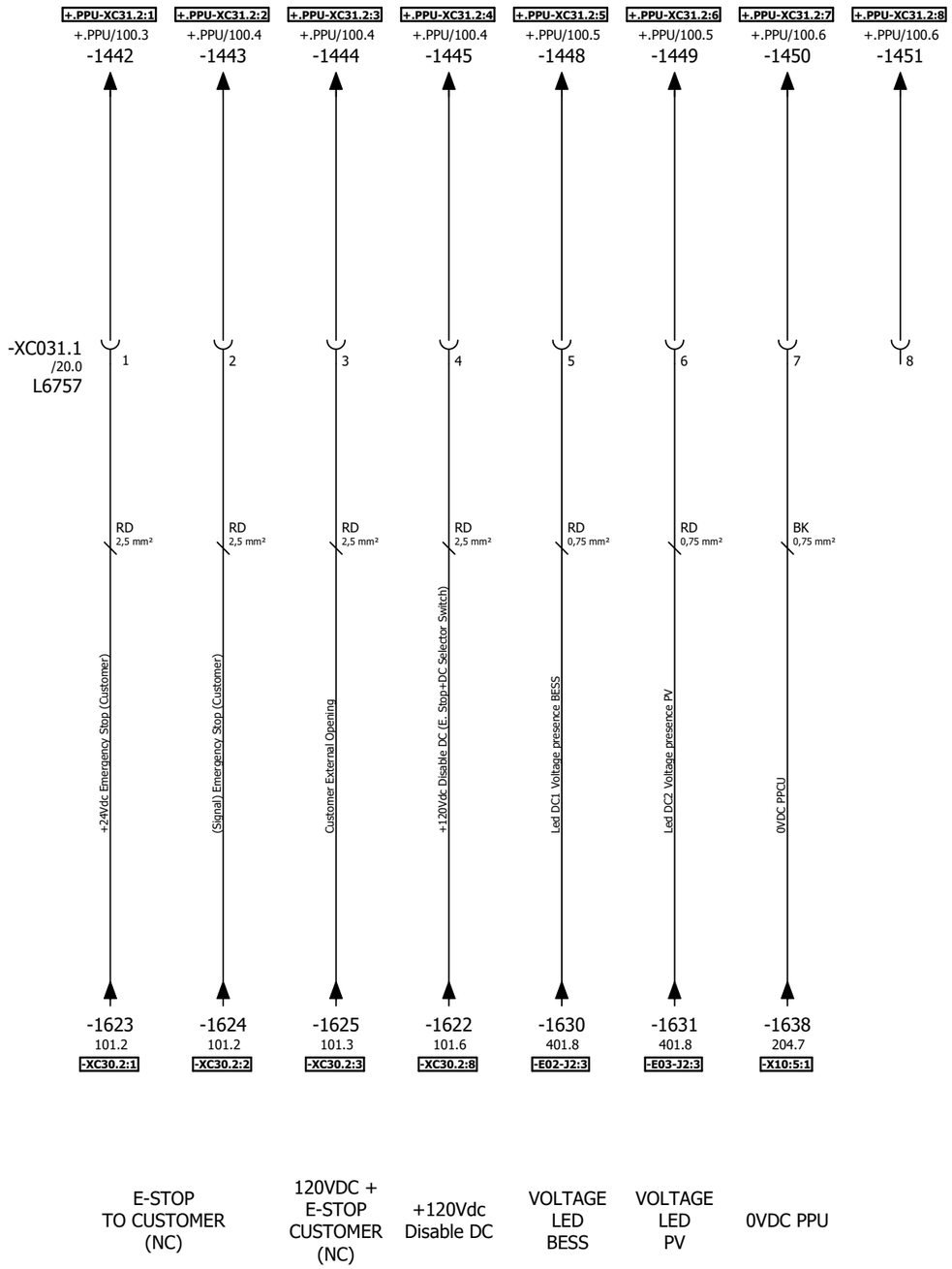
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Interconnections PCL/PSA-PDC	Mounting location: DU CABINET ADU.PCL	Previous page 20 / Next page 102
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 101 / Total number of pages 181

Interconnections PCL
PDC-BESS // PDC-PV
Voltage Measurements



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project ESQ_FD1200U000002_000	Name Interconnections PCL/PDC Voltage Measurement	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PCL	Previous page 101 / Next page 103
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 102 / Total number of pages 181

Interconnections - PCL / PPU

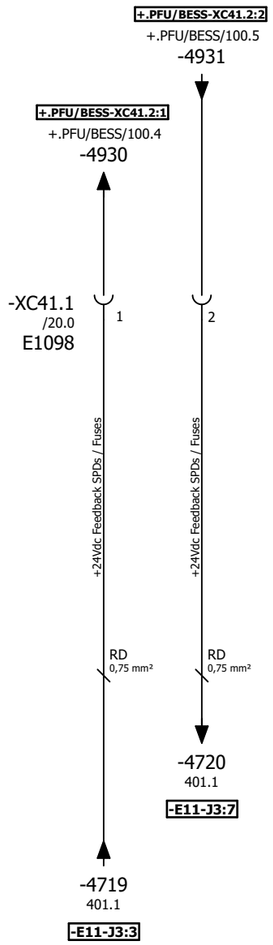


POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Interconnections PCL/PPU	Mounting location: DU CABINET ADU.PCL	Previous page 102 / Next page 104
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 103 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

Interconnections PCL/PFU-BESS

Feedback SPD & Fuses

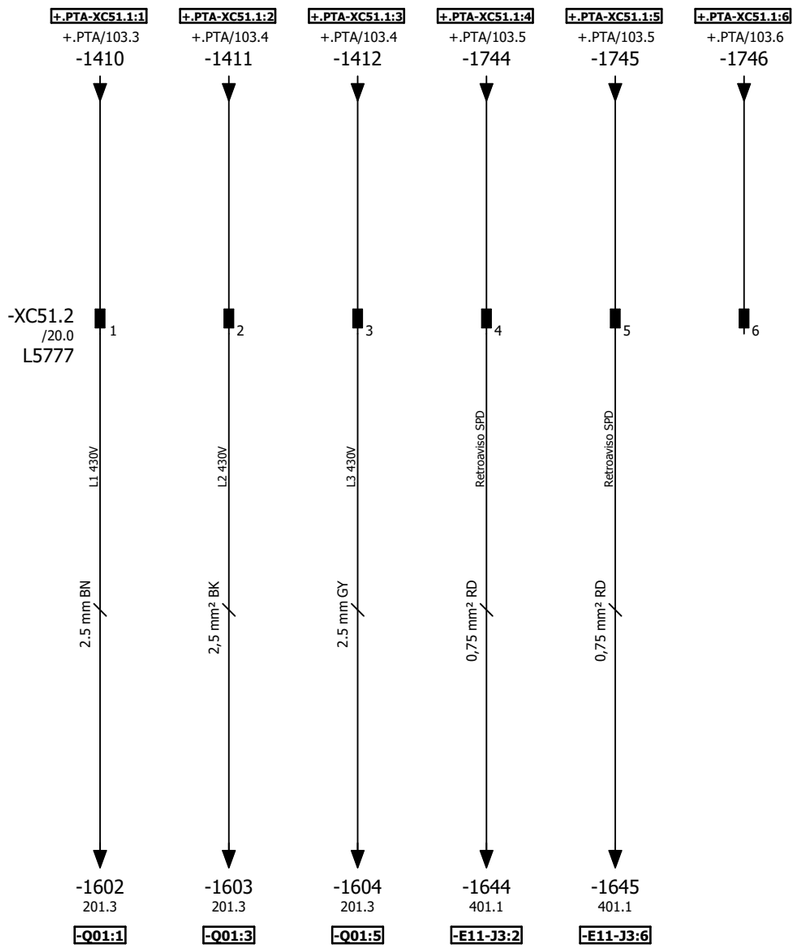


POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project ESQ_FD1200U000002_000	Name Interconnections PCL/PFU (SPD / Fuses)	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PCL	Previous page 103 / Next page 105
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 104 / Total number of pages 181

Interconnection SSAA - PCL

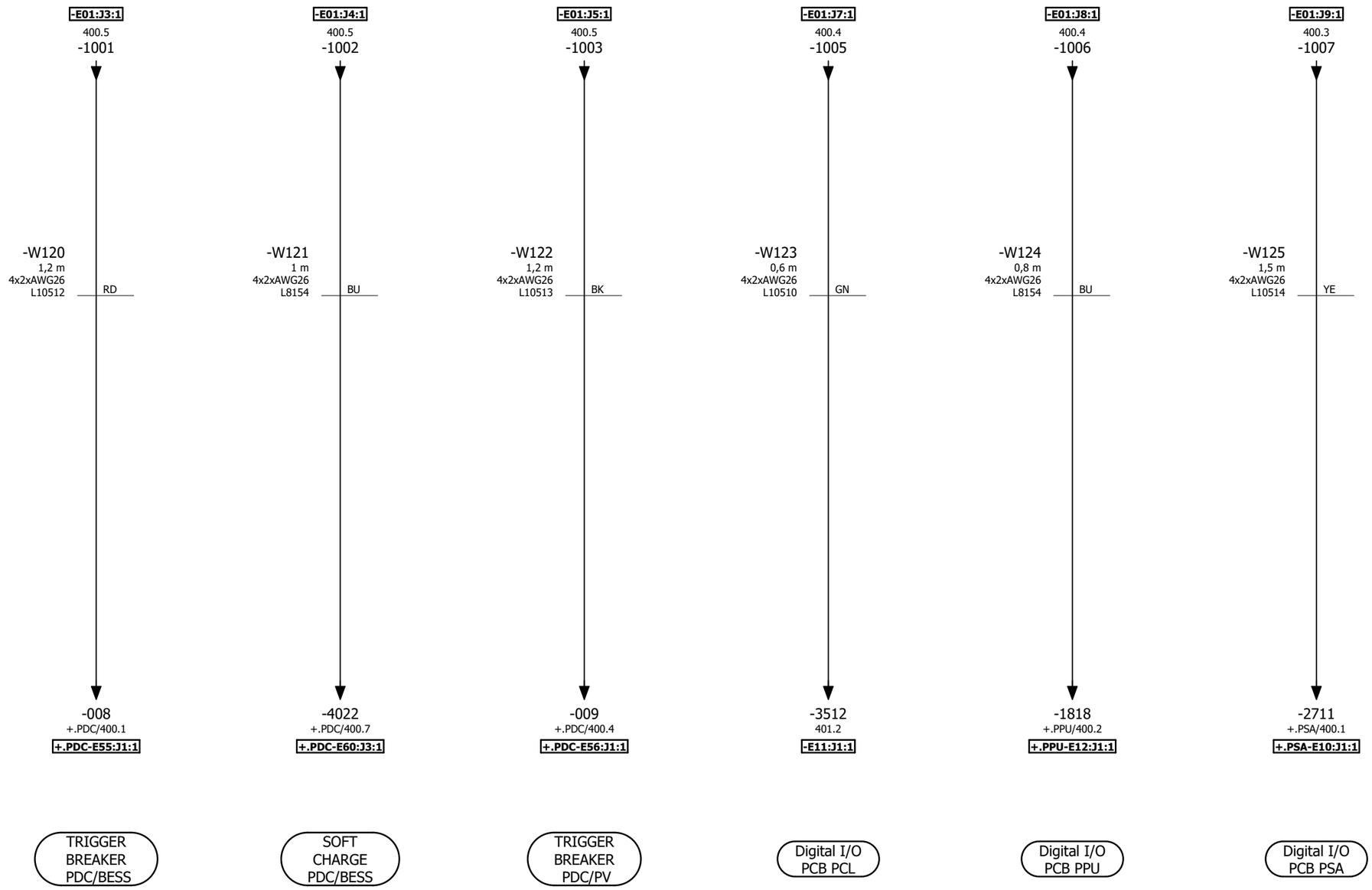
Tap 1 3X430VAC

SPD



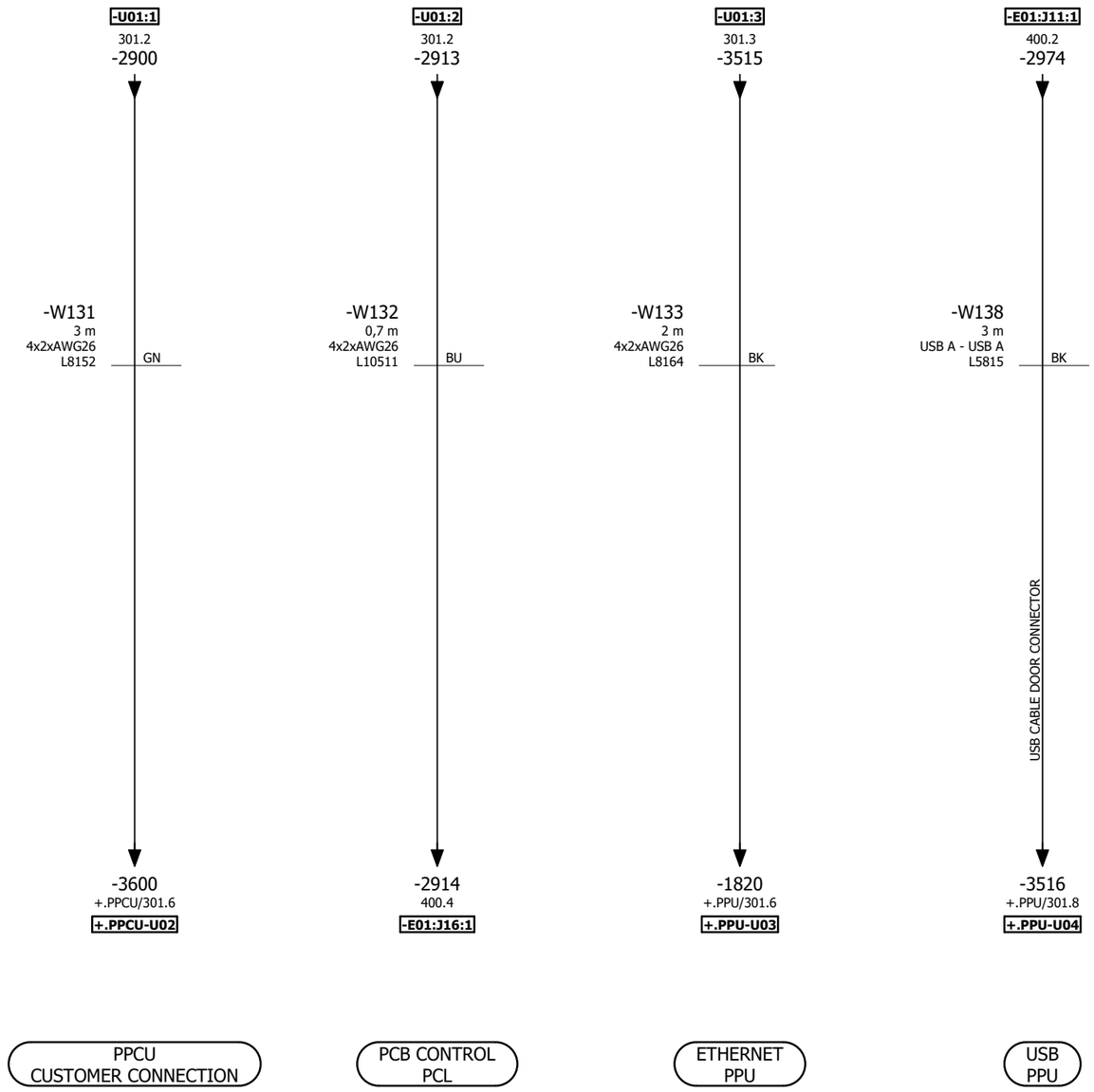
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	REVISED	REVISION	ESQ_FD1200U000002_000	Interconnection PCL/SSAA	Mounting location: DU CABINET ADU.PCL		Previous page 104	Next page 106	
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002		Page 105	Total number of pages 181	

Interconnections
RJ45 - PCB I/O



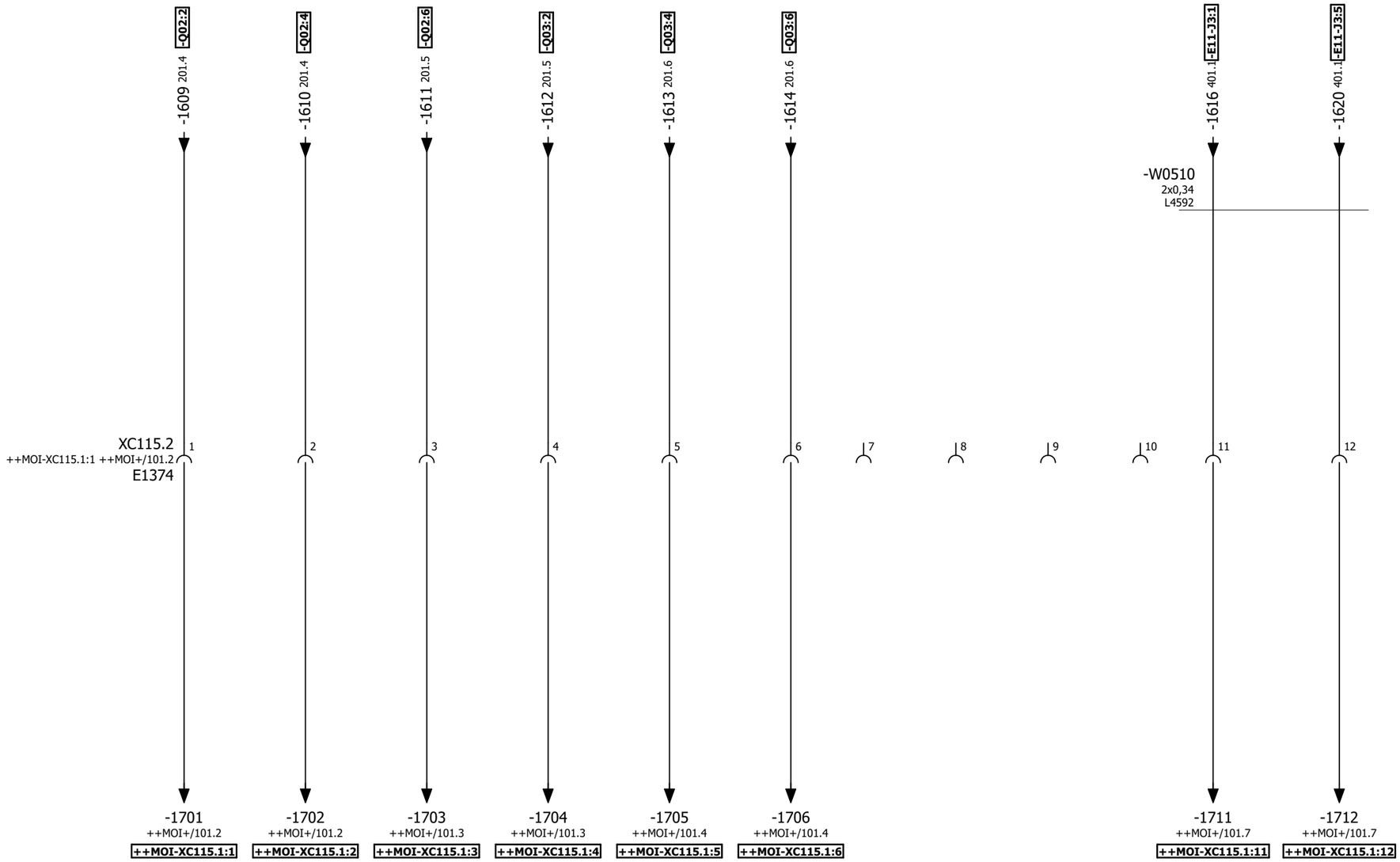
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: Interconnections - RJ45 (I/O)	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PCL	Previous page 105 / Next page 107
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 106 / Total number of pages 181

Interconnections
ETHERNET - USB

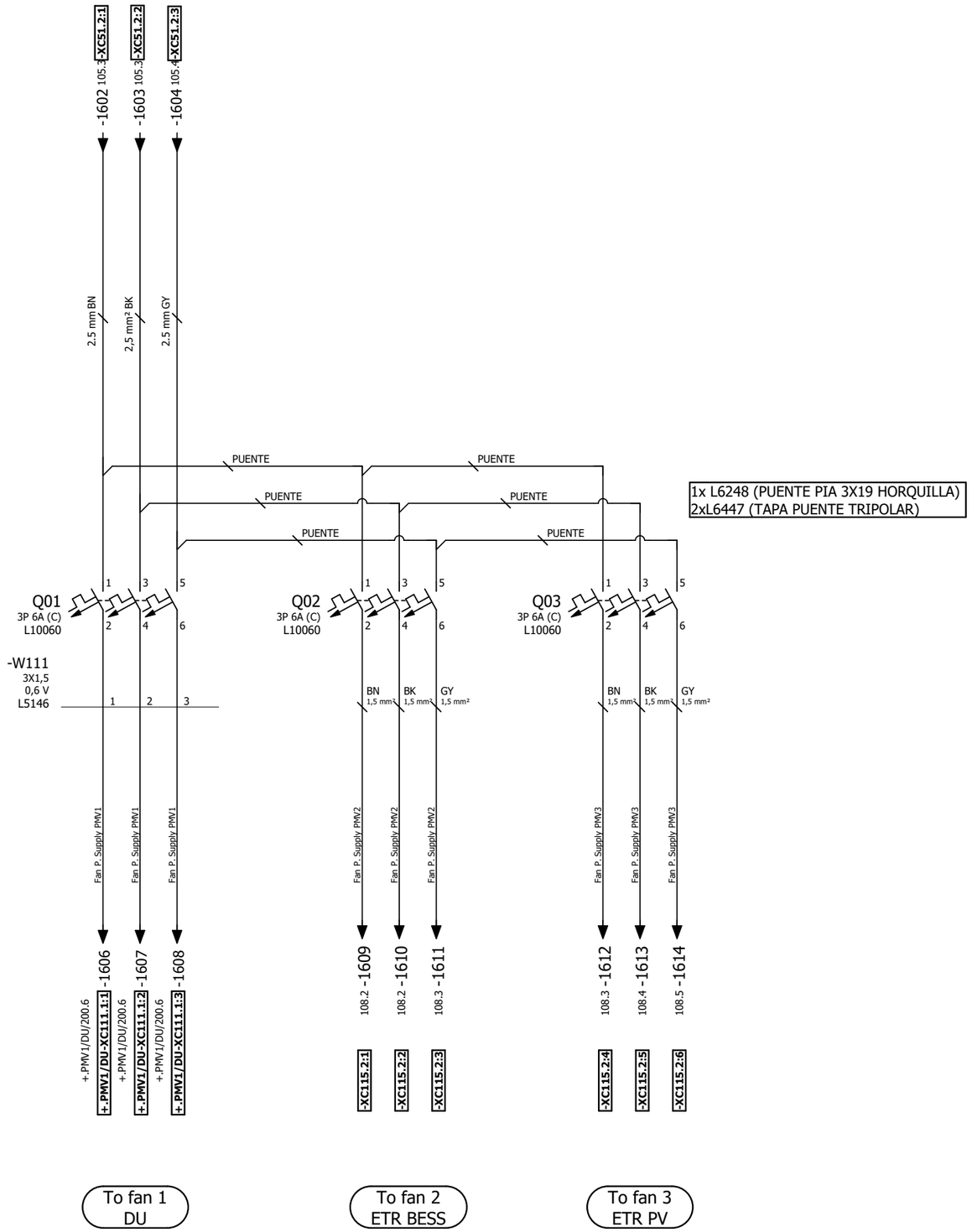


POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	Interconnections - ETHERNET/USB	Mounting location: DU CABINET ADU.PCL	Previous page 106 / Next page 108
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 107 / Total number of pages 181

Interconnections - PCL / MOI (FANS)

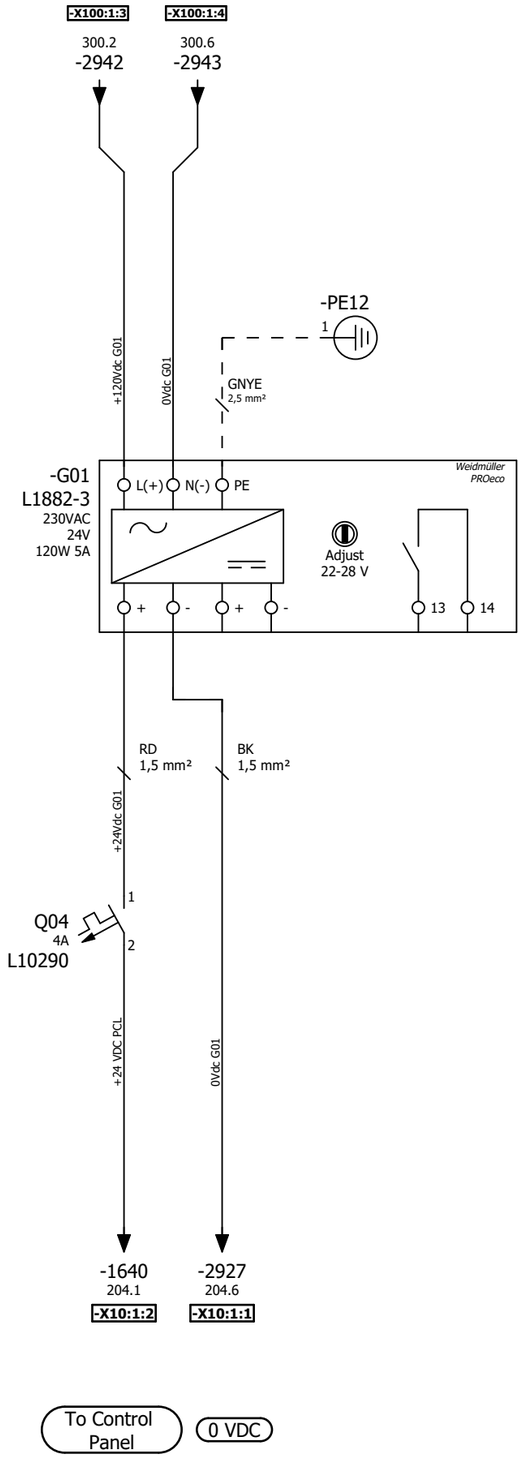


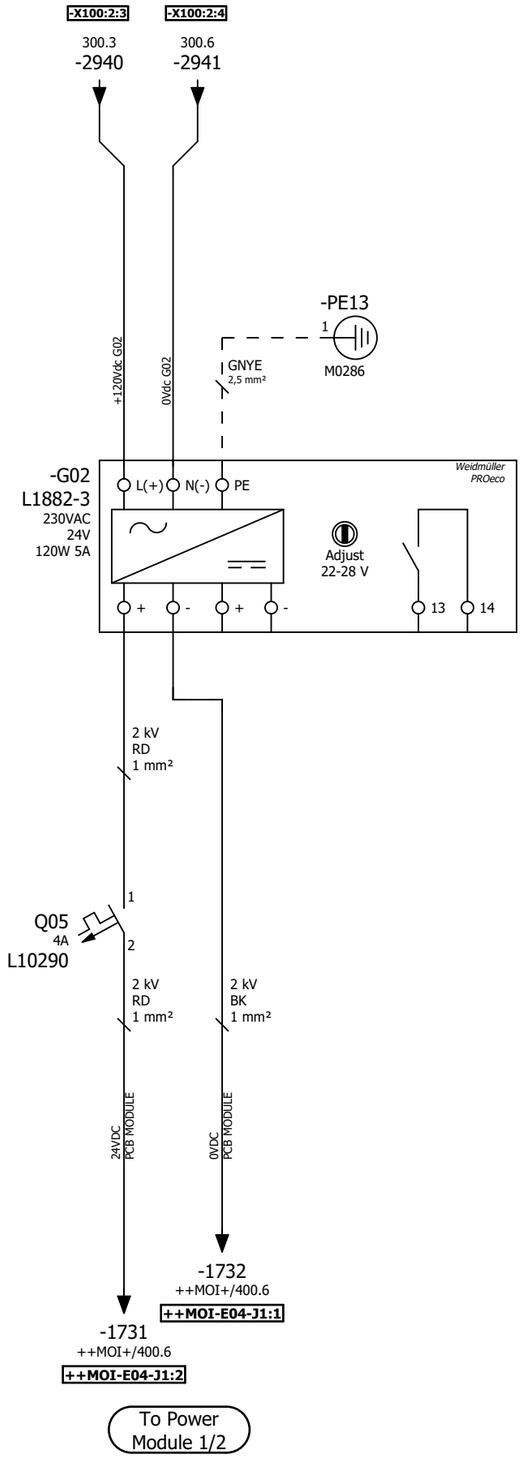
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Interconnections - PCL / MOI (FANS)	Mounting location: DU CABINET ADU.PCL	Previous page 107 / Next page 201
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 108 / Total number of pages 181



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000
	REVISED	REVISION 0.0.0	Name: Fan distribution
	APPROVED A.CABALLERO		

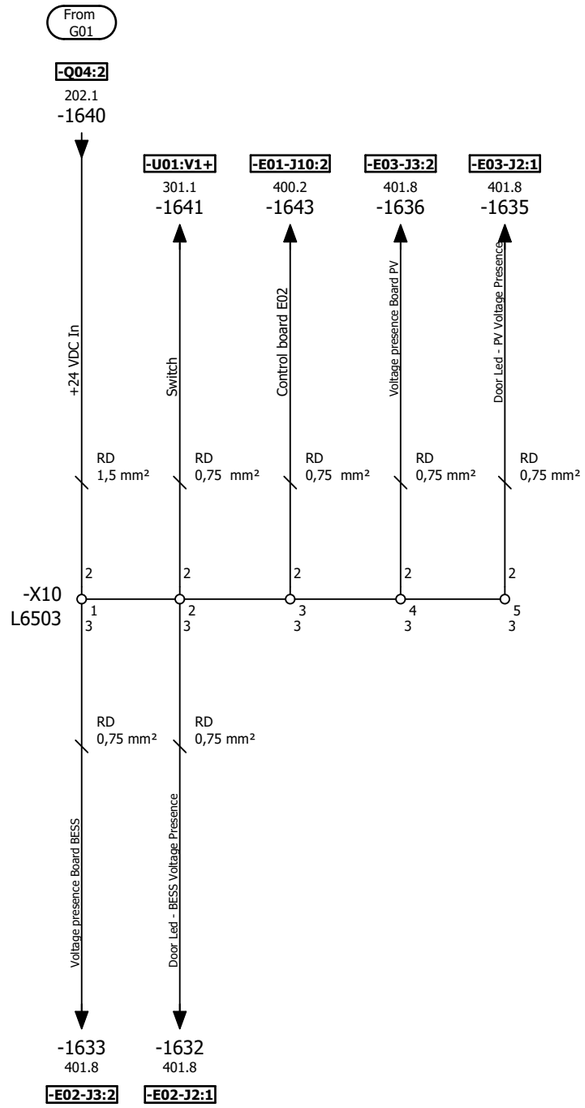
Installation:	DCDC GEN 3 CONVERTER		DCDC	A3
Mounting location:	DU CABINET ADU.PCL	Previous page 108	Next page 202	
Project Code:	FD1200U000002	Page 201	Total number of pages 181	

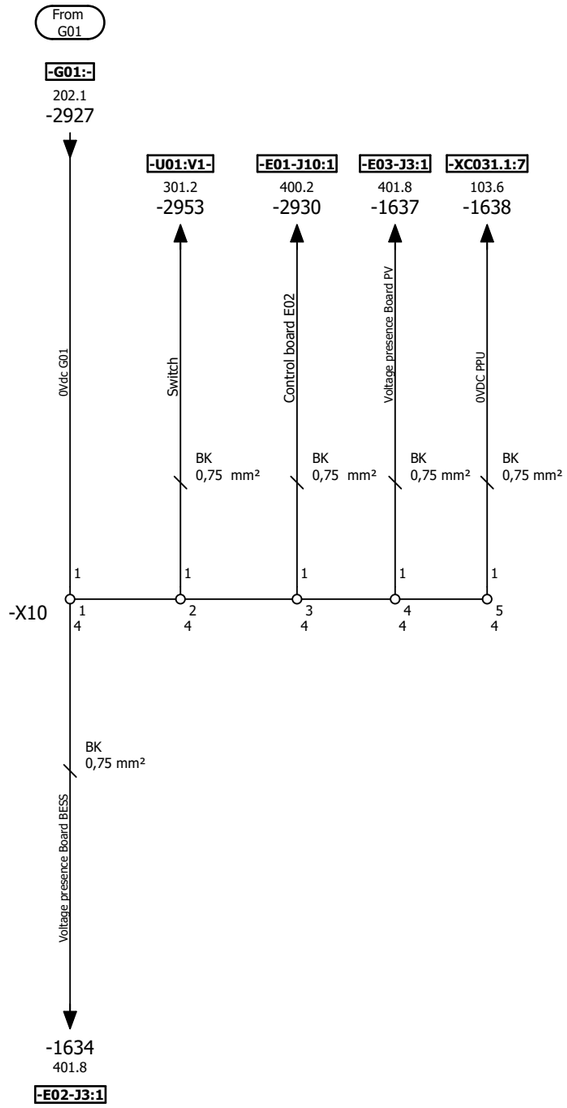




POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: 24Vdc Power Supplies	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PCL	Previous page 201 / Next page 204
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 202 / Total number of pages 181

24VDC power supply





POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	24Vdc distribution	Mounting location: DU CABINET ADU.PCL	Previous page 202 / Next page 250
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 204 / Total number of pages 181

VOLTAGE PRESENCE & MEASUREMENT [BESS]

-E02-J1:1 | -E01-J1:1 | -E02-J1:5 | -E01-J1:5

401.4 | 400.7 | 401.4 | 400.7

-3903 | -3905 | -3902 | -3904

VOLTAGE PRESENCE | VOLTAGE MEASUREMENT | VOLTAGE PRESENCE | VOLTAGE MEASUREMENT

RD (2KV) 1 mm² | RD (2KV) 1 mm² | BK (2KV) 1 mm² | BK (2KV) 1 mm²

1 2

-F11 1500Vdc 4A L4635 | -F12 1500Vdc 4A L4635

-3900 102.3 | -3901 102.4

+.PDC/BESS-BBEP.81:1 | +.PDC/BESS-BBEN.66:1

VOLTAGE PRESENCE & MEASUREMENT [PV]

-E03-J1:1 | -E01-J2:1 | -E03-J1:5 | -E01-J2:6

401.4 | 400.7 | 401.4 | 400.7

-4031 | -4033 | -4030 | -4032

VOLTAGE PRESENCE | VOLTAGE MEASUREMENT | VOLTAGE PRESENCE | VOLTAGE MEASUREMENT

RD (2KV) 1 mm² | RD (2KV) 1 mm² | BK (2KV) 1 mm² | BK (2KV) 1 mm²

1 2

-F13 1500Vdc 4A L4635 | -F14 1500Vdc 4A L4635

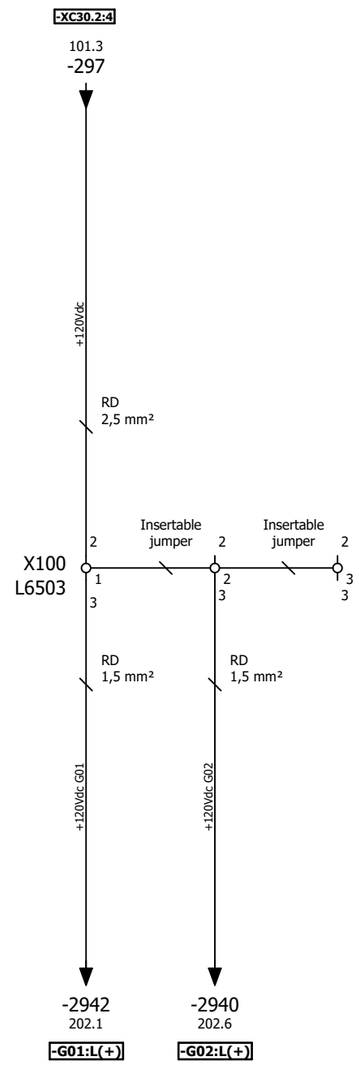
-4130 102.5 | -4131 102.5

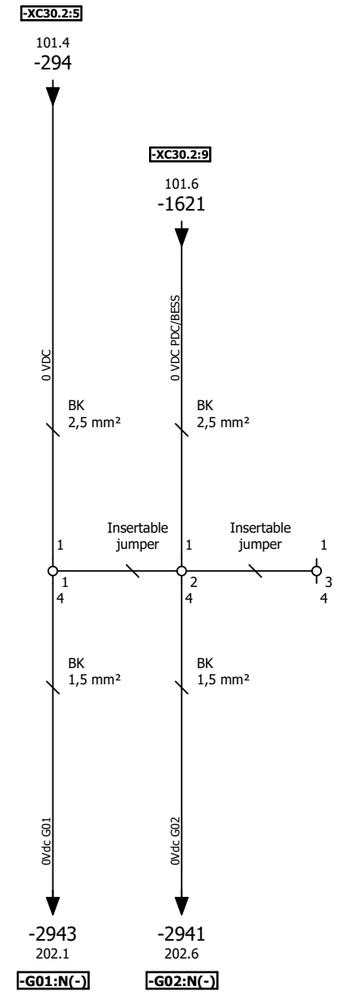
+.PDC/PV-BBEP.79:2 | +.PDC/PV-BBEN.64:1

POWER ELECTRONICS®

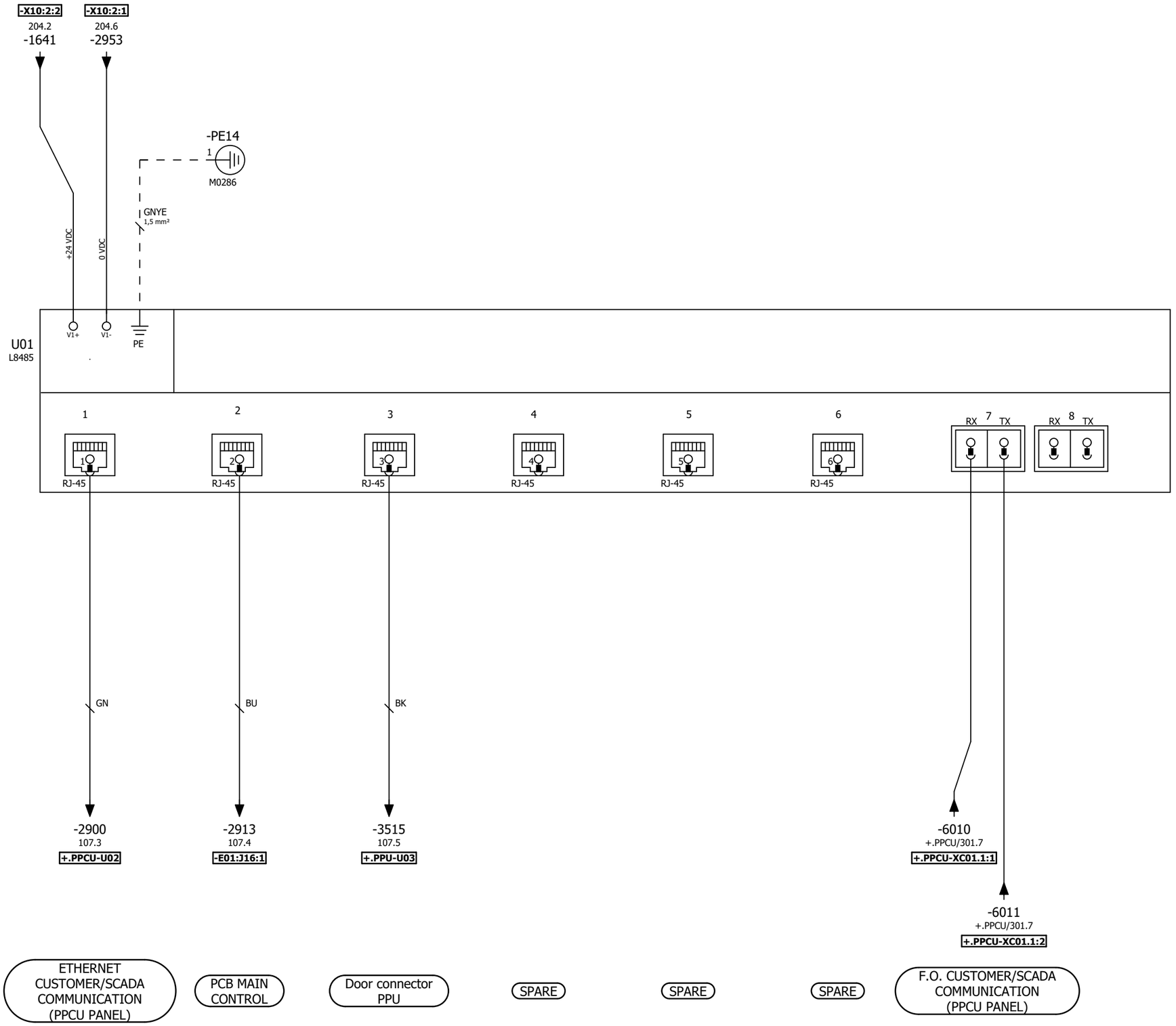
DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	BESS/PV Measure (+/-)	Mounting location: DU CABINET ADU.PCL	Previous page 204 / Next page 300
APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 250 / Total number of pages 181

120Vdc Main Bus Distribution



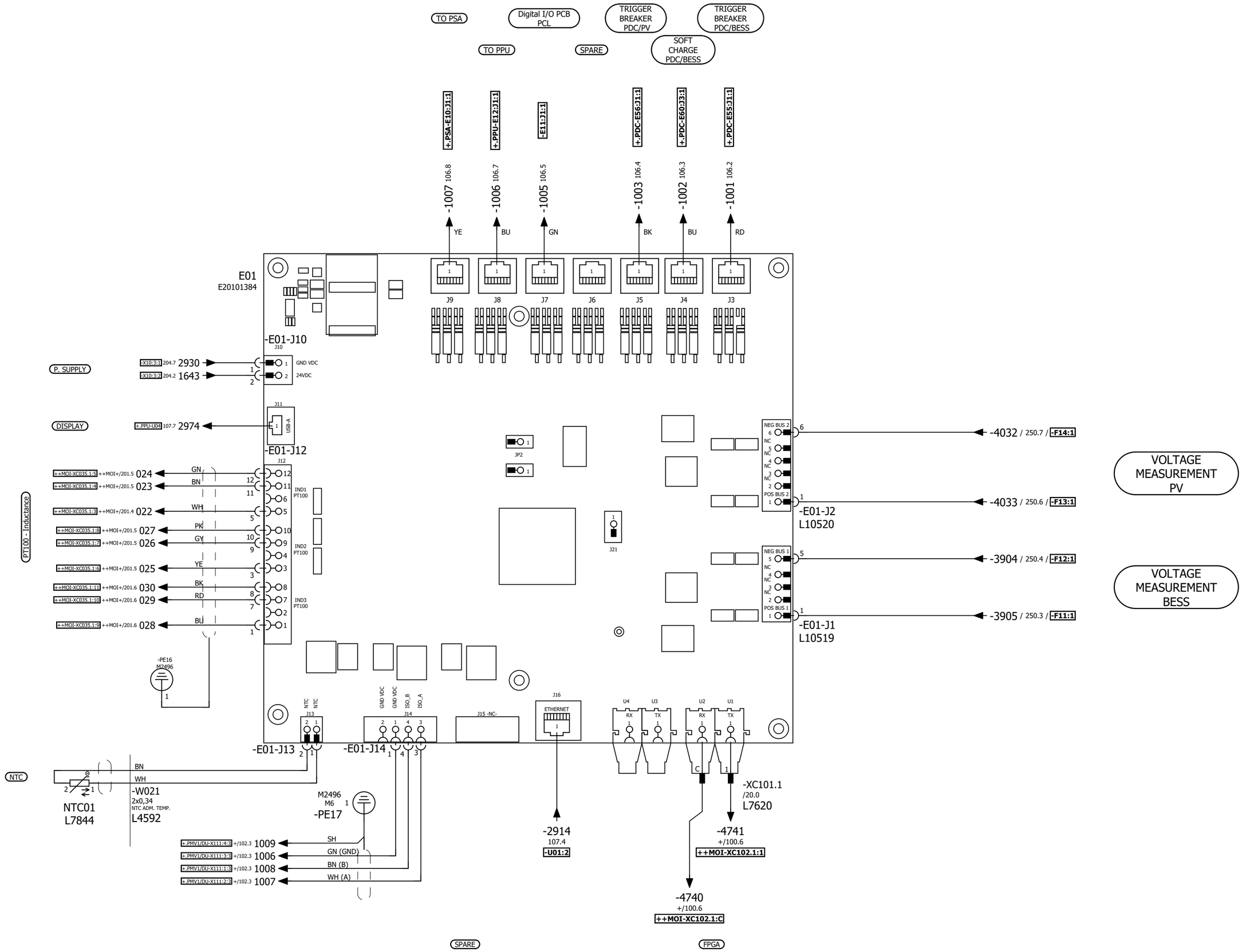


POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	120Vdc Distribution	Mounting location: DU CABINET ADU.PCL	Previous page 250 / Next page 301
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 300 / Total number of pages 181



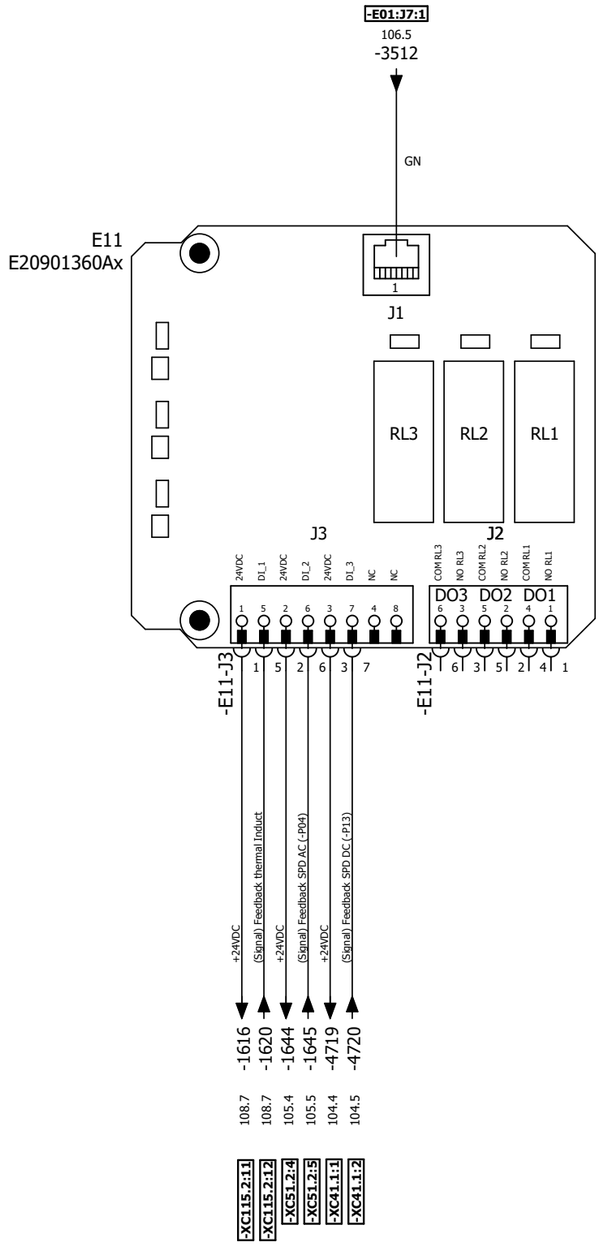
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project ESQ_FD1200U000002_000	Name Ethernet Switch	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PCL	Previous page 300 / Next page 400
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 301 / Total number of pages 181

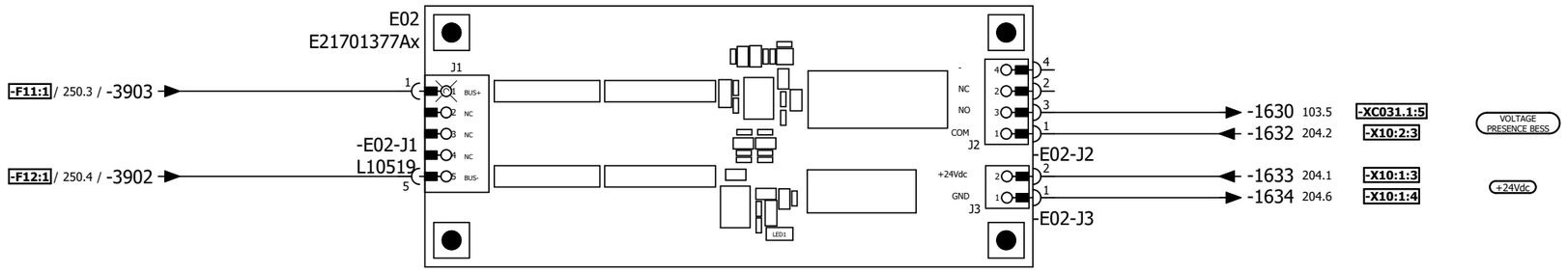
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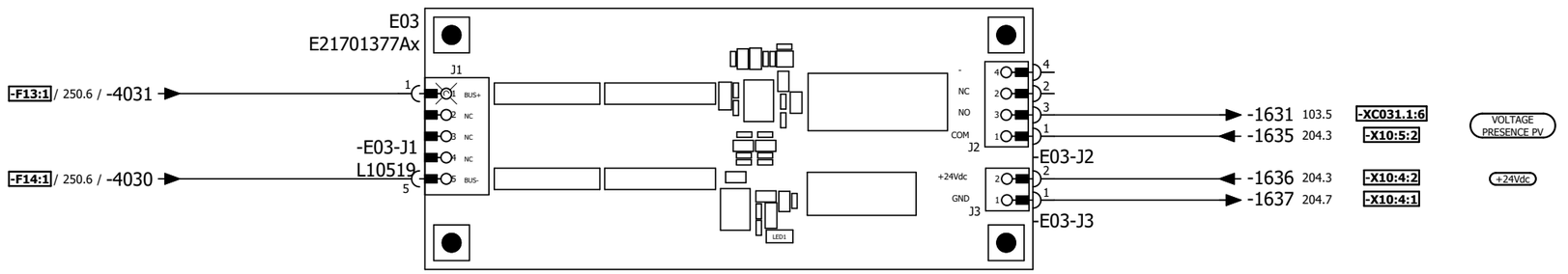


POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	PCB Control Panel	Mounting location: DU CABINET ADU.PCL	Previous page 301 / Next page 401
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 400 / Total number of pages 181

PCB - I/O Control Panel







POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	PCB Inputs/Outputs	Mounting location: DU CABINET ADU.PCL	Previous page 400	Next page &MAT/1000
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 401	Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-C1	L1061	0514500000	DIN RAIL 35x7,5mm	2
-C2	L1061	0514500000	DIN RAIL 35x7,5mm	4
-C3	L1061	0514500000	DIN RAIL 35x7,5mm	2
-E01	E20101384Ax	E20101384AA	TARJETA CONTROL CENTRAL DCDC GEN 3	1
-E01-J1	L10519	2-2178029-5	CONECTOR RAST5 AÉREO 5 VIAS PASO 5	1
-E01-J1	L10521	964202-1	TERMINAL HEMBRA RAST5 AWG13-17 TE	2
-E01-J2	L10520	2322912-6	CONECTOR RAST5 AÉREO 6 VIAS PASO 5	1
-E01-J2	L10521	964202-1	TERMINAL HEMBRA RAST5 AWG13-17 TE	2
-E02	PCB E21701377Ax	PCB E21701377Ax	TARJETA DE SEÑALIZACIÓN DE BUS DCDC GEN3	1
-E02-J1	L10519	2-2178029-5	CONECTOR RAST5 AÉREO 5 VIAS PASO 5	1
-E02-J1	L10521	964202-1	TERMINAL HEMBRA RAST5 AWG13-17 TE	2
-E03	PCB E21701377Ax	PCB E21701377Ax	TARJETA DE SEÑALIZACIÓN DE BUS DCDC GEN3	1
-E03-J1	L10519	2-2178029-5	CONECTOR RAST5 AÉREO 5 VIAS PASO 5	1
-E03-J1	L10521	964202-1	TERMINAL HEMBRA RAST5 AWG13-17 TE	2
-E11	PCB E20901360Ax	PCB E20901360Ax	TARJETA DE ENTRADAS/SALIDAS DIGITALES (DCDC)	2
-E11	L7903	8630740000	End bracket	2
-F11	L4635	492205	FUSE 4A 1500VDC 10KA 10x85 UL DF	2
-F11	L3904-1	485701	FUSE HOLDER 32A 1500V 1P PML 10x85 DF	2
-F11	L7903	8630740000	End bracket	6
-F12	L4635	492205	FUSE 4A 1500VDC 10KA 10x85 UL DF	2
-F12	L3904-1	485701	FUSE HOLDER 32A 1500V 1P PML 10x85 DF	2
-F13	L4635	492205	FUSE 4A 1500VDC 10KA 10x85 UL DF	2
-F13	L3904-1	485701	FUSE HOLDER 32A 1500V 1P PML 10x85 DF	2
-F13	L7903	8630740000	End bracket	1
-F14	L4635	492205	FUSE 4A 1500VDC 10KA 10x85 UL DF	2
-F14	L3904-1	485701	FUSE HOLDER 32A 1500V 1P PML 10x85 DF	2
-G01	L1882-3		PRO ECO 120W 24V 5A	2
-G02	L1882-3		PRO ECO 120W 24V 5A	2
-NTC01	L7844	GA10K3435STM010	NTC 10kOhm 3435K PVC	1
-PE16	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	2
-PE17	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	2
-Q01	L10060	0628032363	BK63H 3P C6A 6kA 480Y/277VAC UL1077	2
-Q01	L7903	8630740000	End bracket	2
-Q02	L10060	0628032363	BK63H 3P C6A 6kA 480Y/277VAC UL1077	2
-Q03	L10060	0628032363	BK63H 3P C6A 6kA 480Y/277VAC UL1077	2
-Q03	L7903	8630740000	End bracket	2
-Q04	L10290	0627049763	BK63HU 1P C4A 10kA 277VAC UL1077	2
-Q05	L10290	0627049763	BK63HU 1P C4A 10kA 277VAC UL1077	2
-Q05	L7903	8630740000	End bracket	2
-U01	L8485	2682170000	SWITCH FO-ETH 2xSC + 6 ETH UNMANAGED	2

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	Bill of materials	Mounting location: DU CABINET ADU.PCL	Previous page &/401	Next page &CAB/2000
APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 1000	Total number of pages 181

Cable overview

Cable designation	Function text	ERP
=DCDC++ZMI+ADU.PCL-W021	NTC ADM. TEMP.	L4592
=DCDC++ZMI+ADU.PCL-W111	Fan P. Supply PMV1	L5146
=DCDC++ZMI+ADU.PCL-W120		L10512
=DCDC++ZMI+ADU.PCL-W121		L8154
=DCDC++ZMI+ADU.PCL-W122		L10513
=DCDC++ZMI+ADU.PCL-W123		L10510
=DCDC++ZMI+ADU.PCL-W124		L8154
=DCDC++ZMI+ADU.PCL-W125		L10514
=DCDC++ZMI+ADU.PCL-W131		L8152
=DCDC++ZMI+ADU.PCL-W132		L10511
=DCDC++ZMI+ADU.PCL-W133		L8164
=DCDC++ZMI+ADU.PCL-W138	USB CABLE DOOR CONNECTOR	L5815
=DCDC++ZMI+ADU.PCL-W0510		L4592

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PCL-E01-J1		L10519	1
		L10521	2
=DCDC++ZMI+ADU.PCL-E01-J2		L10520	1
		L10521	2
=DCDC++ZMI+ADU.PCL-E01-J10			
=DCDC++ZMI+ADU.PCL-E01-J12			
=DCDC++ZMI+ADU.PCL-E01-J13			
=DCDC++ZMI+ADU.PCL-E01-J14			
=DCDC++ZMI+ADU.PCL-E02-J1	(Signal) Feedback SPD DC (-P13)	L10519	1
		L10521	2
=DCDC++ZMI+ADU.PCL-E02-J2	(Signal) Feedback SPD DC (-P13)		
=DCDC++ZMI+ADU.PCL-E02-J3	(Signal) Feedback SPD DC (-P13)		
=DCDC++ZMI+ADU.PCL-E03-J1	(Signal) Feedback SPD DC (-P13)	L10519	1
		L10521	2
=DCDC++ZMI+ADU.PCL-E03-J2	(Signal) Feedback SPD DC (-P13)		
=DCDC++ZMI+ADU.PCL-E03-J3	(Signal) Feedback SPD DC (-P13)		
=DCDC++ZMI+ADU.PCL-E11-J2	(Signal) Feedback SPD DC (-P13)		
=DCDC++ZMI+ADU.PCL-E11-J3	+24VDC		

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PCL-J2	(Signal) Feedback SPD DC (-P13)		
=DCDC++ZMI+ADU.PCL-J12			
=DCDC++ZMI+ADU.PCL-J14			
=DCDC++ZMI+ADU.PCL-XC20.1	CONECTOR AEREO 8P PSA-PCL	L6757	1
		L5709	5
=DCDC++ZMI+ADU.PCL-XC30.1	CONECTOR AEREO 9P PCL-PSA+PDU	L7534	1
		L5709	5
=DCDC++ZMI+ADU.PCL-XC30.2	CONECTOR PANEL 9P PCL	L7535	1
		L5710	8
		L7535	1
=DCDC++ZMI+ADU.PCL-XC031.1	CONECTOR AEREO 8P INTERCONECTION PCL-PPU	L6757	1
		L5709	5
		E1203	3
=DCDC++ZMI+ADU.PCL-XC41.1	CONECTOR AEREO 2P PCL-SPD	E1098	1
		E1203	2
=DCDC++ZMI+ADU.PCL-XC51.2	CONECTOR PANEL 6P PCL - PTA BASE	L5777	1
		L5710	3
		E1204	2
		L5777	1
=DCDC++ZMI+ADU.PCL-XC101.1	CONNECTOR OPTICAL FIBER DUPLEX (CONTROL DCDC PCB)	L7620	1
=DCDC++ZMI+ADU.PCL-XC111.1	CONECTOR AEREO POWER SUPPLY DU FAN	E1096	1
		E1203	3
=DCDC++ZMI+ADU.PCL-XC115.2	M&L SEALED AERIAL POWER SUPPLY DU FAN	E1374	1
		E1203	8
=DCDC++ZMI+ADU.PCL-XC307.1	NTC CONNECTOR - FAN ADMISSION TEMP.	L1047	1
		L1051	2
=DCDC++ZMI+ADU.PCL-XC425.1	CONECTOR AEREO 2P PDC	E1098	1
		L5709	2

POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

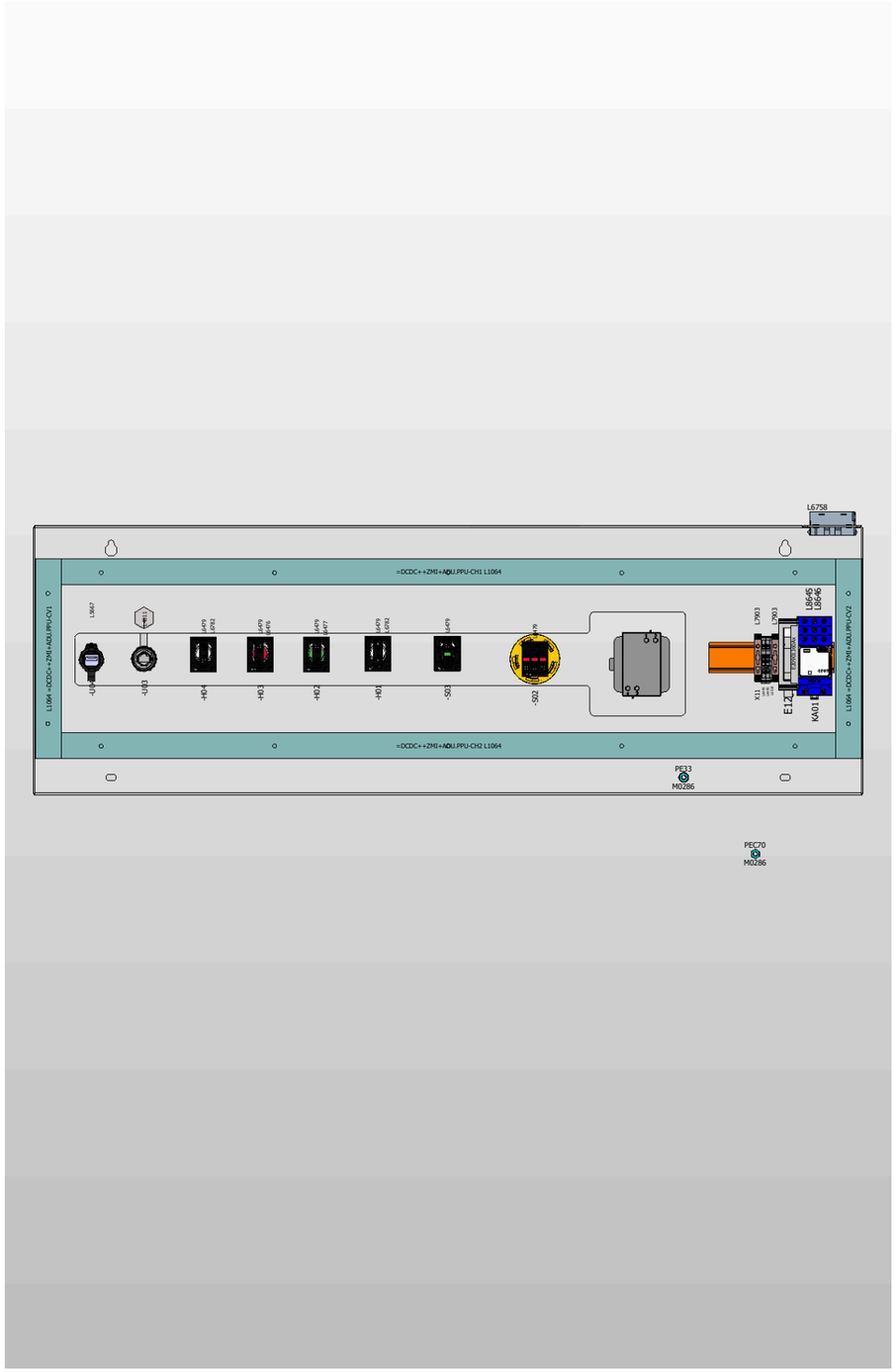
DCDC	DCDC GEN 3 CONVERTER
ZMI	INVERTER ASSEMBLY AREA
ADU.PPU	PUSHBUTTONS PANEL

PPU

PUSHBUTTONS PANEL

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

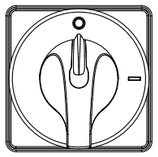
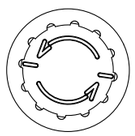
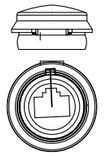
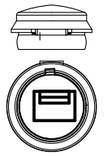
LAYOUT PANEL



Enclosure legend

IME	ERP	Description
E12	PCB E20901360Ax	TARJETA DE ENTRADAS/SALIDAS DIGITALES (DCDC)
H01;H04	L6782	LED MODULE, WHITE
H01-H04;S02;S03	L6479	HOLDER
CA2	L1061	DIN RAIL 35x7,5mm
U03	L4811	INSTALLATION SOCKET RJ45 CAT5E IP65
S03	L6480	KEY-OPERATED SWITCH RONIS, O-I
H02	L6474	INDICATOR LIGHT, GREEN
H03	L6473	INDICATOR LIGHT, RED
H01;H04	L6475	INDICATOR LIGHT, WHITE
S02	L6483	EM. STOP MUSHROOM PUSHBUTTON, 40MM, RED
S02	L6276	PROTECTIVE COLLAR FOR PADLOCKS
H03	L6476	LED MODULE, RED
H02	L6477	LED MODULE, GREEN
KA01	L8646	RELAY BASE "92.03"
S01	L8139	LOTO DC DESCRIPCION
S03	L6481	CONTACT MODULE 1NO
S02	L6275	
CH1;CH2;CV1;CV2	L1064	CABLE DUCT UNEX 42X30 EN U23X
KA01	L8645	24Vdc 3PDT 16A 62.33 SERIES RELAY

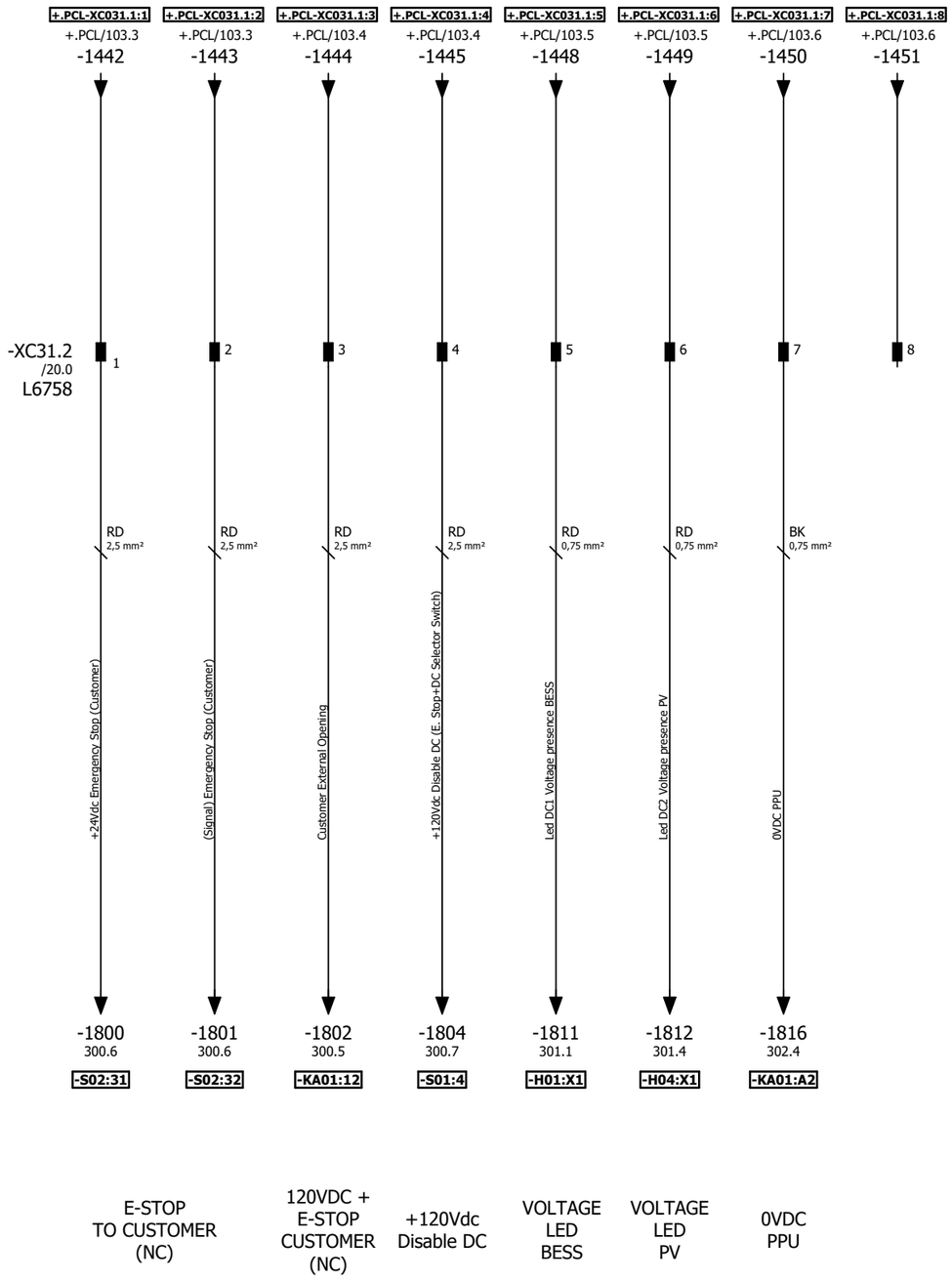
0	1	2	3	4	5	6	7	8	9

S01	S02	S03	H01	H02	H03	H04	U03	U04
								
DC DISCONNECT	EMERGENCY STOP	OFF/ON	DC VOLTAGE PRESENCE BESS	RUN	FAULT	DC VOLTAGE PRESENCE PV	ETHERNET	DISPLAY

0	1	2	3	4	5	6	7	8	9

-X11 = 24VDC Distribution
-XC31.2 = CONECTOR PANEL 8P INTERCONECTION PANEL

Interconnections - PPU/PCL



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Interconnections - PPU/PCL	Mounting location: DU CABINET ADU.PPU	Previous page 20 / Next page 300
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 100 / Total number of pages 181

Start/Stop Selector switch

-E12-J3:1
400.1
-1805

+24Vdc Start/Stop Selector switch

RD
0,75 mm²

13

-S03

14

L6480
L6481
L6479
L6482
Stop/Start

RD
0,75 mm²

(Signal) Start/Stop Selector switch

-1806
400.1
-E12-J3:5

Emergency stop pushbutton

-E12-J3:2
400.1
-1808

+24Vdc Emergency stop pushbutton

-XC31.2:3
100.4
-1802

120VDC + Customer External NC

12 14

-KA01
/302.4

11

RD
2,5 mm²

22 24

-KA01
/302.4

21

RD
2,5 mm²

-XC31.2:1
100.3
-1800

+24Vdc Emergency Stop (Customer)

RD
0,75 mm²

RD
0,75 mm²

RD
2,5 mm²

L6483
L6275
L6479
L6276

-S02

11 21 31
12 22 32

RD
0,75 mm²

(Signal) Emergency stop pushbutton

-1807
400.2
-E12-J3:6

(Signal) Emergency Stop (Customer)

-1801
100.4
-XC31.2:2

-S01
L8139

1 3
2 4

RD
0,75 mm²

RD
2,5 mm²

(Signal) E. Stop+DC Selector Switch

+120Vdc Disable DC (E. Stop+DC Selector Switch)

-1809
400.2
-E12-J3:7

-1804
100.4
-XC31.2:4

S01 in position "0" -> All contacts open.
S01 in position "I" -> All contacts closed.

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC / A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Door Control Maneuver	Mounting location: DU CABINET ADU.PPU	Previous page 100 / Next page 301
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 300 / Total number of pages 181

Led Voltage presence BESS

-XC31.2:5
100.5
-1811

Led DC1 Voltage presence BESS

X1
-H01
24VDC
WH
Control Supply
L6782
L6475
L6479
X2

BK
0,75 mm²

0Vdc Leds

-1819
302.5
-KA01:A2

Run Led

-E12-J2:1
400.3
-1813

Led Run

RD
0,75 mm²

X1
-H02
24VDC
GN
Started
L6477
L6474
L6479
X2

BK
0,75 mm²

Fault led

-E12-J2:3
400.2
-1814

Led Warning/Fault

RD
0,75 mm²

X1
-H03
24VDC
RD
Fault
L6476
L6473
L6479
X2

BK
0,75 mm²

Led Voltage presence PV

-XC31.2:6
100.5
-1812

Led DC2 Voltage presence PV

X1
-H04
24VDC
WH
Control Supply
L6782
L6475
L6479
X2

BK
0,75 mm²

Ethernet Connection

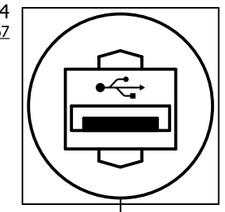
-U03
L4811

ETHERNET CABLE DOOR CONNECTOR

-1820
+.PCL/107.5
+.PCL-U01:3

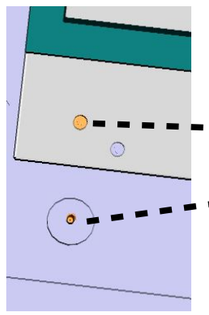
Display Connection

-U04
L5667



USB CABLE DOOR CONNECTOR

-3516
+.PCL/107.7
+.PCL-E01:J11:1



GNYE
2,5 mm²

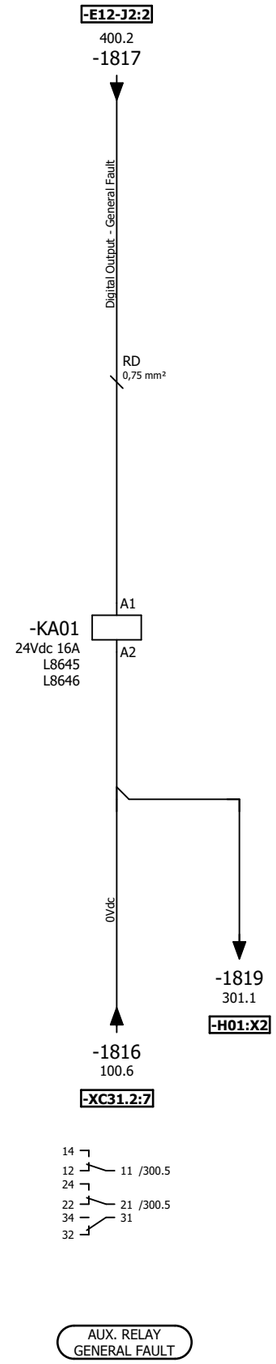
-PE33
M6
M0286

-PEC70
M6
M0286

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: Door Control Maneuver	Installation: DCDC GEN 3 CONVERTER	DCDC / A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PPU	Previous page 300 / Next page 302
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 301 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

Digital Output
General Fault

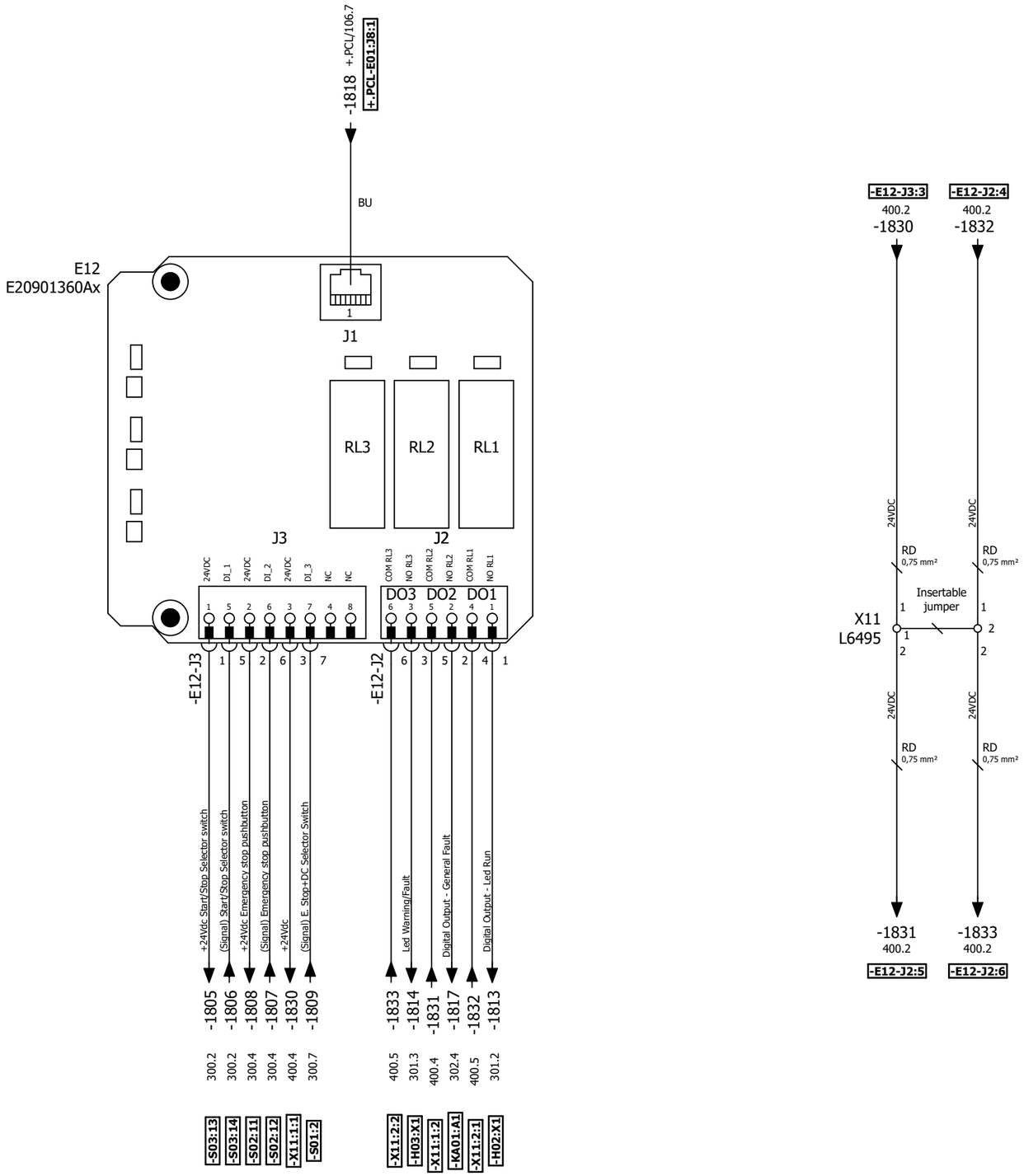


AUX. RELAY
GENERAL FAULT

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	General Fault Maneuver	Mounting location: DU CABINET ADU.PPU	Previous page 301 / Next page 400
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 302 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

PCB - I/O Door Pushbuttons



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: PCB´s - Inputs/Outputs	Installation: DCDC GEN 3 CONVERTER DCDC A3
	REVISED	REVISION 0.0.0			Mounting location: DU CABINET ADU.PPU — Previous page 302 — Next page &MAT/1000
	APPROVED A.CABALLERO				Project Code: FD1200U000002 — Page 400 — Total number of pages 181

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA2	L1061	0514500000	DIN RAIL 35x7,5mm	2
-E12	PCB E20901360Ax	PCB E20901360Ax	TARJETA DE ENTRADAS/SALIDAS DIGITALES (DCDC)	1
-E12	L7903	8630740000	End bracket	2
-H01	L6782	3SU1401-1BB60-1AA0	LED MODULE, WHITE	2
-H01	L6475	3SU1001-6AA60-0AA0	INDICATOR LIGHT, WHITE	2
-H01	L6479	3SU1500-0AA10-0AA0	HOLDER	2
-H02	L6477	3SU1401-1BB40-1AA0	LED MODULE, GREEN	2
-H02	L6474	3SU1001-6AA40-0AA0	INDICATOR LIGHT, GREEN	2
-H02	L6479	3SU1500-0AA10-0AA0	HOLDER	2
-H03	L6476	3SU1401-1BB20-1AA0	LED MODULE, RED	2
-H03	L6473	3SU1001-6AA20-0AA0	INDICATOR LIGHT, RED	2
-H03	L6479	3SU1500-0AA10-0AA0	HOLDER	2
-H04	L6782	3SU1401-1BB60-1AA0	LED MODULE, WHITE	2
-H04	L6475	3SU1001-6AA60-0AA0	INDICATOR LIGHT, WHITE	2
-H04	L6479	3SU1500-0AA10-0AA0	HOLDER	2
-KA01	L8645	62.33.9.024.0074	24Vdc 3PDT 16A 62.33 SERIES RELAY	2
-KA01	L8646	92.03	RELAY BASE "92.03"	2
-S01	L8139	ZFC16E2H1A2	LOTO DC DESCRIPCION	2
-S02	L6483	3SU1000-1HB20-0AA0	EM. STOP MUSHROOM PUSHBUTTON, 40MM, RED	2
-S02	L6275	3SU1400-1AA10-1CA0		6
-S02	L6479	3SU1500-0AA10-0AA0	HOLDER	2
-S02	L6276	3SU1950-0DX30-0AA0	PROTECTIVE COLLAR FOR PADLOCKS	2
-S03	L6480	3SU1000-4CF11-0AA0	KEY-OPERATED SWITCH RONIS, O-I	2
-S03	L6481	3SU1400-1AA10-1BA0	CONTACT MODULE 1NO	2
-S03	L6479	3SU1500-0AA10-0AA0	HOLDER	2
-S03	L6482	3SU1900-0EB10-0AA0	DUST COVER	1
-U03	L4811	4000-73000-0010000	INSTALLATION SOCKET RJ45 CAT5E IP65	2
-U04	L5667	1018840000	CABLE GLAND USB A - USB A IP65	2

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++ZMI+ADU.PPU-E12-J2	Digital Output - Led Run		
=DCDC++ZMI+ADU.PPU-E12-J3	+24Vdc Start/Stop Selector switch		
=DCDC++ZMI+ADU.PPU-J2			
=DCDC++ZMI+ADU.PPU-J3			
=DCDC++ZMI+ADU.PPU-XC31.2	CONECTOR PANEL 8P INTERCONECTION PANEL	L6758	1
		L5710	4
		E1204	3

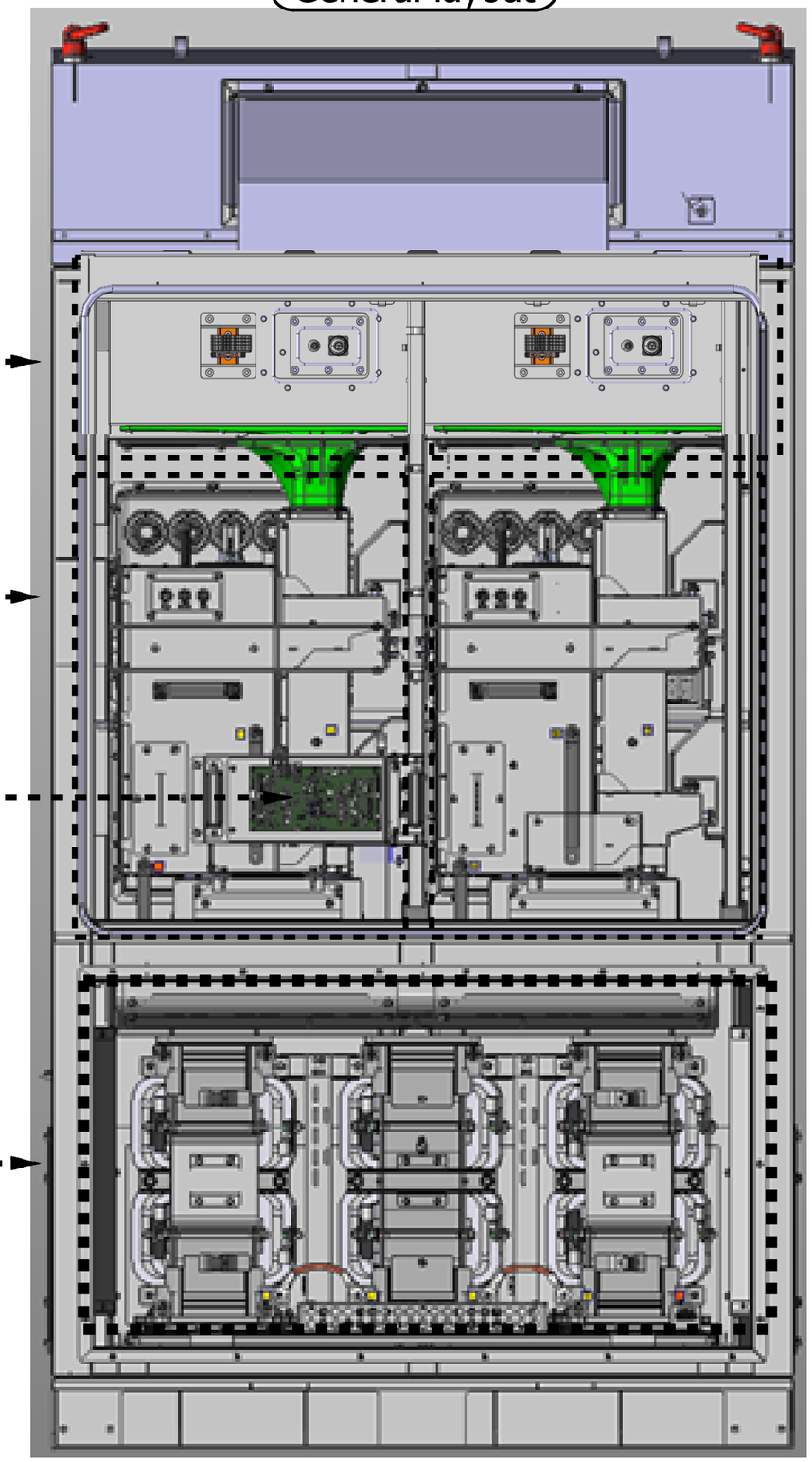
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

DCDC GEN 3 CONVERTER

MOI INVERTER MODULE AREA

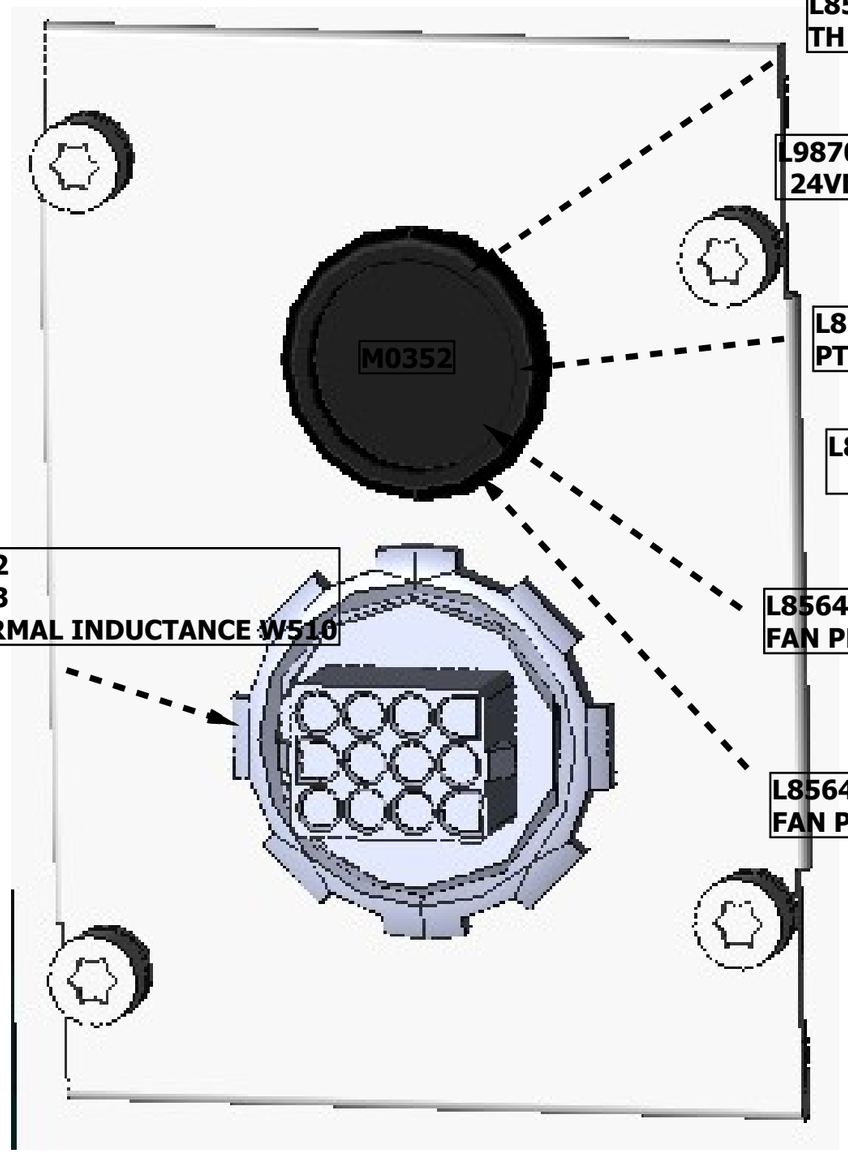
Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

General layout



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Layout	Mounting location:	Previous page 1 / Next page 11
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 10 / Total number of pages 181

Interconnection Areas
ZMI - MOI



L8565 (Manguito 5-6)
TH IND (W051)

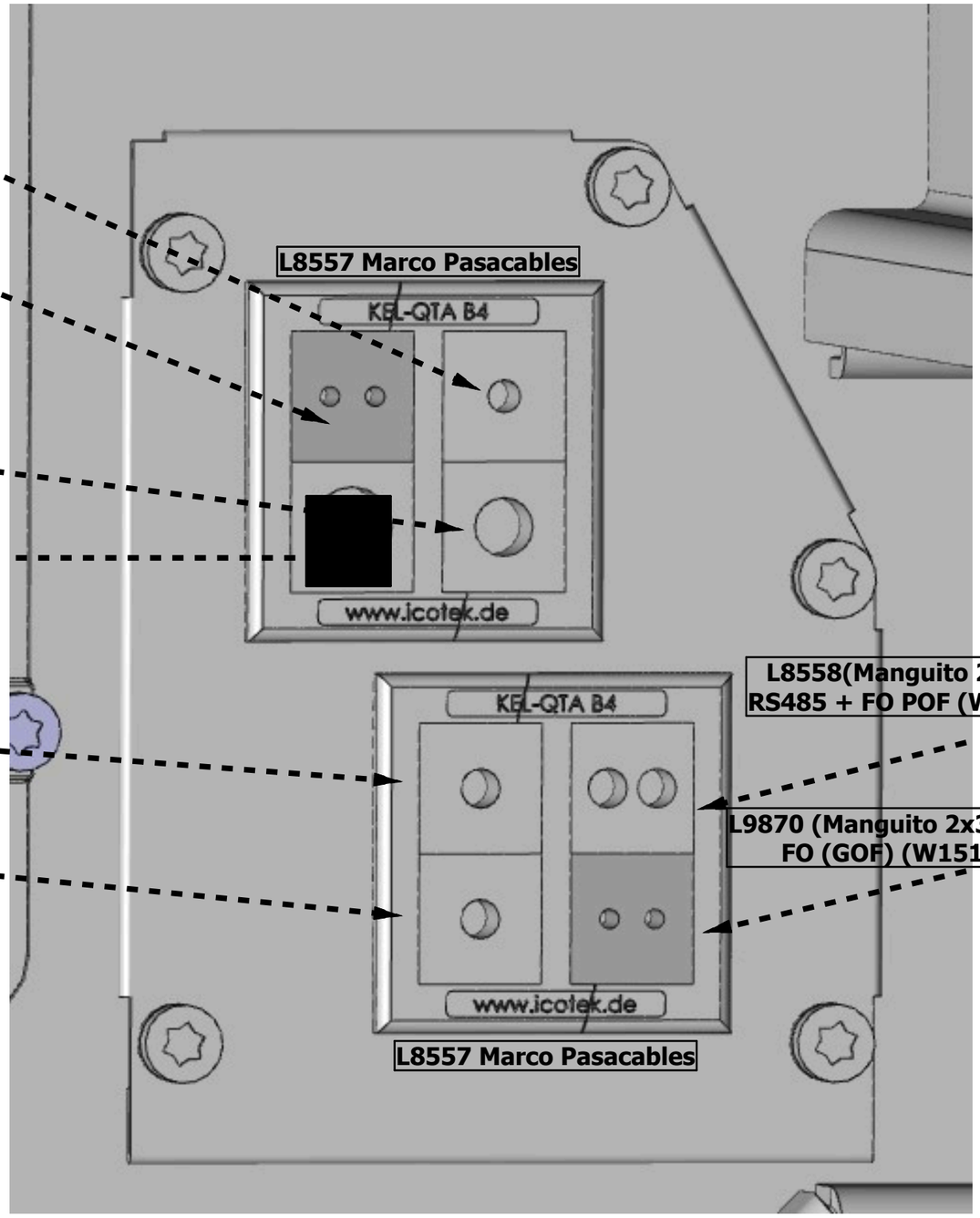
L9870 (Manguito 2x3mm)
24VDC 2kV PCB MODULO

L8625 (Manguito 9-10)
PT100 INDC.1-2-3 (W050)

L8562 (Manguito Ciego)
SPARE

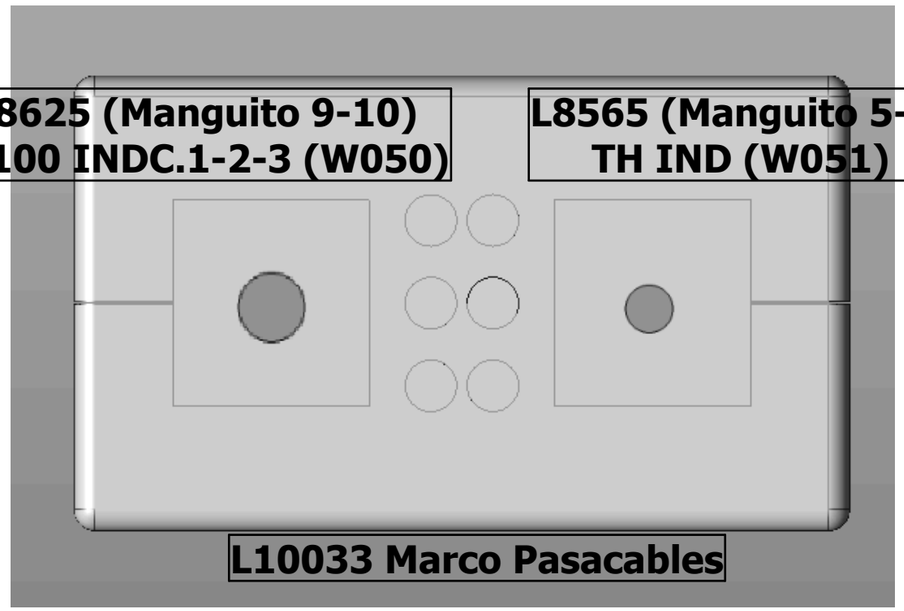
L8564 (Manguito 6-7)
FAN PMV2 W112

L8564 (Manguito 6-7)
FAN PMV3 W113



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	Interconnection Areas ZMI - MOI	Mounting location: Previous page 10 Next page 12
	APPROVED A.CABALLERO				Project Code: FD1200U000002 Page 11 Total number of pages 181

Interconnection Areas
ZMI - Inductance

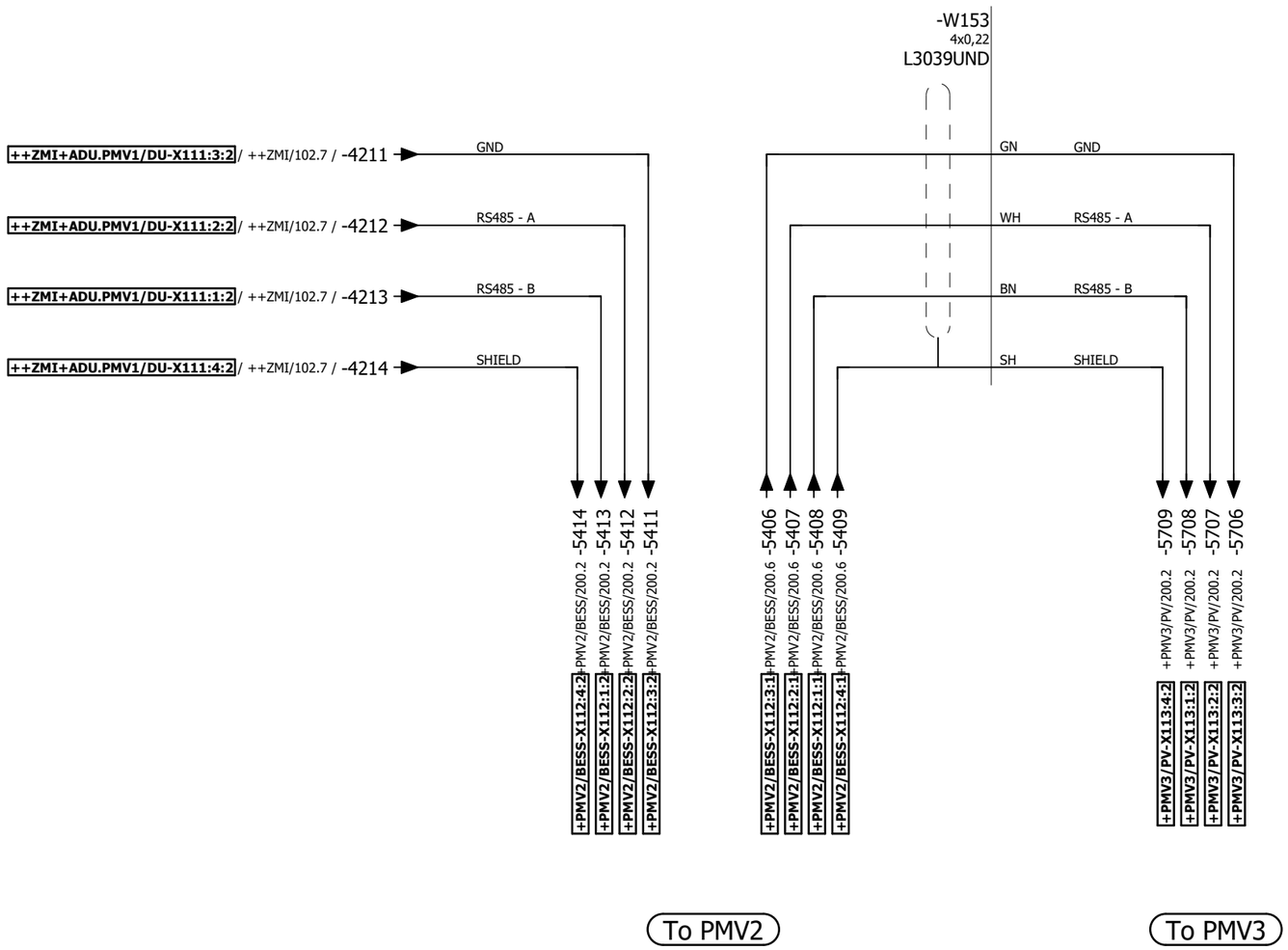


0	1	2	3	4	5	6	7	8	9

-XC035.1 = MATE-N-LOK 12P AEREAL INDUCTANCE

-XC051.1 = CONNECTOR FLAT 50 VIAS
-XC052.1 = CONNECTOR FLAT 40 VIAS
-XC053.1 = CONNECTOR FLAT 50 VIAS
-XC054.1 = CONNECTOR FLAT 40 VIAS

-XC102.1 = CONNECTOR OPTICAL FIBER DUPLEX (CONTROL ETR PCB)
-XC115.1 = M&L SEALED BASE POWER SUPPLY DU FAN

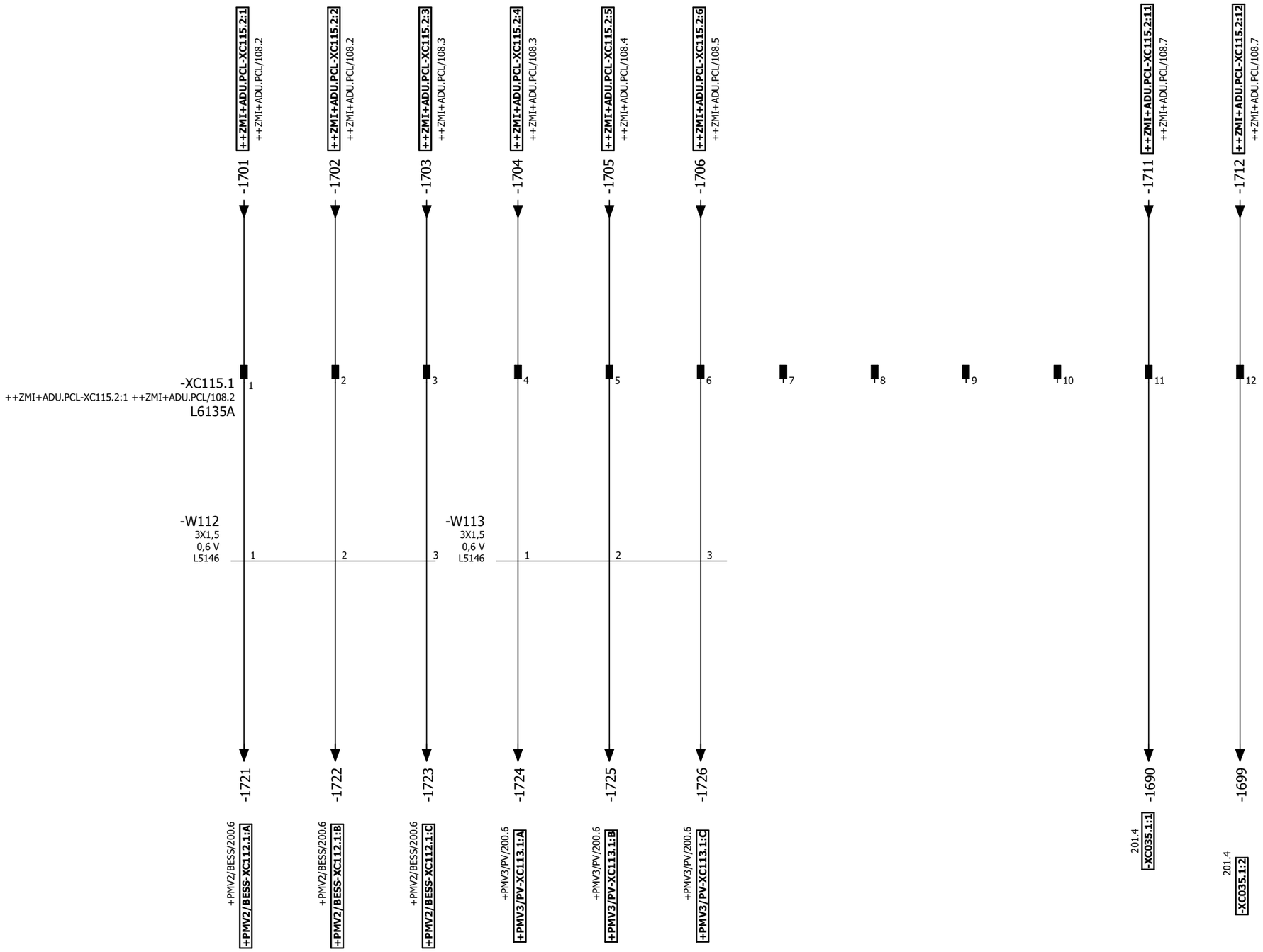


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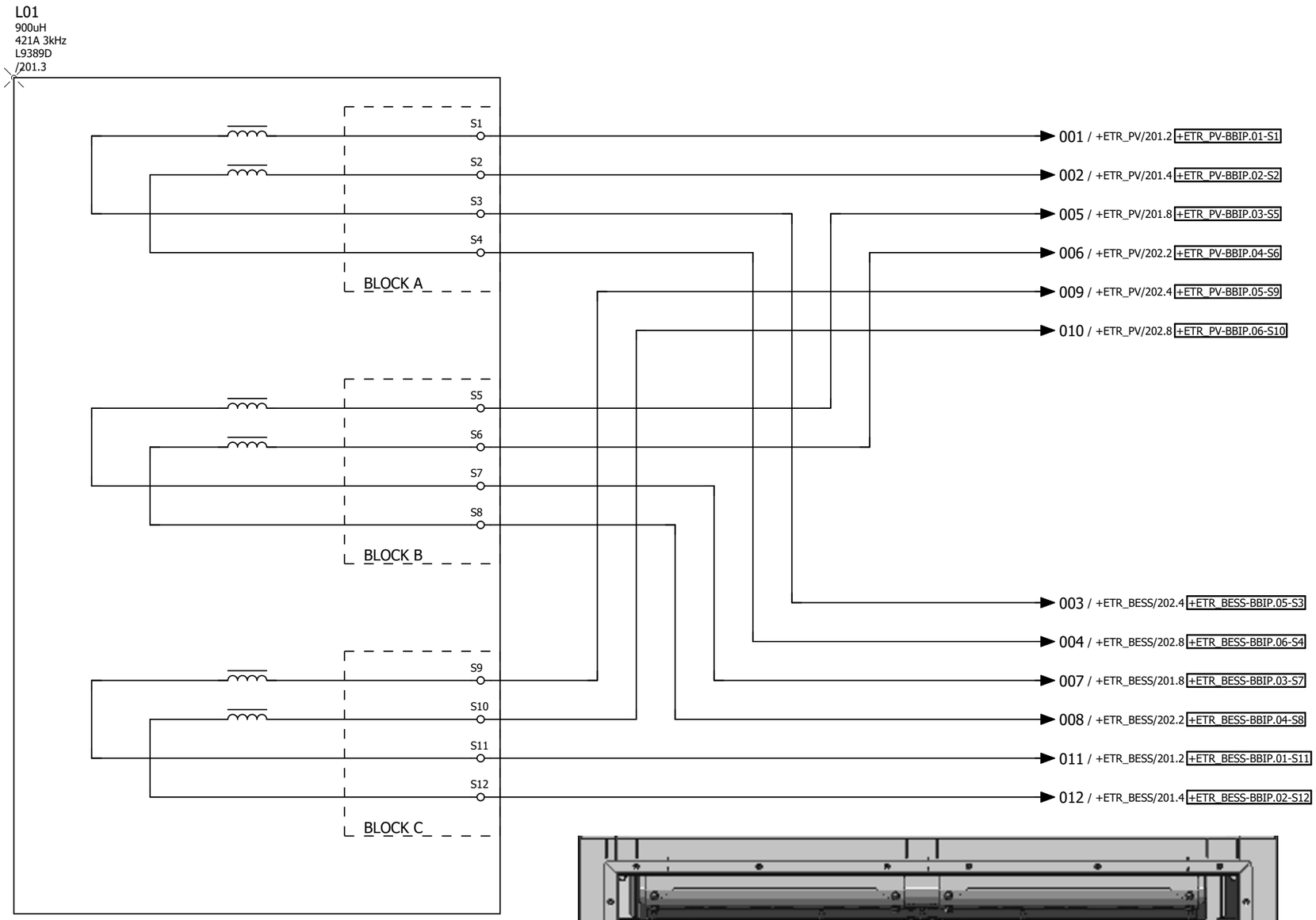
POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
REVISED		ESQ_FD1200U000002_000	Interconnections - Comms RS485 Fans	Mounting location:	Previous page 20 / Next page 101
APPROVED A.CABALLERO	REVISION 0.0.0			Project Code: FD1200U000002	Page 100 / Total number of pages 181

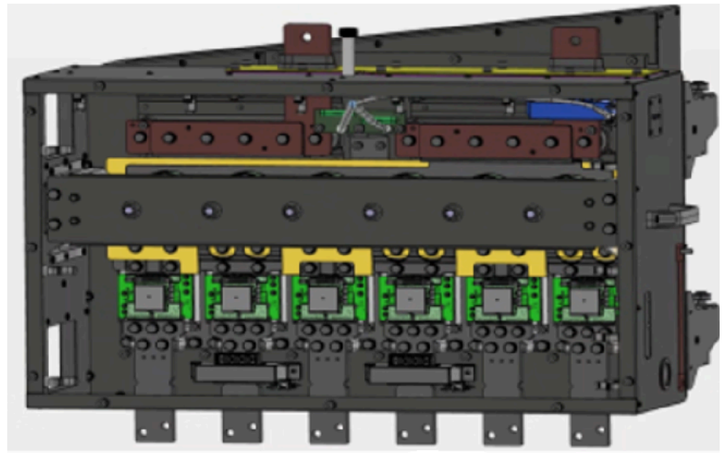
Interconnections - PCL / MOI (FANS)



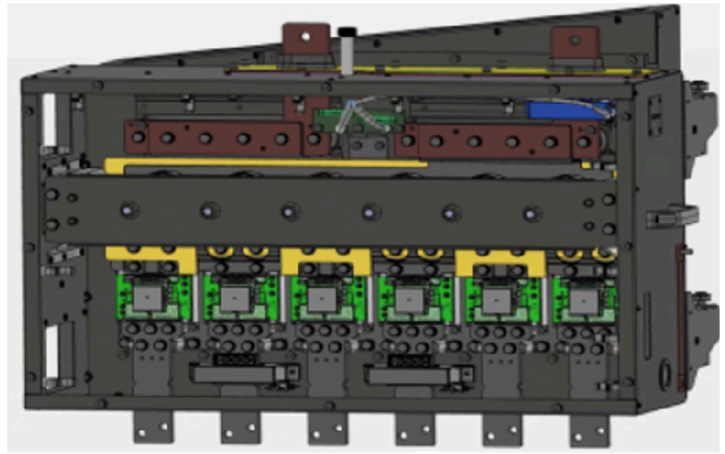
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Interconnections - PCL / MOI (FANS)	Mounting location:	Previous page 100 / Next page 200
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 101 / Total number of pages 181

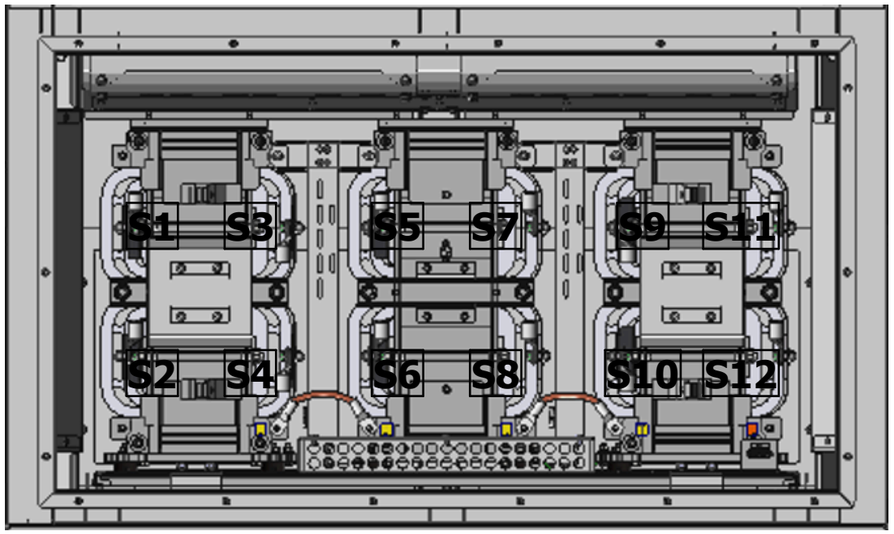


ETR_PV

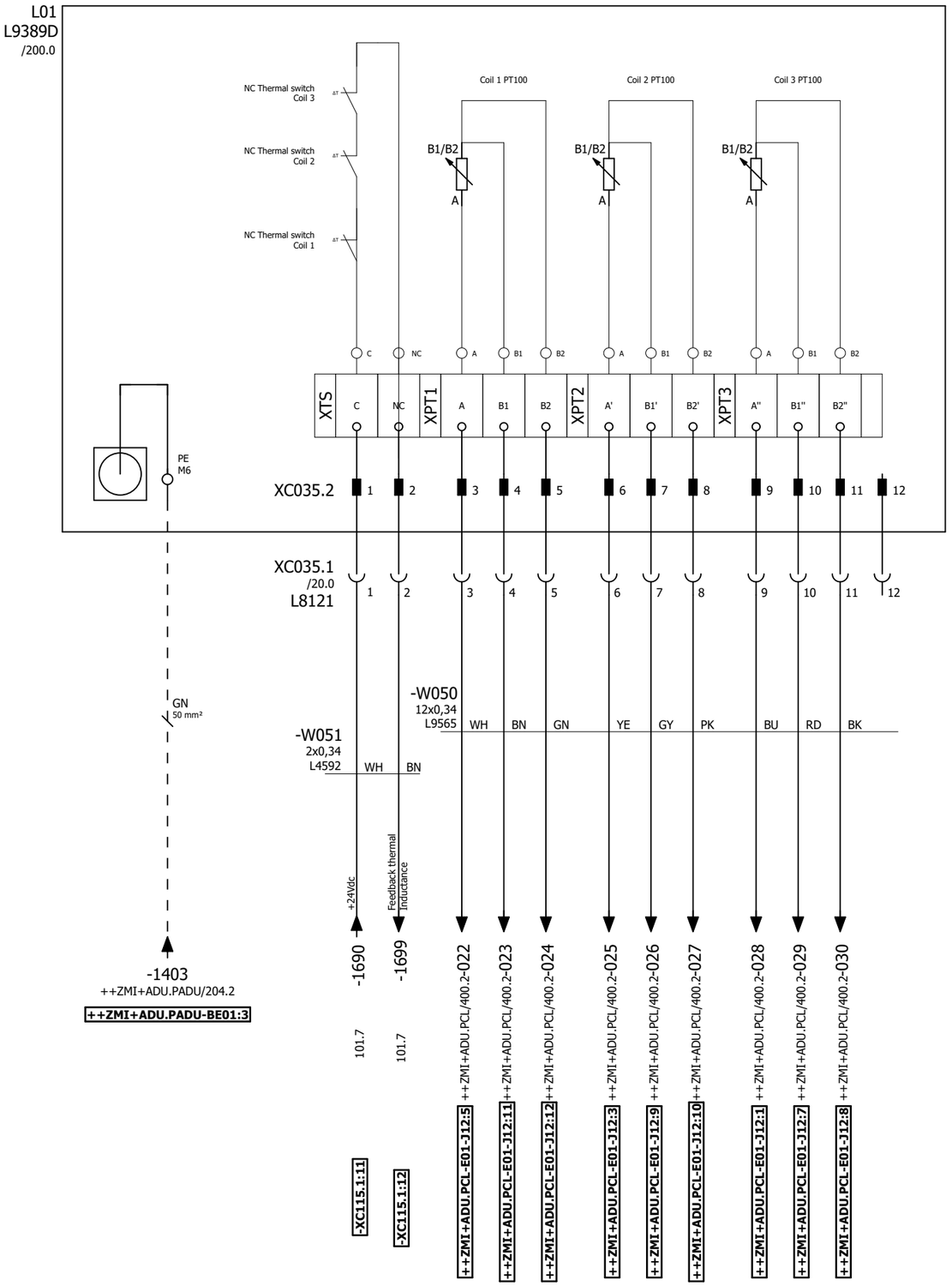


ETR_BESS





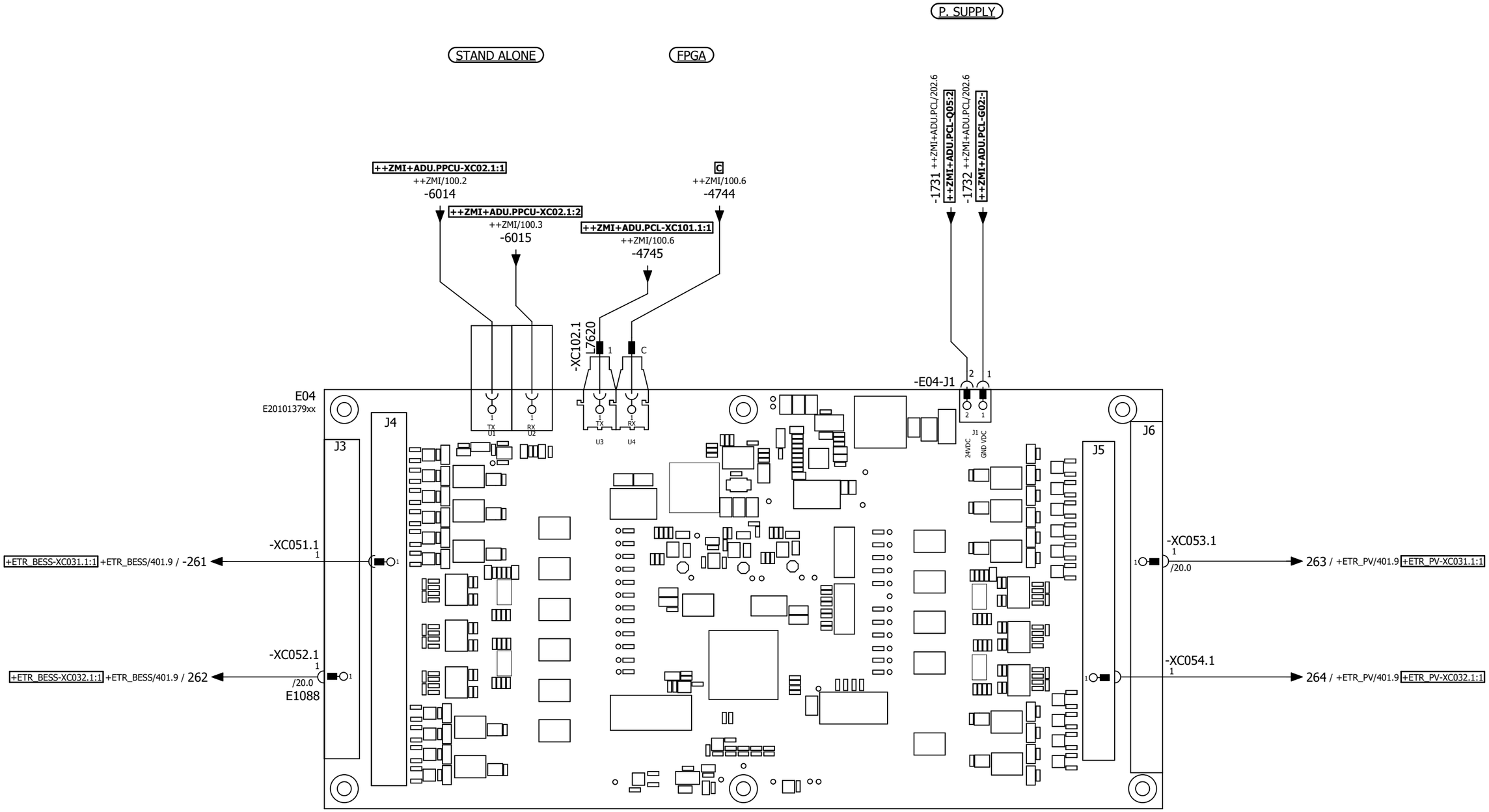
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION	ESQ_FD1200U000002_000	AC Inductance (Power busbar)	Mounting location:	Previous page 101 / Next page 201
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 200 / Total number of pages 181



0	1	2	3	4	5	6	7	8	9

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	AC Inductance (Terminals block)	Mounting location:	Previous page 200	Next page 400
APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 201	Total number of pages 181



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	PCB control and measures	Mounting location:	Previous page 201 / Next page &MAT/1000
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 400 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-E04				1
-L01	L9389D	L9389D	INDUCTANCIA DC x3 900uH 421A 3kHz	1

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER		DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Bill of materials	Mounting location:	Previous page &/400	Next page &CAB/2000	
APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 1000	Total number of pages 181	

Cable overview

Cable designation	Function text	ERP
=DCDC++MOI+-W050	Feedback thermal Inductance	L9565
=DCDC++MOI+-W051	+24Vdc	L4592
=DCDC++MOI+-W112		L5146
=DCDC++MOI+-W113		L5146
=DCDC++MOI+-W153	SHIELD	L3039UND

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++MOI+-E04-J1			
=DCDC++MOI+-E04-XC051.1			
=DCDC++MOI+-E04-XC054.1			
=DCDC++MOI+-XC035.1	MATE-N-LOK 12P AEREAL INDUCTANCE	L8121	1
		L8914	2
		E1062	9
=DCDC++MOI+-XC051.1	CONNECTOR FLAT 50 VIAS	L8031	1
		L8029	1
=DCDC++MOI+-XC052.1	CONNECTOR FLAT 40 VIAS	E1088	1
		E1089	1
=DCDC++MOI+-XC053.1	CONNECTOR FLAT 50 VIAS	L8031	1
		L8029	1
=DCDC++MOI+-XC054.1	CONNECTOR FLAT 40 VIAS	E1088	1
		E1089	1
=DCDC++MOI+-XC102.1	CONNECTOR OPTICAL FIBER DUPLEX (CONTROL ETR PCB)	L7620	1
=DCDC++MOI+-XC115.1	M&L SEALED BASE POWER SUPPLY DU FAN	L6135A	1
		E1204	8

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CV35 SALIDA 30

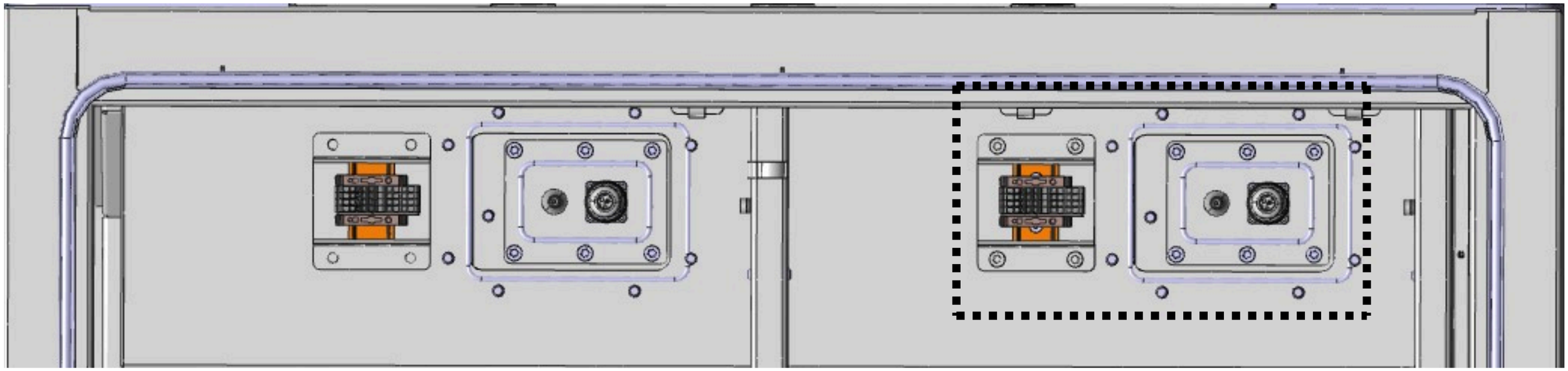
DCDC	DCDC GEN 3 CONVERTER
MOI	INVERTER MODULE AREA
PMV2/BESS	FAN PRE-ASSEMBLY 2 BESS

PMV2/BESS

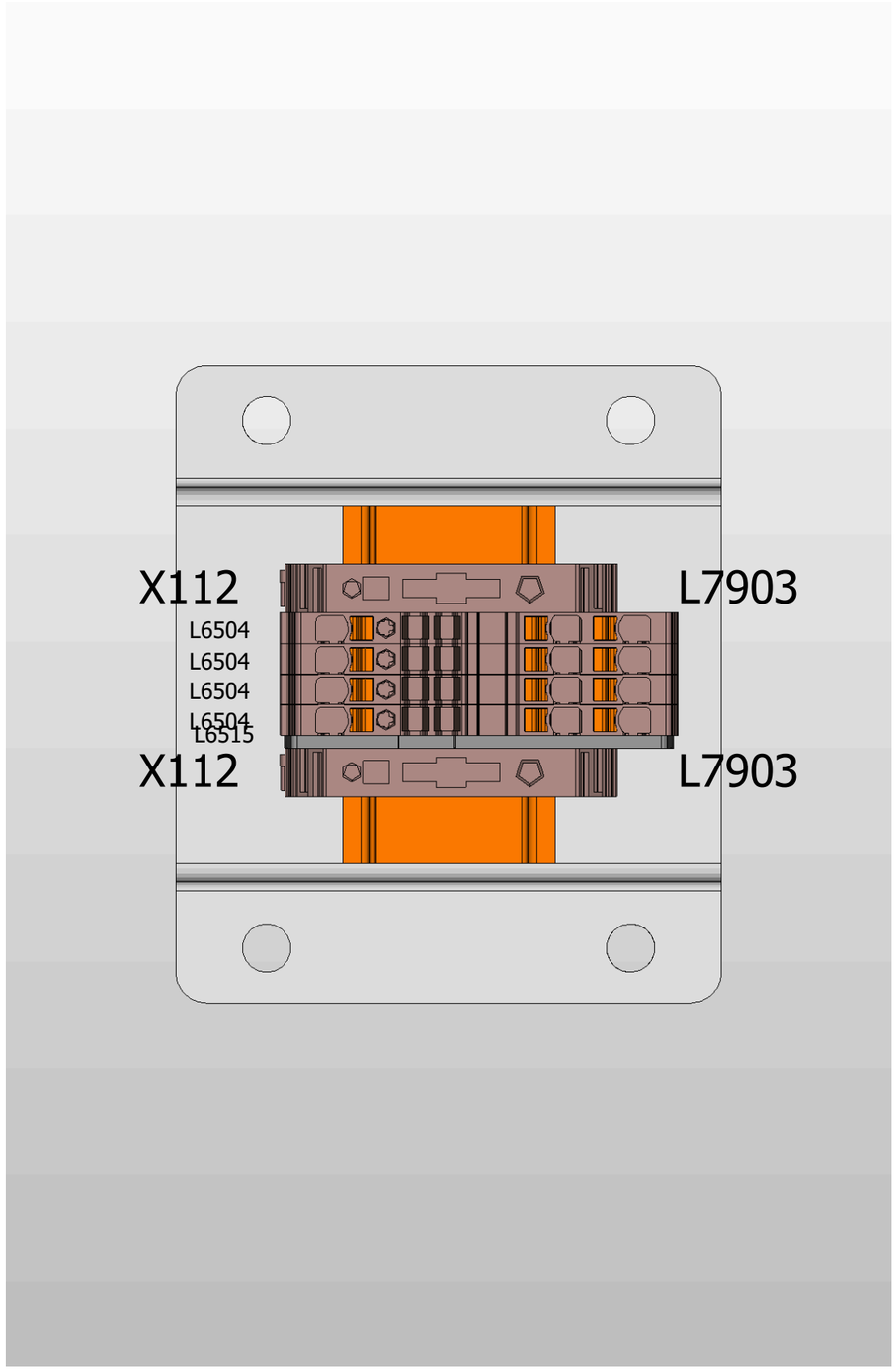
FAN PRE-ASSEMBLY 2 BESS

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

PRE-ASSEMBLY LAYOUT PMV2 BESS



LAYOUT PANEL



Enclosure legend

IME	ERP	Description
	L1061	DIN RAIL 35x7,5mm

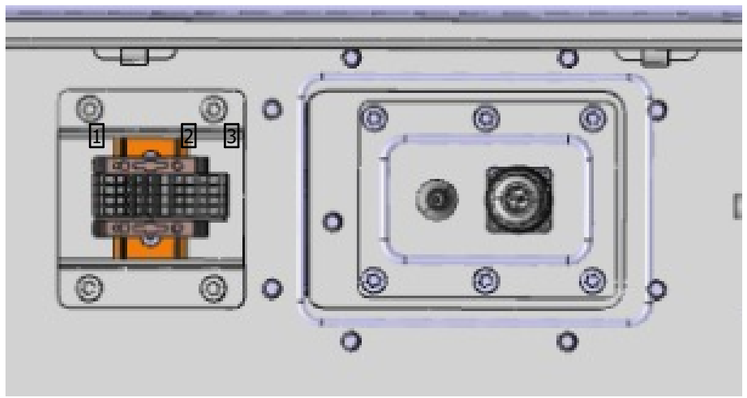
0	1	2	3	4	5	6	7	8	9

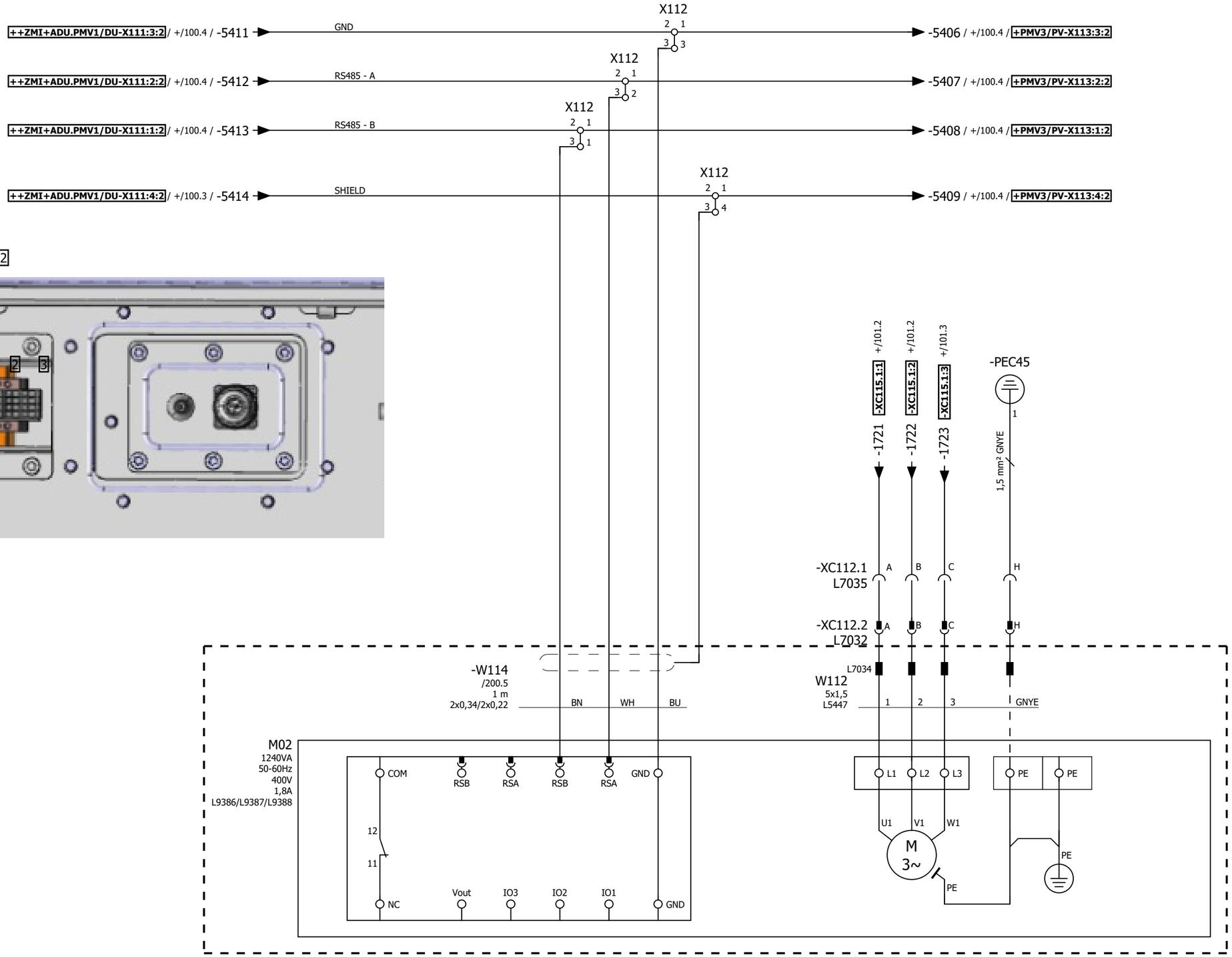
-X112 = RS485 TERMINAL PMV2
-XC112.1 = AEREAL CONNECTOR POWER SUPPLY FAN 2 (ETR/BESS)
-XC112.2 = SEALED BASE CONNECTOR POWER SUPPLY FAN 2 (ETR/BESS)

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Designation terminals and connectors	Mounting location: FAN PRE-ASSEMBLY 2 BESS PMV2/BESS	Previous page 11	Next page 200
APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 20	Total number of pages 181

0	1	2	3	4	5	6	7	8	9





POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Power module BESS - Fan 2	Mounting location: FAN PRE-ASSEMBLY 2 BESS PMV2/BESS	Previous page 20 / Next page &CAB/2000
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 200 / Total number of pages 181

Cable overview

Cable designation	Function text	ERP
=DCDC++MOI+PMV2/BESS-W112	SHIELD	L5447
=DCDC++MOI+PMV2/BESS-W114	SHIELD	L8023

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++MOI+PMV2/BESS-X112	RS485 TERMINAL PMV2		
=DCDC++MOI+PMV2/BESS-XC112.1	AEREAL CONNECTOR POWER SUPPLY FAN 2 (ETR/BESS)	L7035	1
		L7037	4
=DCDC++MOI+PMV2/BESS-XC112.2	SEALED BASE CONNECTOR POWER SUPPLY FAN 2 (ETR/BESS)	L7032	1

POWER ELECTRONICS
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CV35 SALIDA 30

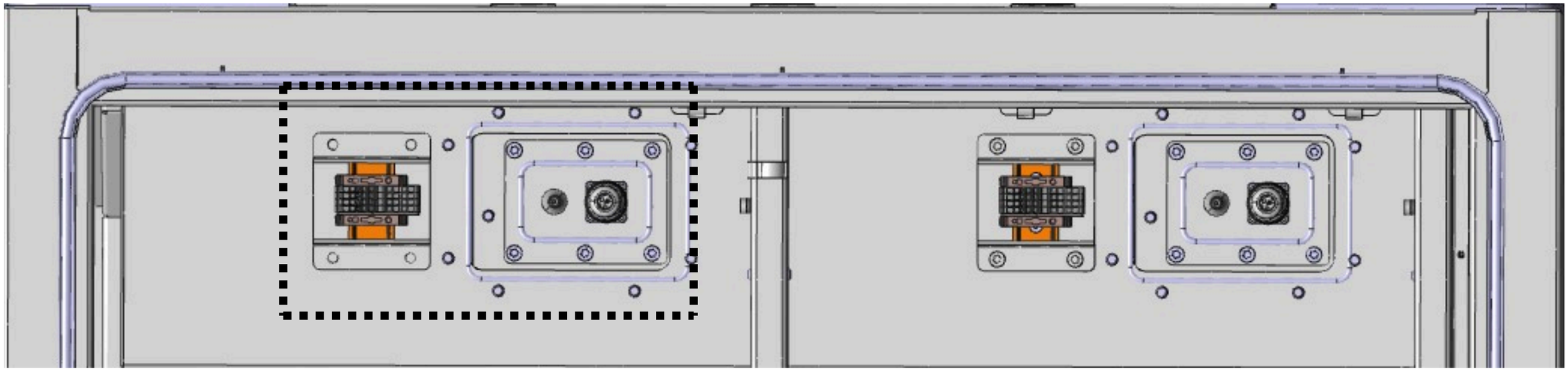
DCDC	DCDC GEN 3 CONVERTER
MOI	INVERTER MODULE AREA
PMV3/PV	FAN PRE-ASSEMBLY 3 PV

PMV3/PV

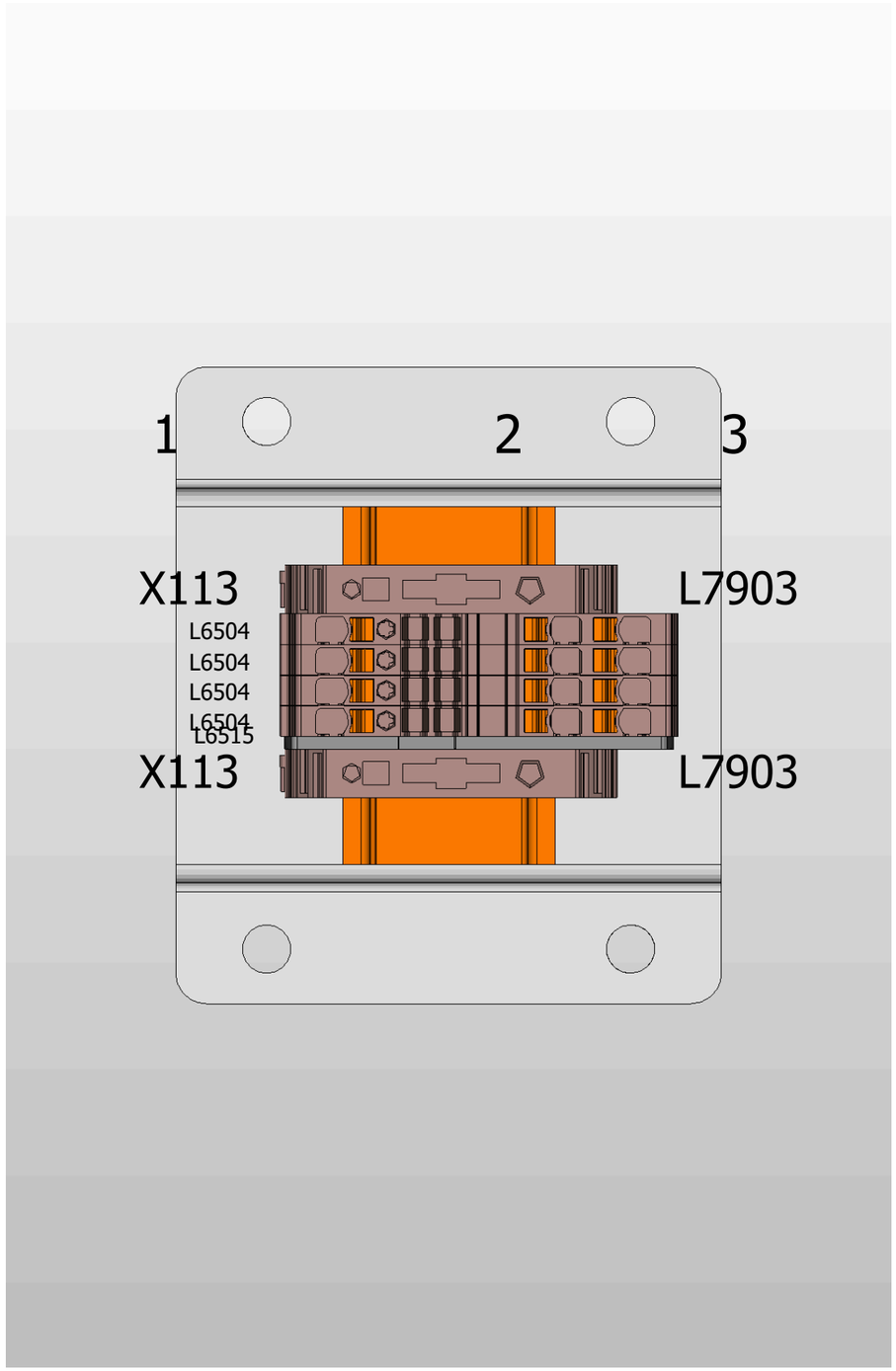
FAN PRE-ASSEMBLY 3 PV

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

PRE-ASSEMBLY LAYOUT PMV3 PV



LAYOUT PANEL



Enclosure legend

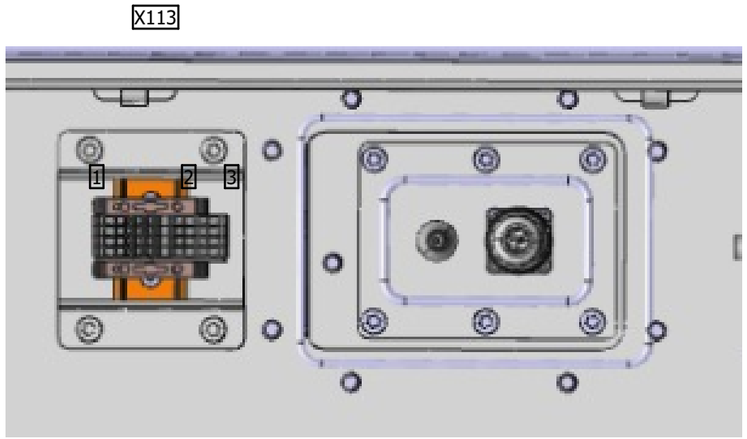
IME	ERP	Description
	L1061	DIN RAIL 35x7,5mm

0	1	2	3	4	5	6	7	8	9

-X113 = RS485 TERMINAL PMV3
-XC113.1 = AEREAL CONNECTOR POWER SUPPLY FAN 2 (ETR/PV)
-XC113.2 = SEALED BASE POWER SUPPLY FAN 3 (ETR/PV)

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Designation terminals and connectors	Mounting location:	FAN PRE-ASSEMBLY 3 PV PMV3/PV	Previous page 11	Next page 200
APPROVED A.CABALLERO	0.0.0			Project Code:	FD1200U000002	Page 20	Total number of pages 181



+PMV2/BESS-X112:3:1 / +/100.6 / -5706 — GND — X113 2 1 3 3
+PMV2/BESS-X112:2:1 / +/100.6 / -5707 — RS485 - A — X113 2 1 3 2 — -R1 x1 x2 120Ω 0.5W L9025
+PMV2/BESS-X112:1:1 / +/100.6 / -5708 — RS485 - B — X113 2 1 3 1
+PMV2/BESS-X112:4:1 / +/100.6 / -5709 — SHIELD — X113 2 1 3 4

+/101.3 -XC115.1:4 -1724
+/101.4 -XC115.1:5 -1725
+/101.4 -XC115.1:6 -1726

-PEC46 1 — 1,5 mm² GNYE

-XC113.1 L7035 A B C H
-XC113.2 L7032 A B C H
L7034

-W115 /200.5 1 m 2x0,34/2x0,22 — BN WH BU
W113 5x1,5 L5447 — 1 2 3 GNYE

M03
1240VA
50-60Hz
400V
1,8A
L9386/L9387/L9388

COM RSB RSA RSB RSA GND
12
11
NC Vout IO3 IO2 IO1 GND

L1 L2 L3 PE PE
U1 V1 W1
M 3~
PE
PE

POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: Power module PV - Fan 3	Installation: DCDC GEN 3 CONVERTER	DCDC / A3
	REVISED	REVISION 0.0.0			Mounting location: FAN PRE-ASSEMBLY 3 PV PMV3/PV	Previous page 20 / Next page &MAT/1000
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 200 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-R1	L9025	707-8154	Resistencia 120ohm 0.5W, Cierre bus RS485	1

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Bill of materials	Mounting location:	FAN PRE-ASSEMBLY 3 PV PMV3/PV	Previous page &/200	Next page &CAB/2000
APPROVED A.CABALLERO	0.0.0			Project Code:	FD1200U000002	Page 1000	Total number of pages 181

Cable overview

Cable designation	Function text	ERP
=DCDC++MOI+PMV3/PV-W113	SHIELD	L5447
=DCDC++MOI+PMV3/PV-W115	SHIELD	L8023

0	1	2	3	4	5	6	7	8	9

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++MOI+PMV3/PV-X113	RS485 TERMINAL PMV3		
=DCDC++MOI+PMV3/PV-XC113.1	AEREAL CONNECTOR POWER SUPPLY FAN 2 (ETR/PV)	L7035	1
		L7037	4
=DCDC++MOI+PMV3/PV-XC113.2	SEALED BASE POWER SUPPLY FAN 3 (ETR/PV)	L7032	1

POWER ELECTRONICS®

DESIGNED L. PUERTES	DATE	Project	Name	Installation:	DCDC GEN 3 CONVERTER	DCDC	A3
REVISED	REVISION	ESQ_FD1200U000002_000	Plug overview : =DCDC++MOI+PMV3/PV-X113 - =DCDC++MOI+PMV3/PV-XC113.2	Mounting location:	FAN PRE-ASSEMBLY 3 PV PMV3/PV	Previous page &CAB/2000	Next page +ETR_BESS&/1
APPROVED A.CABALLERO	0.0.0			Project Code:	FD1200U000002	Page 3000	Total number of pages 181

POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

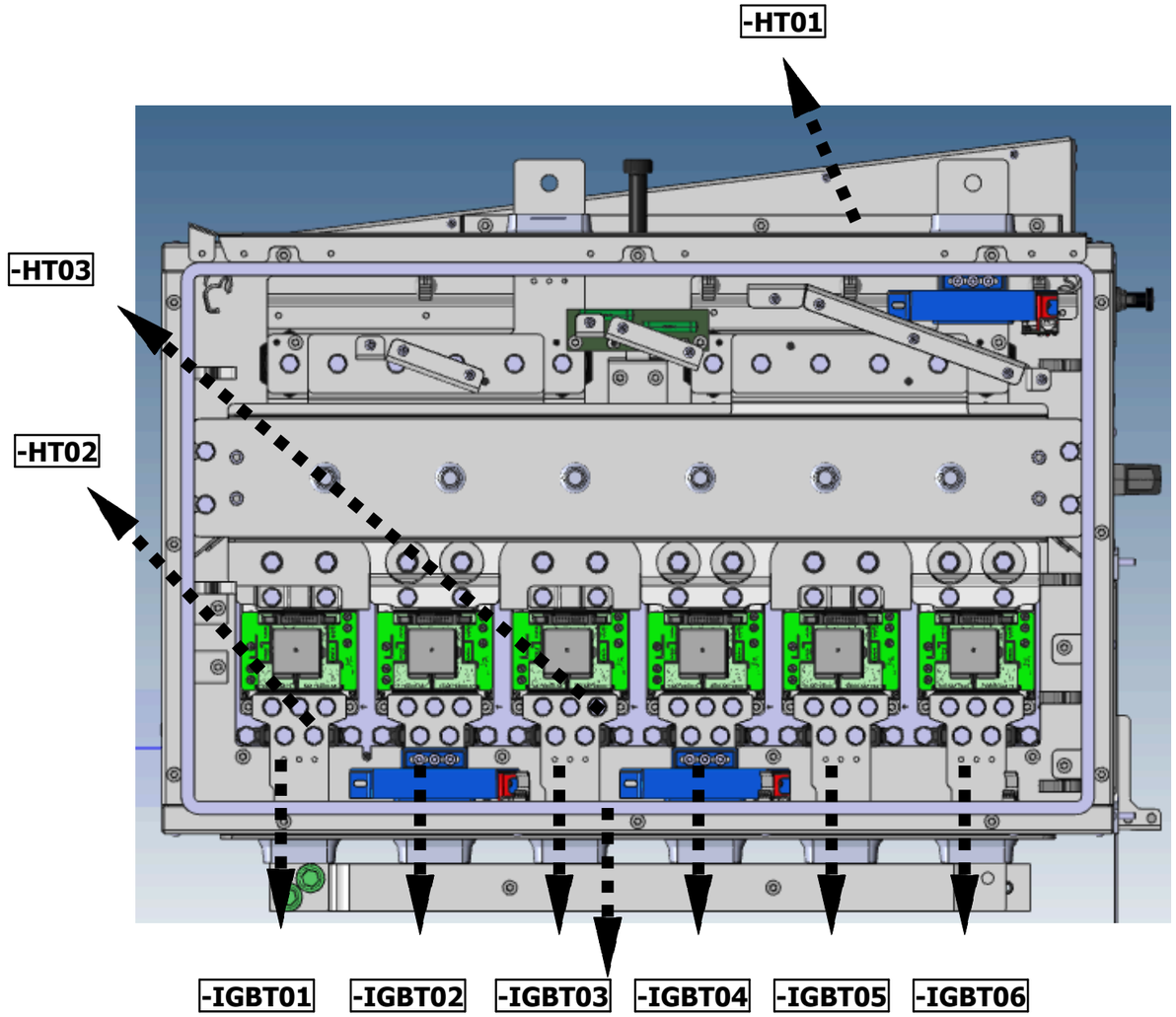
DCDC	DCDC GEN 3 CONVERTER
MOI	INVERTER MODULE AREA
ETR_BESS	REMOVABLE DC/AC STAGE BESS

ETR_BESS

REMOVABLE DC/AC STAGE BESS

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

PULL OUT ASSEMBLY (+ETR BESS)



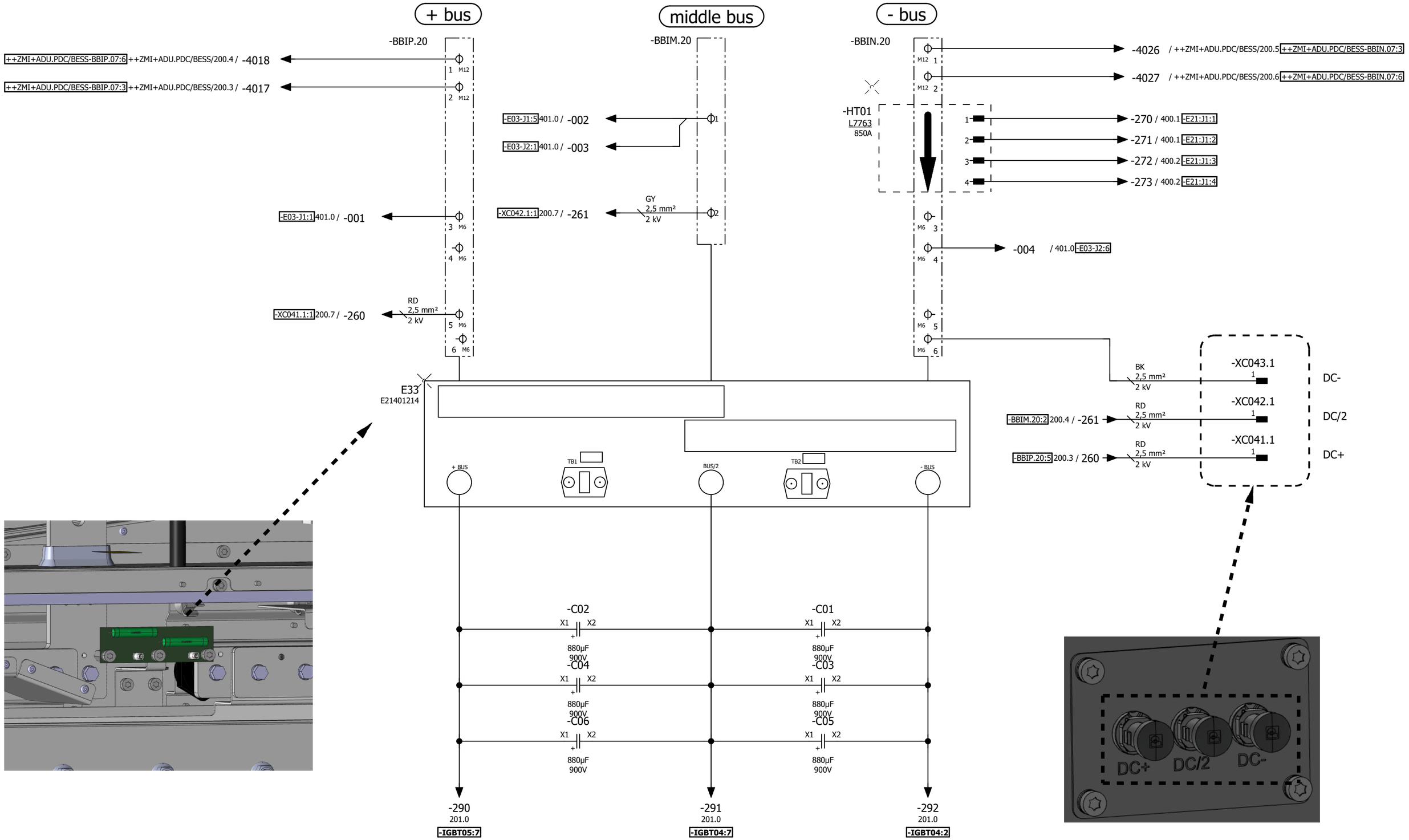
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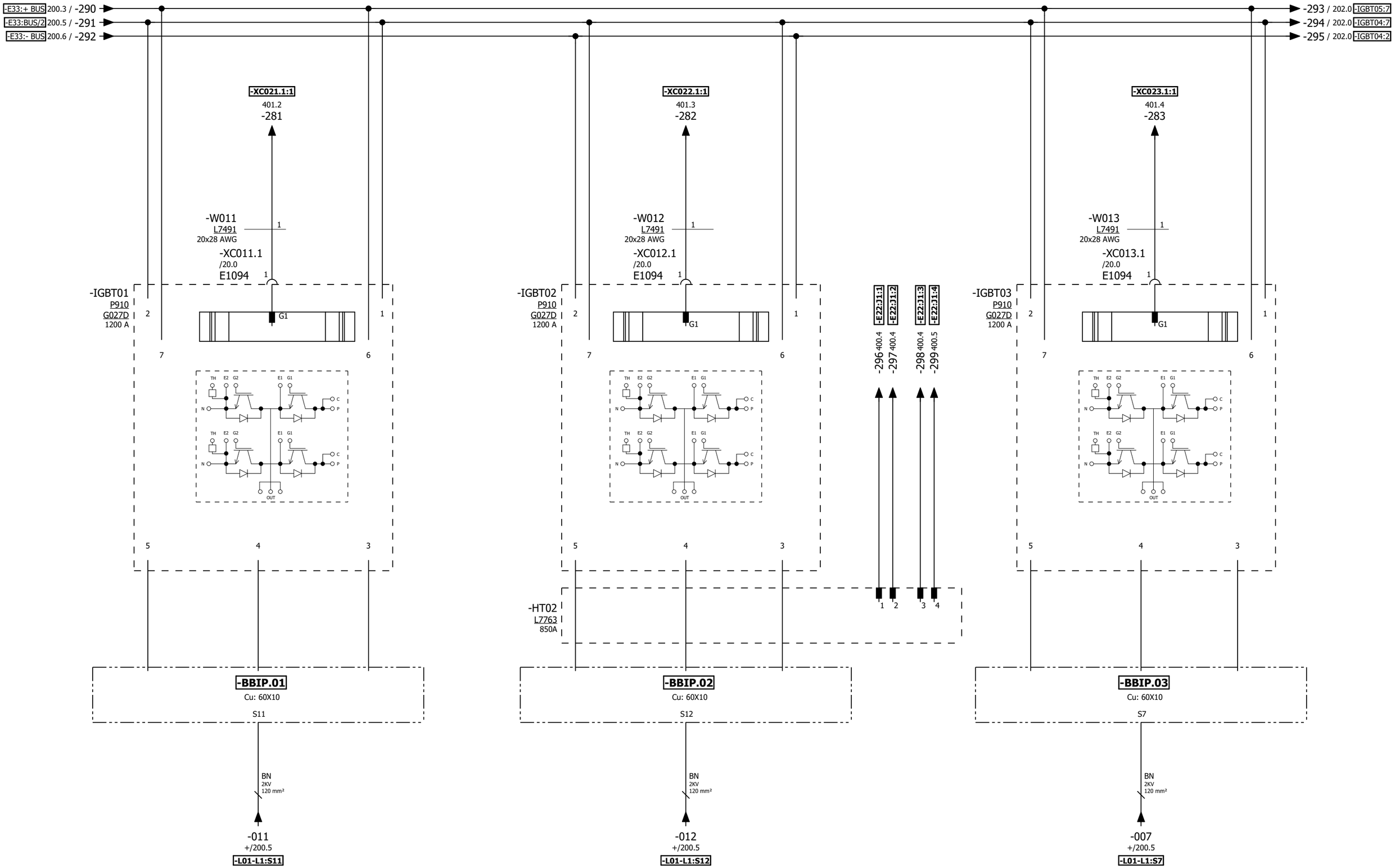
-XC011.1 = CONNECTOR FLAT IGBT 1
-XC012.1 = CONNECTOR FLAT IGBT 2
-XC013.1 = CONNECTOR FLAT IGBT 3
-XC014.1 = CONNECTOR FLAT IGBT 4
-XC015.1 = CONNECTOR FLAT IGBT 5
-XC016.1 = CONNECTOR FLAT IGBT 6

-XC021.1 = CONNECTOR FLAT IGBT 1
-XC022.1 = CONNECTOR FLAT IGBT 2
-XC023.1 = CONNECTOR FLAT IGBT 3
-XC024.1 = CONNECTOR FLAT IGBT 4
-XC025.1 = CONNECTOR FLAT IGBT 5
-XC026.1 = CONNECTOR FLAT IGBT 6

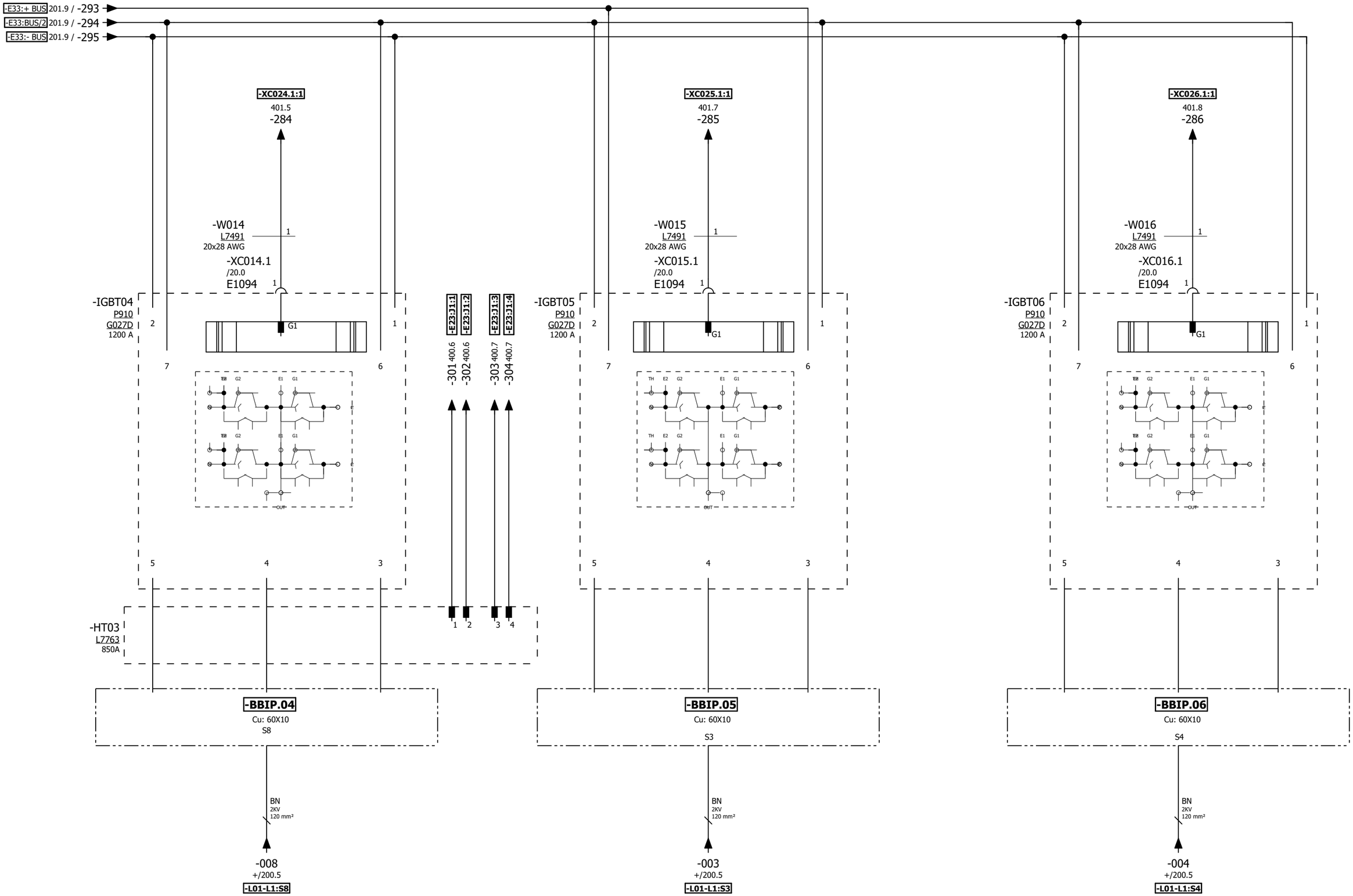
-XC031.1 = CONNECTOR FLAT 50 VIAS
-XC032.1 = CONNECTOR FLAT 40 VIAS

-XC041.1 = CONNECTOR DC+ SAT
-XC042.1 = CONNECTOR DC/2 SAT
-XC043.1 = CONNECTOR DC- SAT


+ bus
middle bus
- bus
-BBIP.20
-BBIM.20
-BBIN.20
++ZMI+ADU.PDC/BESS-BBIP.07:6 ++ZMI+ADU.PDC/BESS/200.4 / -4018
++ZMI+ADU.PDC/BESS-BBIP.07:3 ++ZMI+ADU.PDC/BESS/200.3 / -4017
-E03-J1:5 401.0 / -002
-E03-J2:1 401.0 / -003
-E03-J1:1 401.0 / -001
-XC042.1:1 200.7 / -261
GY 2,5 mm² 2 kV
-XC041.1:1 200.7 / -260
RD 2,5 mm² 2 kV
-4026 / ++ZMI+ADU.PDC/BESS/200.5 ++ZMI+ADU.PDC/BESS-BBIN.07:3
-4027 / ++ZMI+ADU.PDC/BESS/200.6 ++ZMI+ADU.PDC/BESS-BBIN.07:6
-HT01
L7763
850A
-270 / 400.1 -E21:J1:1
-271 / 400.1 -E21:J1:2
-272 / 400.2 -E21:J1:3
-273 / 400.2 -E21:J1:4
-004 / 401.0 -E03-J2:6
E33
E21401214
+ BUS
TB1
BUS/2
TB2
- BUS
BK 2,5 mm² 2 kV
RD 2,5 mm² 2 kV
-BBIM.20:2 200.4 / -261
RD 2,5 mm² 2 kV
-BBIP.20:5 200.3 / 260
-XC043.1
-XC042.1
-XC041.1
DC-
DC/2
DC+
-C02
-C01
-C04
-C03
-C06
-C05
X1 X2
880µF 900V
-290 201.0 -IGBT05:7
-291 201.0 -IGBT04:7
-292 201.0 -IGBT04:2
DC+ DC/2 DC-



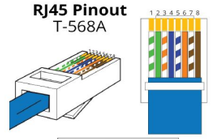
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	REVISED	REVISION 0.0.0			Mounting location: REMOVABLE DC/AC STAGE BESS ETR_BESS	Previous page 200 / Next page 202
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 201 / Total number of pages 181

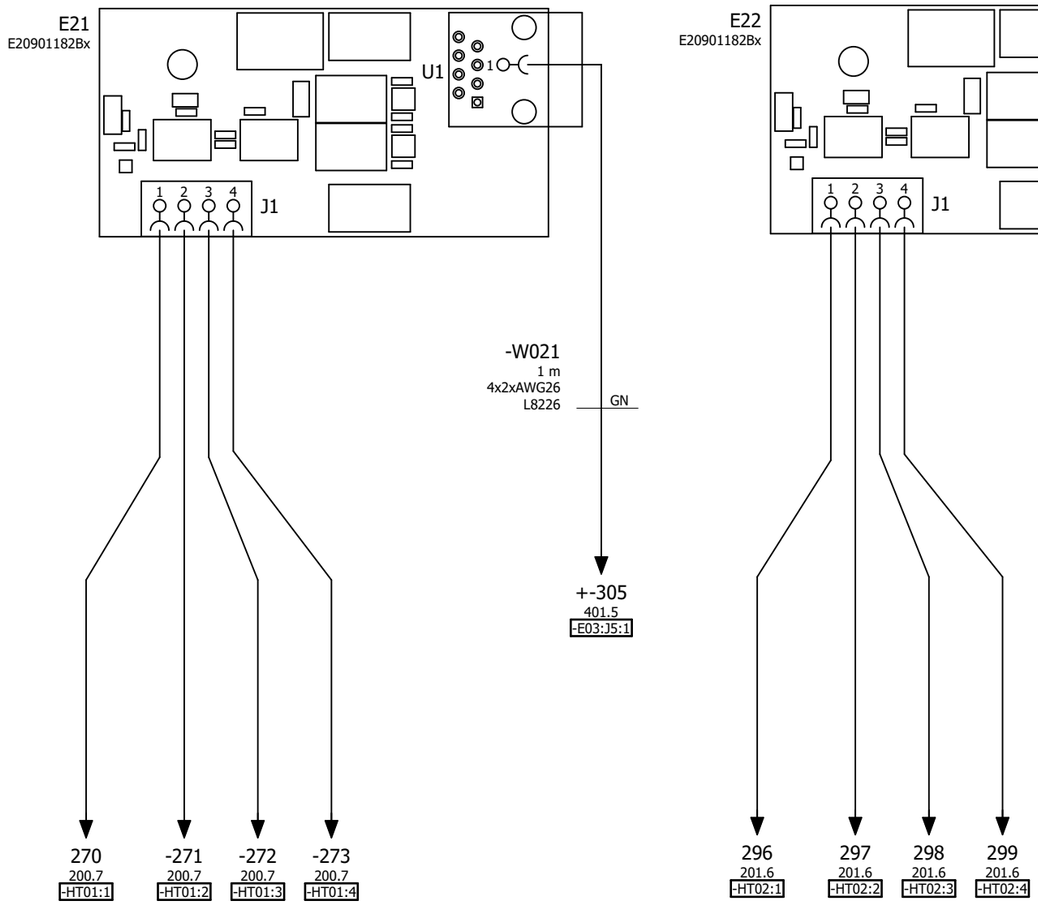


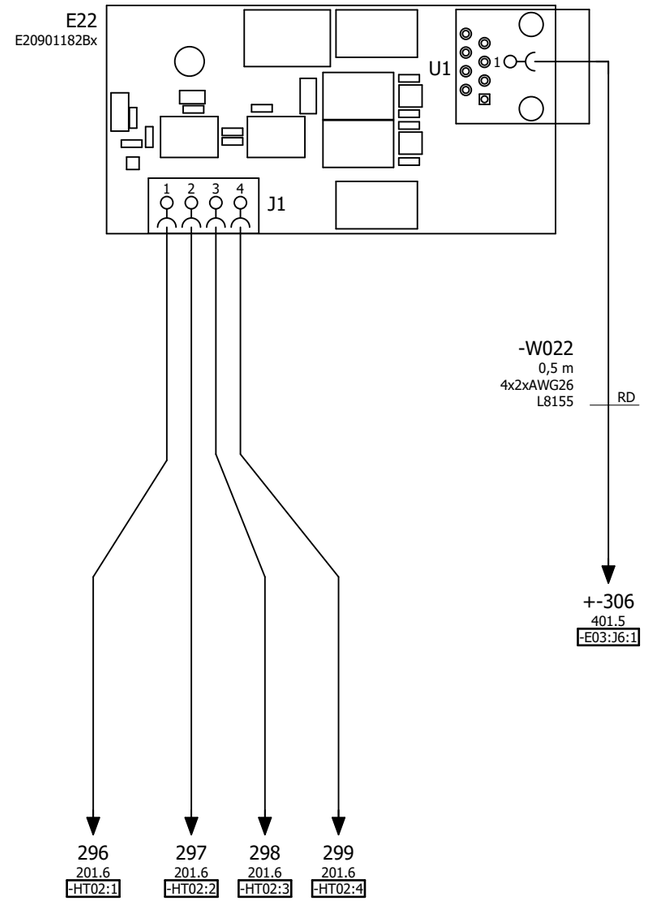
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	Igbt assembly BESS	Mounting location: REMOVABLE DC/AC STAGE BESS ETR_BESS	Previous page 201 / Next page 400
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 202 / Total number of pages 181

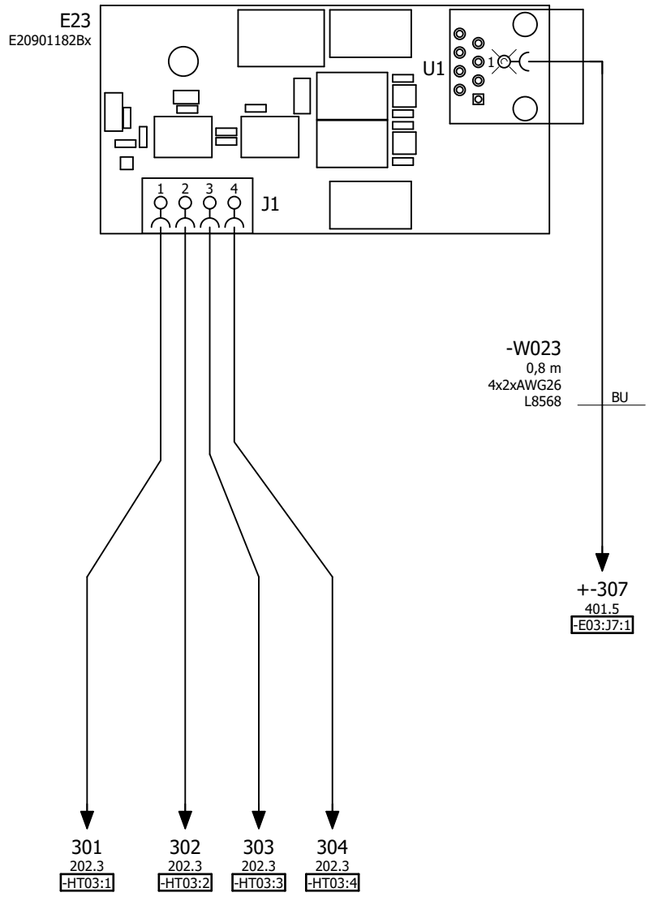
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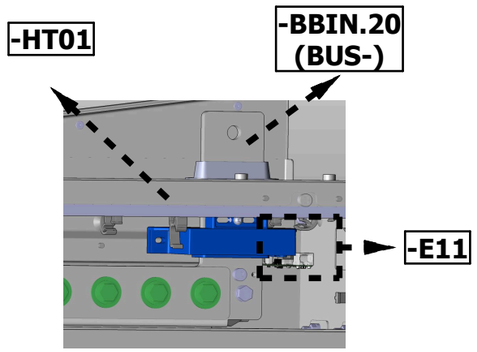
RJ45 Pinout
T-568A

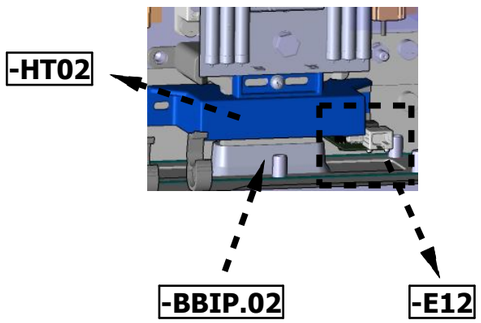


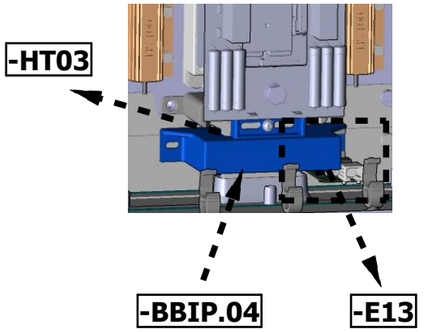


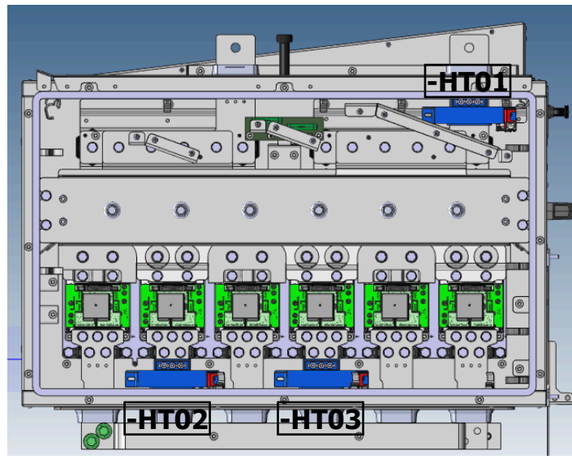




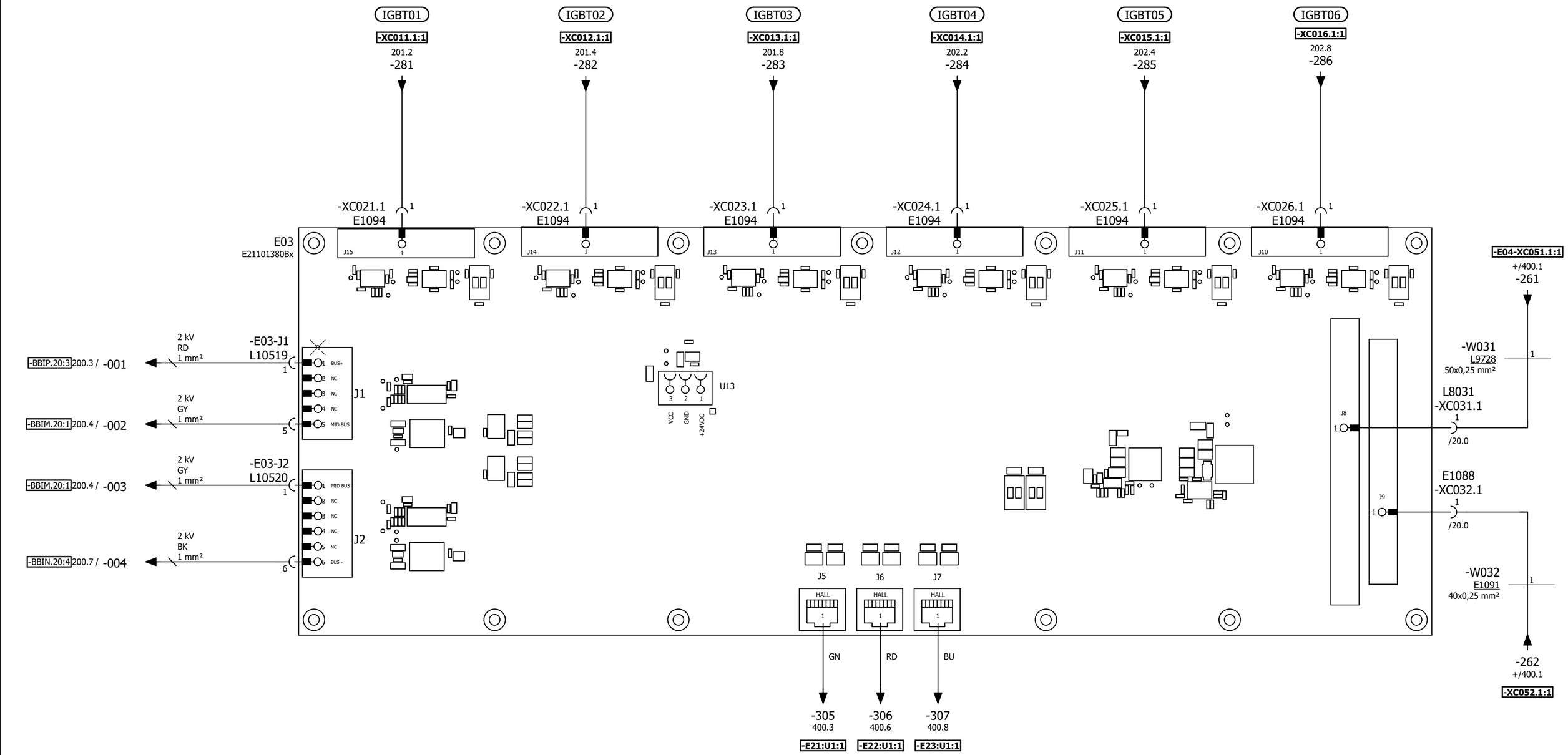








POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	V/I conversor	Mounting location: REMOVABLE DC/AC STAGE BESS ETR_BESS	Previous page 202 / Next page 401
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 400 / Total number of pages 181



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: PCB control and measures	Installation: DCDC GEN 3 CONVERTER	DCDC / A3
	REVISED	REVISION 0.0.0			Mounting location: REMOVABLE DC/AC STAGE BESS ETR_BESS	Previous page 400 / Next page &MAT/1000
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 401 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-C01	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C02	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C03	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C04	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C05	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C06	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-E03				1
-E03-J1	L10519	2-2178029-5	CONECTOR RAST5 AÉREO 5 VIAS PASO 5	1
-E03-J1	L10521	964202-1	TERMINAL HEMBRA RAST5 AWG13-17 TE	2
-E03-J2	L10520	2322912-6	CONECTOR RAST5 AÉREO 6 VIAS PASO 5	1
-E03-J2	L10521	964202-1	TERMINAL HEMBRA RAST5 AWG13-17 TE	2
-E21	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E22	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E23	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E33	PCB E21401214	PCB E21401214	HEM GEN3 BUS BALANCING PCB BOARD	1
-HT01	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-HT02	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-HT03	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-IGBT01	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT01	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT02	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT02	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT03	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT03	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT04	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT04	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT05	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT05	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT06	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT06	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1

Cable overview

Cable designation	Function text	ERP
=DCDC++MOI+ETR_BESS-W011	TWISTED WIRE TRIP IGBT 1	L7491
=DCDC++MOI+ETR_BESS-W012	TWISTED WIRE TRIP IGBT 3	L7491
=DCDC++MOI+ETR_BESS-W013	TWISTED WIRE TRIP IGBT 2	L7491
=DCDC++MOI+ETR_BESS-W014	TWISTED WIRE TRIP IGBT 1	L7491
=DCDC++MOI+ETR_BESS-W015	TWISTED WIRE TRIP IGBT 3	L7491
=DCDC++MOI+ETR_BESS-W016	TWISTED WIRE TRIP IGBT 2	L7491
=DCDC++MOI+ETR_BESS-W021		L8226
=DCDC++MOI+ETR_BESS-W022		L8155
=DCDC++MOI+ETR_BESS-W023		L8568
=DCDC++MOI+ETR_BESS-W031	TWISTED WIRE TRIP IGBT 6	L9728
=DCDC++MOI+ETR_BESS-W032	TWISTED WIRE TRIP IGBT 6	E1091

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++MOI+ETR_BESS-E03-J1		L10519	1
		L10521	2
=DCDC++MOI+ETR_BESS-E03-J2		L10520	1
		L10521	2
=DCDC++MOI+ETR_BESS-XC011.1	CONNECTOR FLAT IGBT 1	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC012.1	CONNECTOR FLAT IGBT 2	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC013.1	CONNECTOR FLAT IGBT 3	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC014.1	CONNECTOR FLAT IGBT 4	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC015.1	CONNECTOR FLAT IGBT 5	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC016.1	CONNECTOR FLAT IGBT 6	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC021.1	CONNECTOR FLAT IGBT 1	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC022.1	CONNECTOR FLAT IGBT 2	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC023.1	CONNECTOR FLAT IGBT 3	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC024.1	CONNECTOR FLAT IGBT 4	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC025.1	CONNECTOR FLAT IGBT 5	E1094	1
		E1095	1
=DCDC++MOI+ETR_BESS-XC026.1	CONNECTOR FLAT IGBT 6	E1094	1
		E1095	1

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++MOI+ETR_BESS-XC031.1	CONNECTOR FLAT 50 VIAS	L8031	1
		L8029	1
=DCDC++MOI+ETR_BESS-XC032.1	CONNECTOR FLAT 40 VIAS	E1088	1
		E1089	1
=DCDC++MOI+ETR_BESS-XC041.1	CONNECTOR DC+ SAT	L9570	1
		L9571	1
		L9572	1
		L9573	1
=DCDC++MOI+ETR_BESS-XC042.1	CONNECTOR DC/2 SAT	L9570	1
		L9571	1
		L9572	1
		L9573	1
=DCDC++MOI+ETR_BESS-XC043.1	CONNECTOR DC- SAT	L9570	1
		L9571	1
		L9572	1
		L9573	1

POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

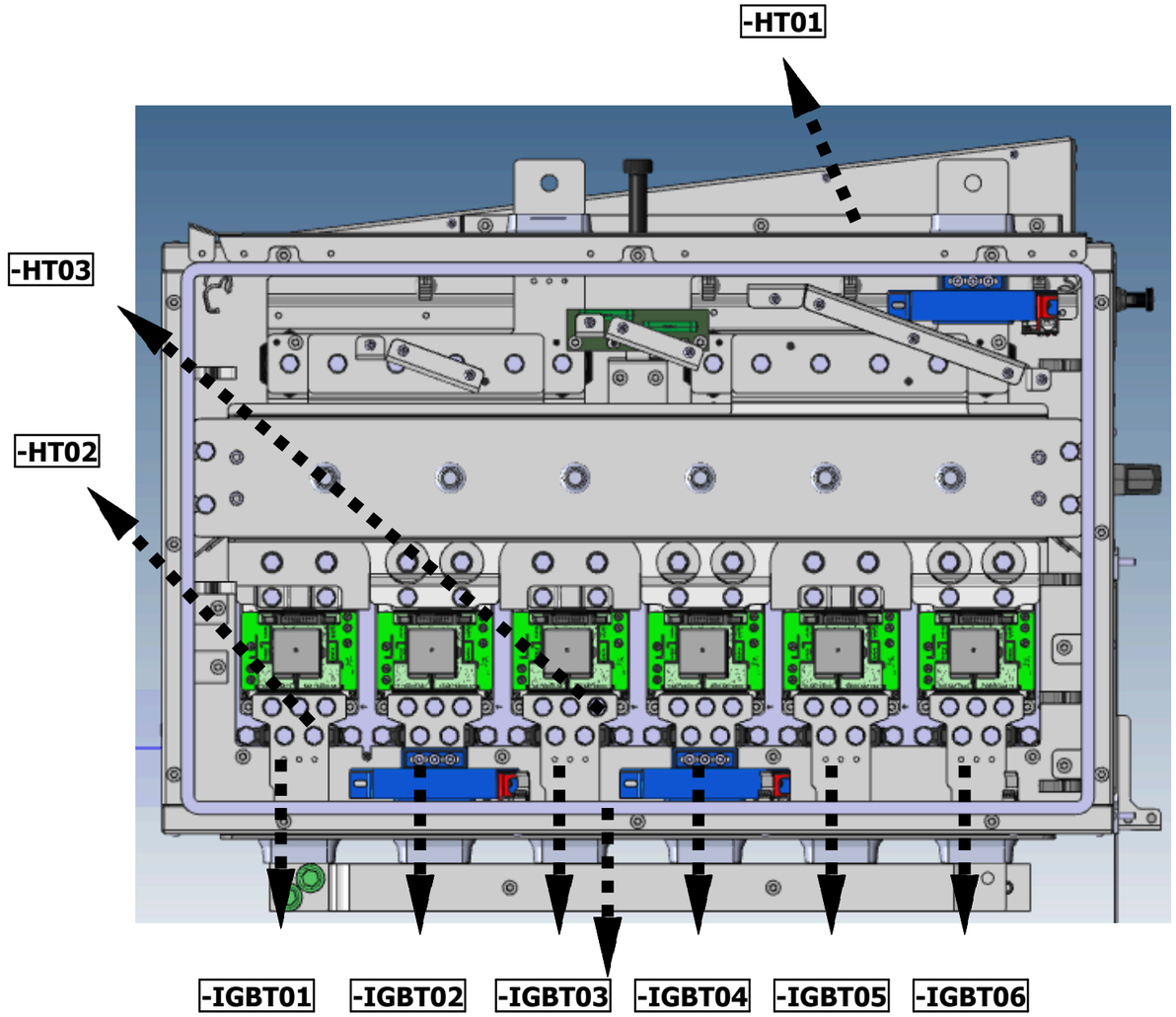
DCDC	DCDC GEN 3 CONVERTER
MOI	INVERTER MODULE AREA
ETR_PV	REMOVABLE DC/AC STAGE PV

ETR_PV

REMOVABLE DC/AC STAGE PV

Created on	29/05/2024	Creator	L. PUERTES
Modification date	30/05/2024		

PULL OUT ASSEMBLY (+ETR PV)



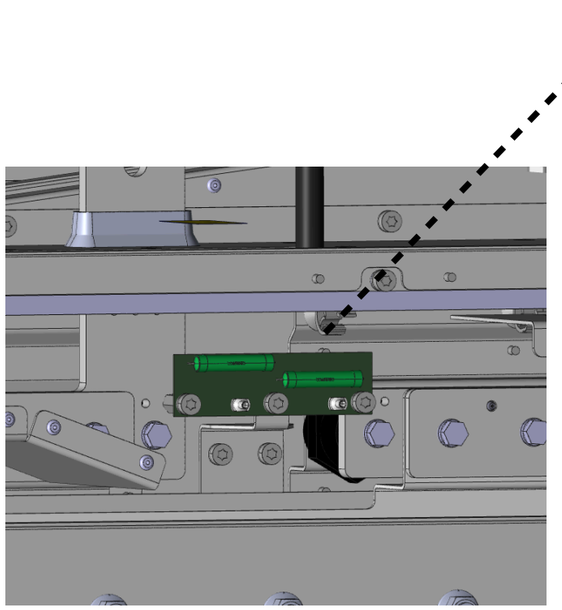
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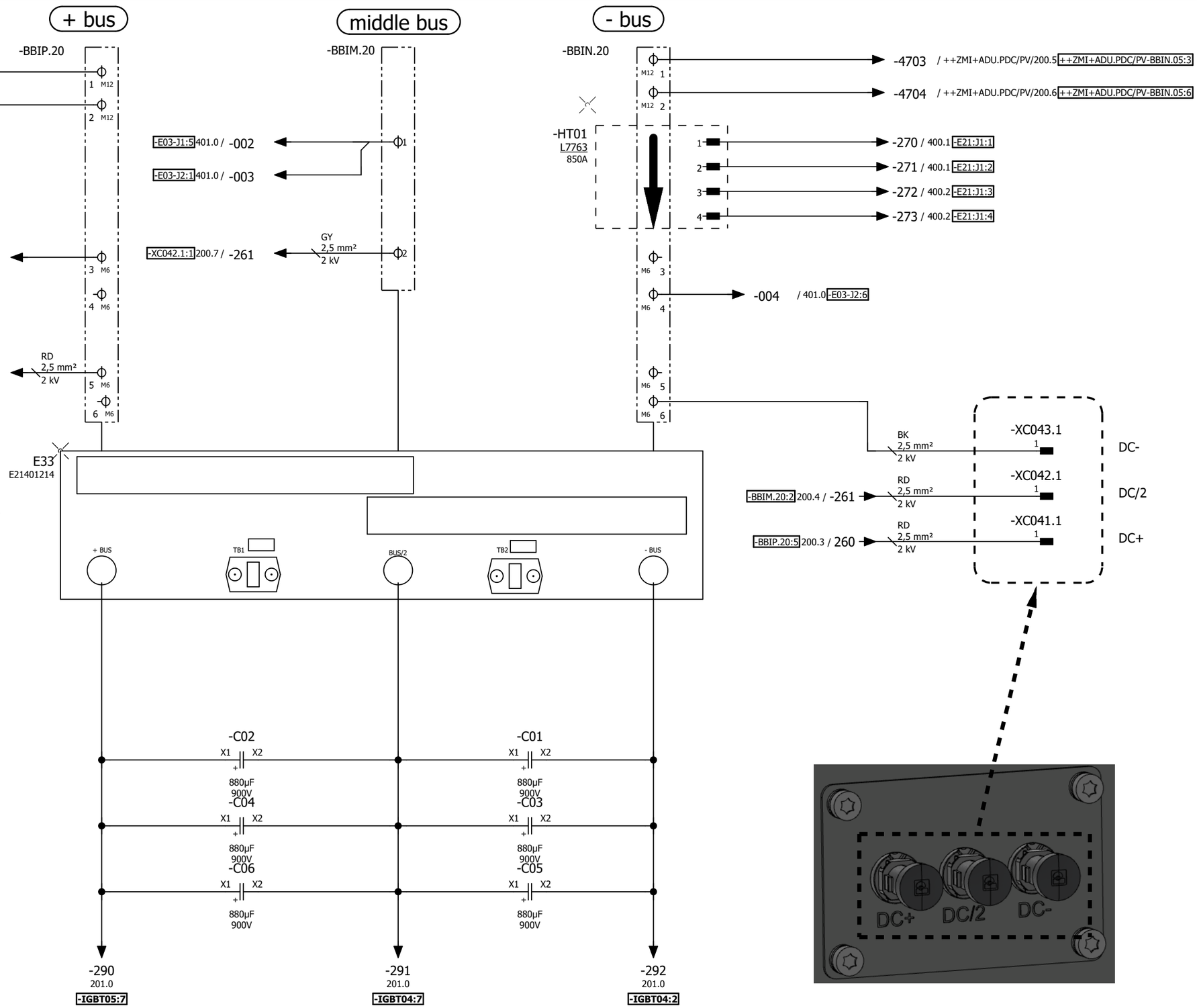
-XC011.1 = CONNECTOR FLAT IGBT 1
-XC012.1 = CONNECTOR FLAT IGBT 2
-XC013.1 = CONNECTOR FLAT IGBT 3
-XC014.1 = CONNECTOR FLAT IGBT 4
-XC015.1 = CONNECTOR FLAT IGBT 5
-XC016.1 = CONNECTOR FLAT IGBT 6

-XC021.1 = CONNECTOR FLAT IGBT 1
-XC022.1 = CONNECTOR FLAT IGBT 2
-XC023.1 = CONNECTOR FLAT IGBT 3
-XC024.1 = CONNECTOR FLAT IGBT 4
-XC025.1 = CONNECTOR FLAT IGBT 5
-XC026.1 = CONNECTOR FLAT IGBT 6

-XC031.1 = CONNECTOR FLAT 50 VIAS
-XC032.1 = CONNECTOR FLAT 40 VIAS

-XC041.1 = CONNECTOR DC+ SAT
-XC042.1 = CONNECTOR DC/2 SAT
-XC043.1 = CONNECTOR DC- SAT





POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER DCDC	A3
	REVISED	REVISION	ESQ_FD1200U000002_000	Capacitor bus dc-link PV	Mounting location: REMOVABLE DC/AC STAGE PV ETR_PV	Previous page 20 / Next page 201
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 200 / Total number of pages 181

-E33:+ BUS 200.3 / -290
-E33:BUS/2 200.5 / -291
-E33:- BUS 200.6 / -292

-293 / 202.0 -IGBT05:7
-294 / 202.0 -IGBT04:7
-295 / 202.0 -IGBT04:2

-XC021.1:1
401.2
-281

-W011
L7491
20x28 AWG
1
-XC011.1
/20.0
E1094
1

-XC022.1:1
401.3
-282

-W012
L7491
20x28 AWG
1
-XC012.1
/20.0
E1094
1

-XC023.1:1
401.4
-283

-W013
L7491
20x28 AWG
1
-XC013.1
/20.0
E1094
1

-IGBT01
P910
G027D
1200 A
2 1 7 6
G1
TH E2 G2 E1 G1
OUT
5 4 3

-IGBT02
P910
G027D
1200 A
2 1 7 6
G1
TH E2 G2 E1 G1
OUT
5 4 3

-IGBT03
P910
G027D
1200 A
2 1 7 6
G1
TH E2 G2 E1 G1
OUT
5 4 3

-E22:J1:1 -296 400.4
-E22:J1:2 -297 400.4
-E22:J1:3 -298 400.4
-E22:J1:4 -299 400.5

-HT02
L7763
850A
1 2 3 4

-BBIP.01
Cu: 60X10
S1

-BBIP.02
Cu: 60X10
S2

-BBIP.03
Cu: 60X10
S5

BN
2KV
120 mm²

-001
+/200.5
-L01-L1:S1

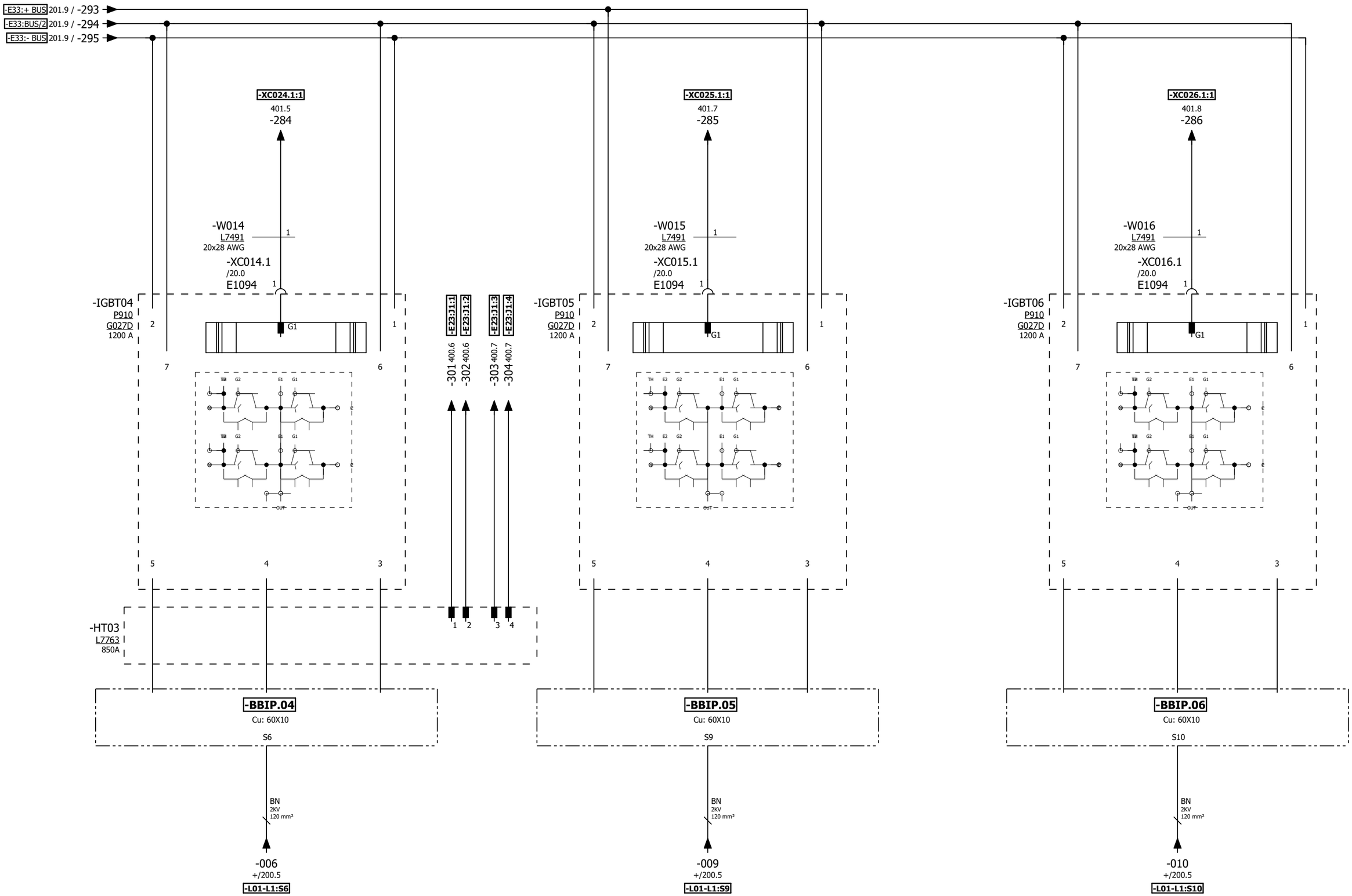
BN
2KV
120 mm²

-002
+/200.5
-L01-L1:S2

BN
2KV
120 mm²

-005
+/200.5
-L01-L1:S5

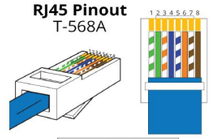
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC / A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	Igbt assembly PV	Mounting location: REMOVABLE DC/AC STAGE PV ETR_PV	Previous page 200 / Next page 202
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 201 / Total number of pages 181

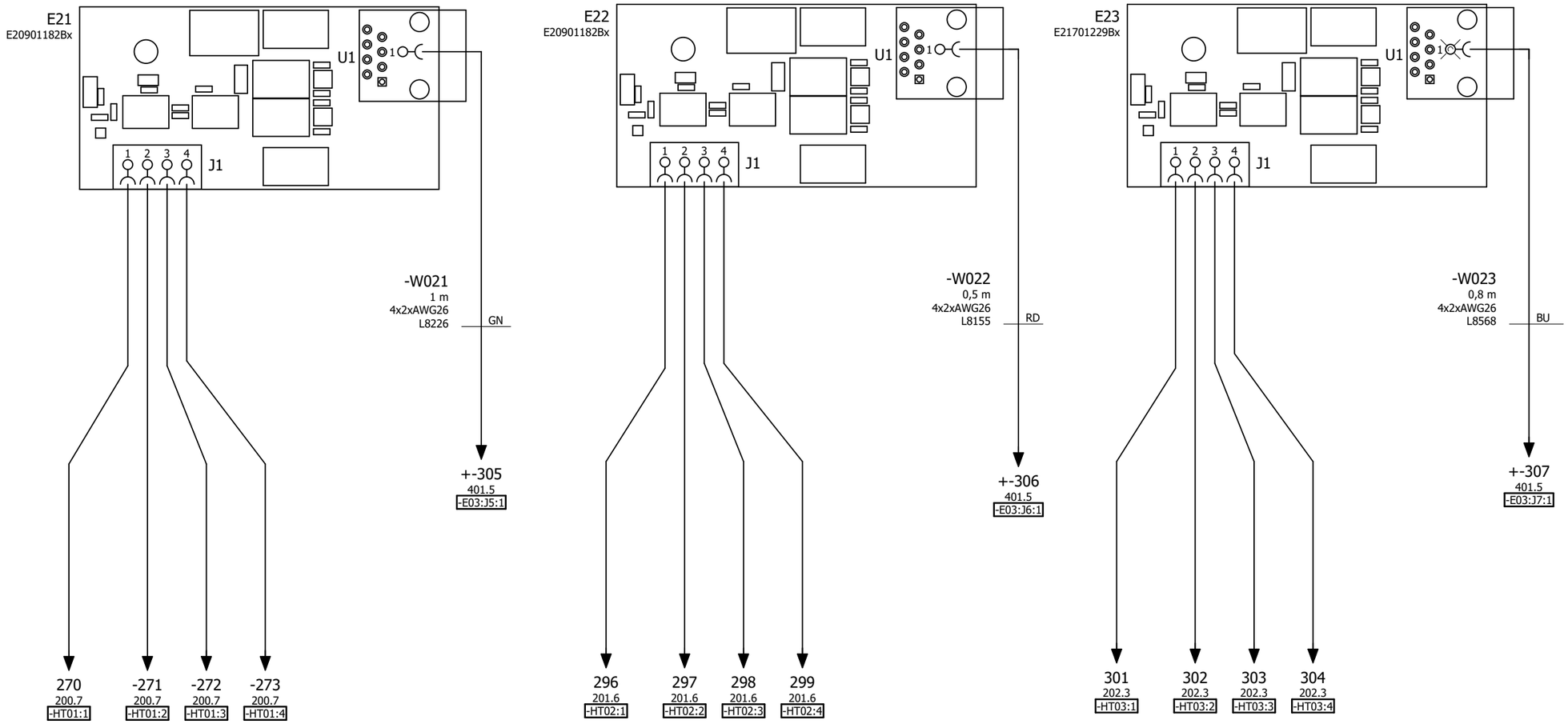


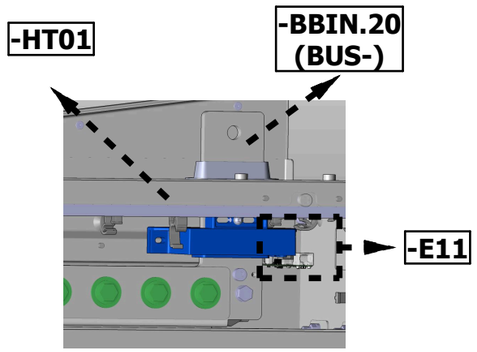
POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION 0.0.0	ESQ_FD1200U000002_000	Igbt assembly PV	Mounting location: REMOVABLE DC/AC STAGE PV ETR_PV	Previous page 201 / Next page 400
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 202 / Total number of pages 181

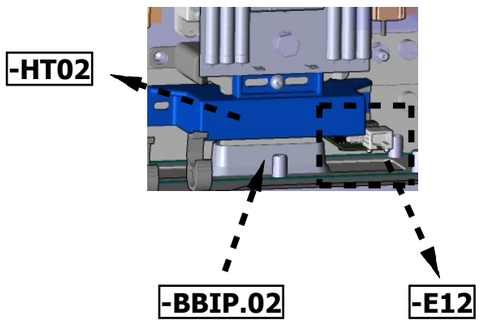
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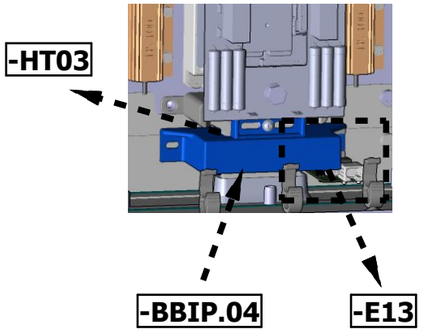
RJ45 Pinout
T-568A

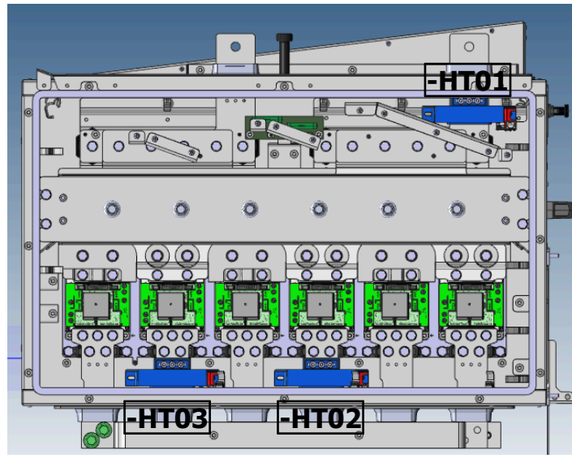






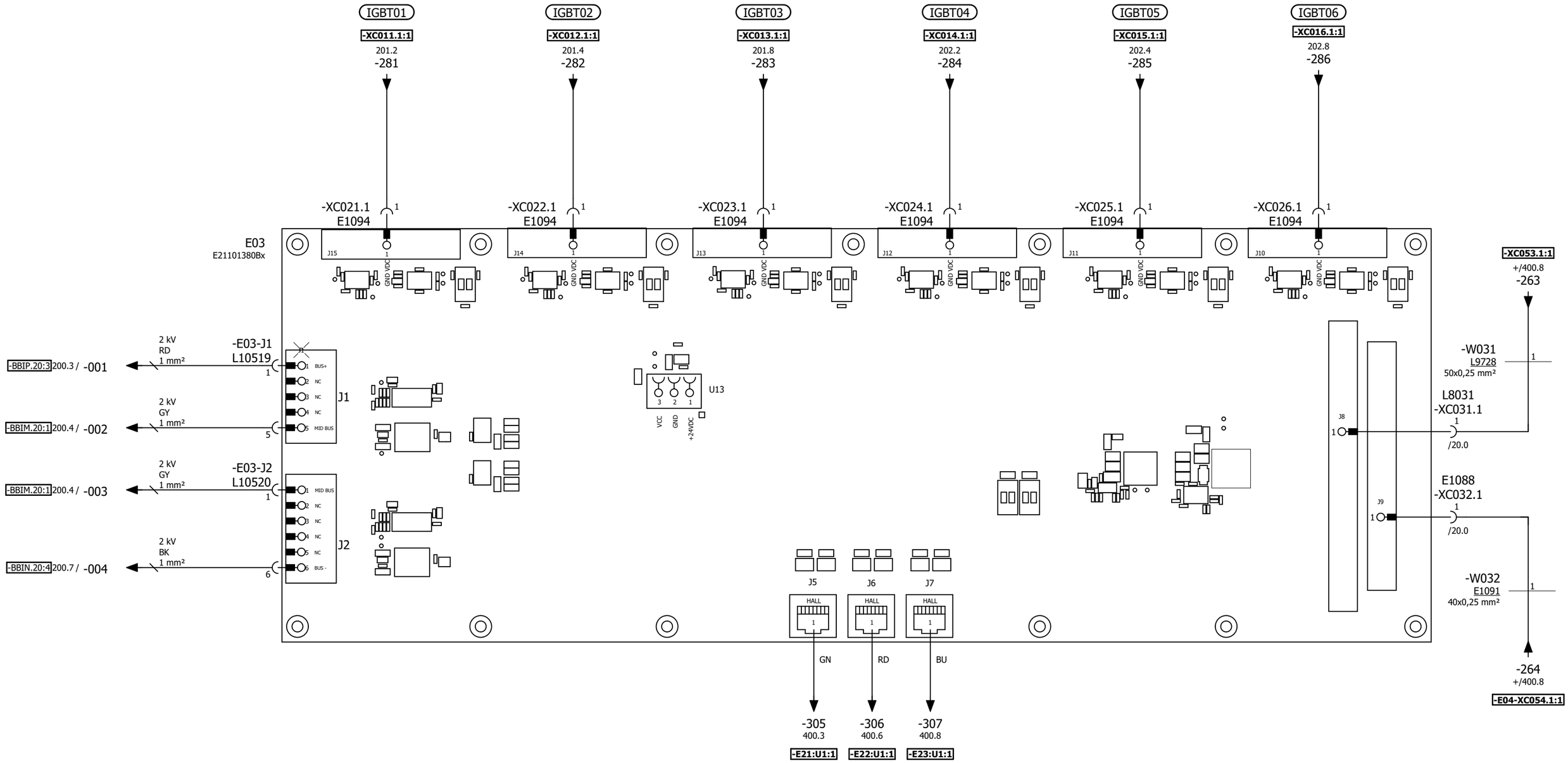






POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project	Name	Installation: DCDC GEN 3 CONVERTER	DCDC A3
	REVISED	REVISION	ESQ_FD1200U000002_000	V/I conversor	Mounting location: REMOVABLE DC/AC STAGE PV ETR_PV	Previous page 202 / Next page 401
	APPROVED A.CABALLERO	0.0.0			Project Code: FD1200U000002	Page 400 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9



POWER ELECTRONICS®	DESIGNED L. PUERTES	DATE	Project: ESQ_FD1200U000002_000	Name: PCB control and measures	Installation: DCDC GEN 3 CONVERTER	DCDC / A3
	REVISED	REVISION 0.0.0			Mounting location: REMOVABLE DC/AC STAGE PV ETR_PV	Previous page 400 / Next page &MAT/1000
	APPROVED A.CABALLERO				Project Code: FD1200U000002	Page 401 / Total number of pages 181

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-C01	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C02	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C03	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C04	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C05	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C06	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-E03				1
-E03-J1	L10519	2-2178029-5	CONECTOR RAST5 AÉREO 5 VIAS PASO 5	1
-E03-J1	L10521	964202-1	TERMINAL HEMBRA RAST5 AWG13-17 TE	2
-E03-J2	L10520	2322912-6	CONECTOR RAST5 AÉREO 6 VIAS PASO 5	1
-E03-J2	L10521	964202-1	TERMINAL HEMBRA RAST5 AWG13-17 TE	2
-E21	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E22	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E23	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E33	PCB E21401214	PCB E21401214	HEM GEN3 BUS BALANCING PCB BOARD	1
-HT01	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-HT02	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-HT03	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-IGBT01	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT01	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT02	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT02	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT03	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT03	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT04	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT04	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT05	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT05	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT06	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT06	G027D		G-Drive LV100 Tamura 2LG03ACDC11N-01	1

Cable overview

Cable designation	Function text	ERP
=DCDC++MOI+ETR_PV-W011	TWISTED WIRE TRIP IGBT 1	L7491
=DCDC++MOI+ETR_PV-W012	TWISTED WIRE TRIP IGBT 3	L7491
=DCDC++MOI+ETR_PV-W013	TWISTED WIRE TRIP IGBT 2	L7491
=DCDC++MOI+ETR_PV-W014	TWISTED WIRE TRIP IGBT 1	L7491
=DCDC++MOI+ETR_PV-W015	TWISTED WIRE TRIP IGBT 3	L7491
=DCDC++MOI+ETR_PV-W016	TWISTED WIRE TRIP IGBT 2	L7491
=DCDC++MOI+ETR_PV-W021		L8226
=DCDC++MOI+ETR_PV-W022		L8155
=DCDC++MOI+ETR_PV-W023		L8568
=DCDC++MOI+ETR_PV-W031	TWISTED WIRE TRIP IGBT 6	L9728
=DCDC++MOI+ETR_PV-W032	TWISTED WIRE TRIP IGBT 6	E1091

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++MOI+ETR_PV-E03-J1		L10519	1
		L10521	2
=DCDC++MOI+ETR_PV-E03-J2		L10520	1
		L10521	2
=DCDC++MOI+ETR_PV-XC011.1	CONNECTOR FLAT IGBT 1	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC012.1	CONNECTOR FLAT IGBT 2	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC013.1	CONNECTOR FLAT IGBT 3	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC014.1	CONNECTOR FLAT IGBT 4	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC015.1	CONNECTOR FLAT IGBT 5	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC016.1	CONNECTOR FLAT IGBT 6	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC021.1	CONNECTOR FLAT IGBT 1	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC022.1	CONNECTOR FLAT IGBT 2	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC023.1	CONNECTOR FLAT IGBT 3	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC024.1	CONNECTOR FLAT IGBT 4	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC025.1	CONNECTOR FLAT IGBT 5	E1094	1
		E1095	1
=DCDC++MOI+ETR_PV-XC026.1	CONNECTOR FLAT IGBT 6	E1094	1
		E1095	1

0	1	2	3	4	5	6	7	8	9

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=DCDC++MOI+ETR_PV-XC031.1	CONNECTOR FLAT 50 VIAS	L8031	1
		L8029	1
=DCDC++MOI+ETR_PV-XC032.1	CONNECTOR FLAT 40 VIAS	E1088	1
		E1089	1
=DCDC++MOI+ETR_PV-XC041.1	CONNECTOR DC+ SAT	L9570	1
		L9571	1
		L9572	1
		L9573	1
=DCDC++MOI+ETR_PV-XC042.1	CONNECTOR DC/2 SAT	L9570	1
		L9571	1
		L9572	1
		L9573	1
=DCDC++MOI+ETR_PV-XC043.1	CONNECTOR DC- SAT	L9570	1
		L9571	1
		L9572	1
		L9573	1